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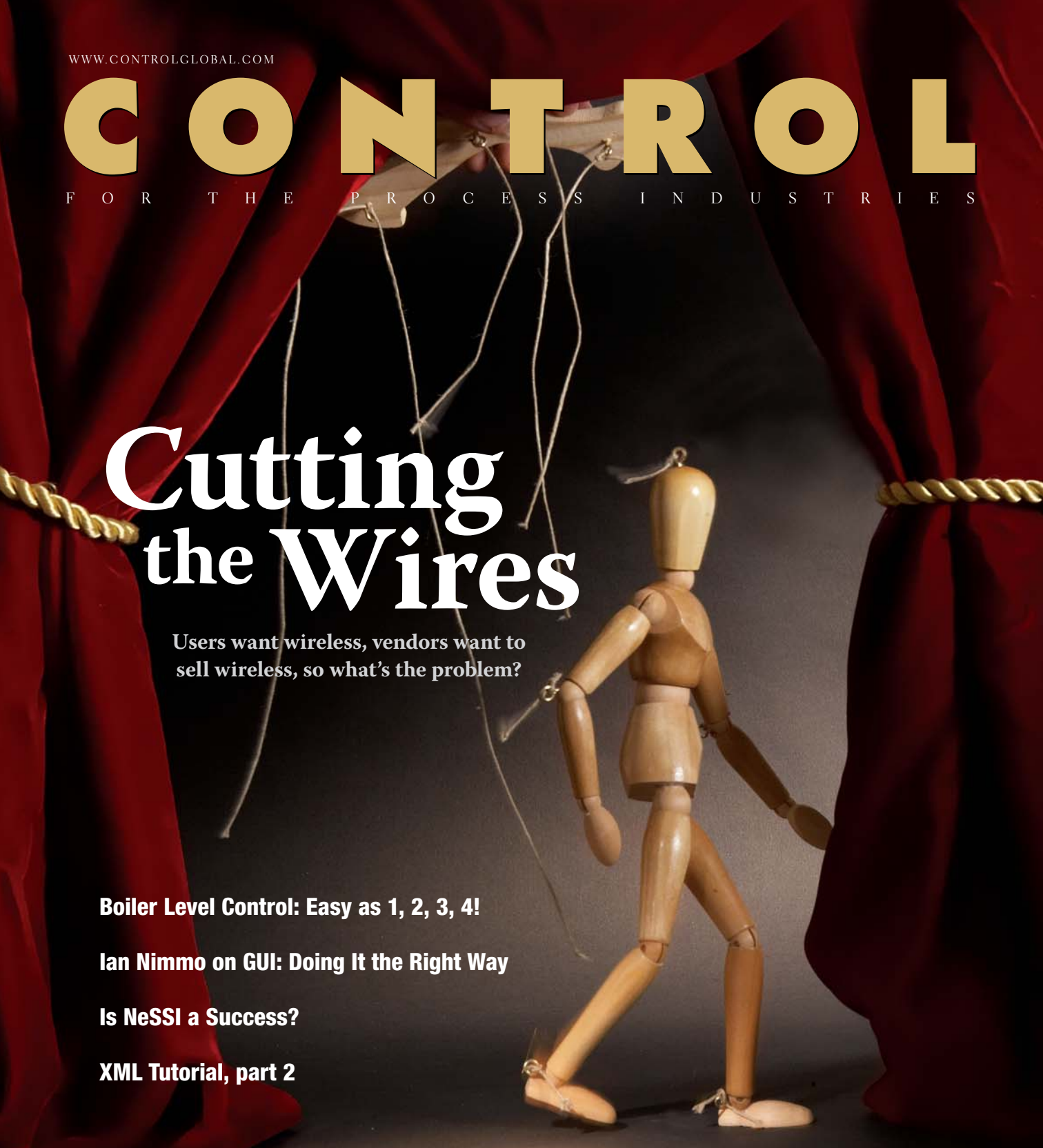

Cutting the Wires

Users want wireless, vendors want to sell wireless, so what's the problem?

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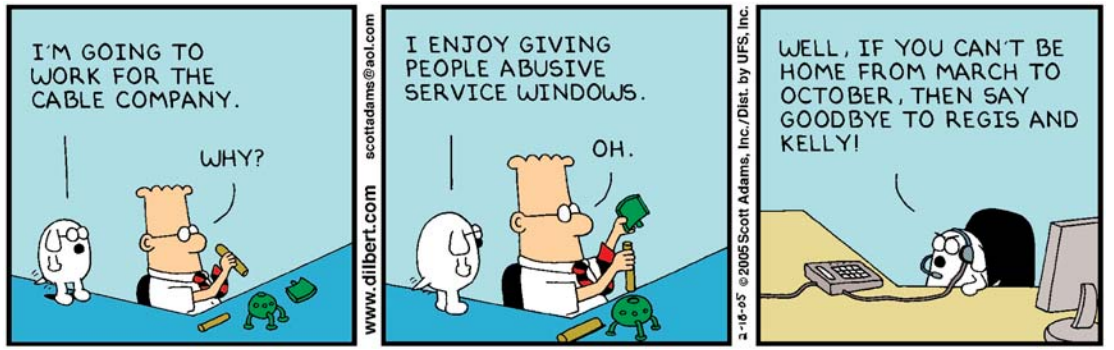

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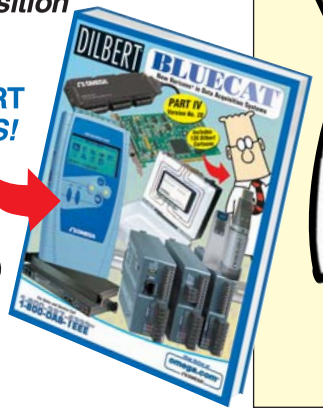

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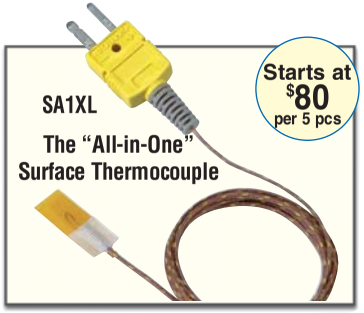

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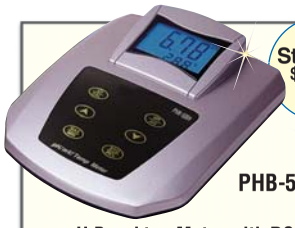

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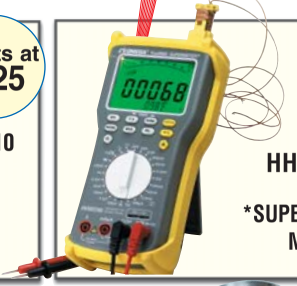

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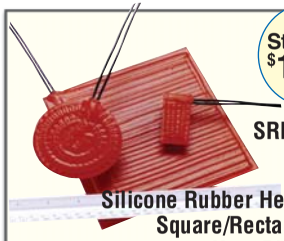

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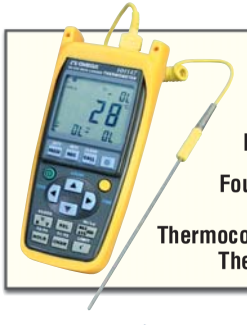

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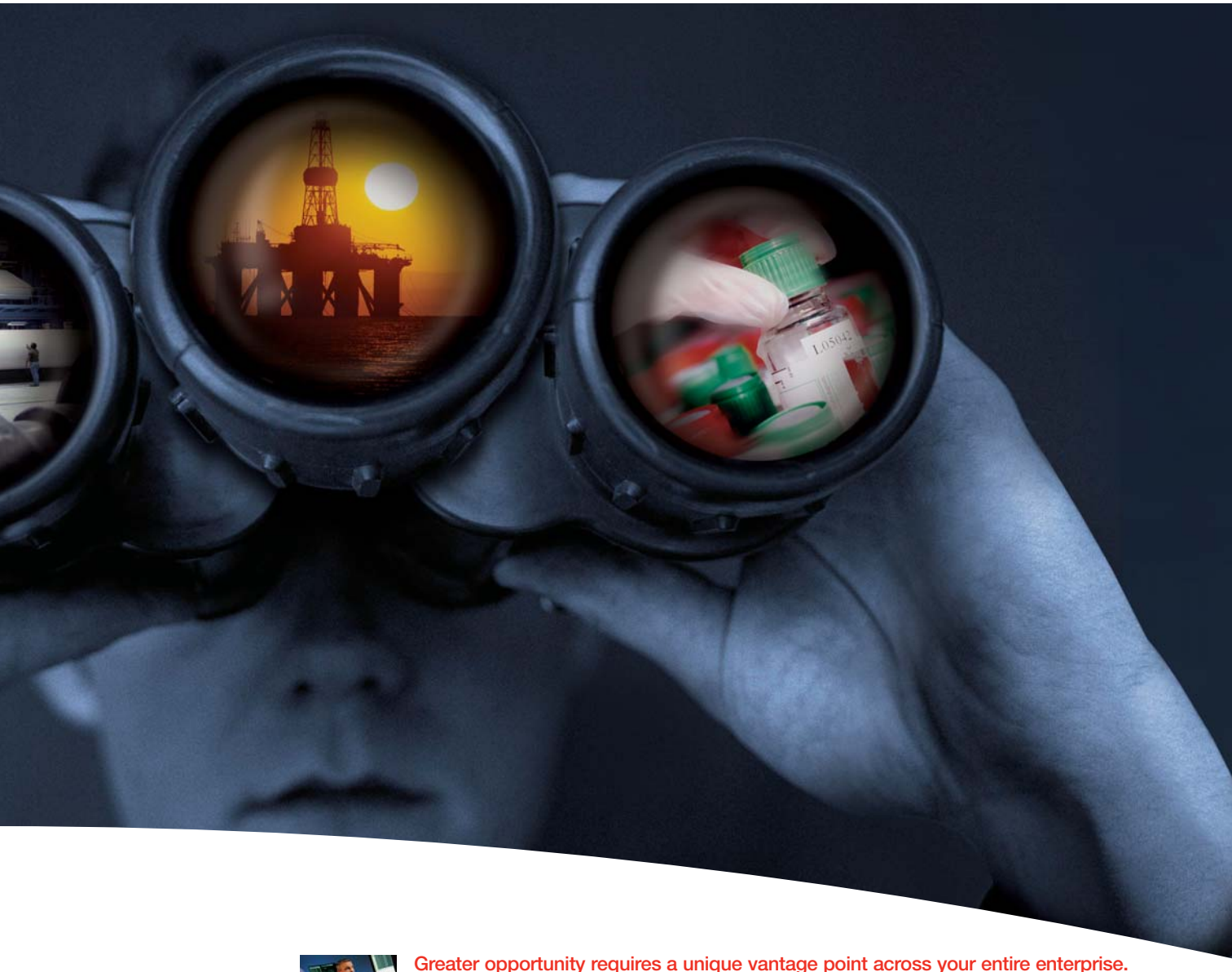

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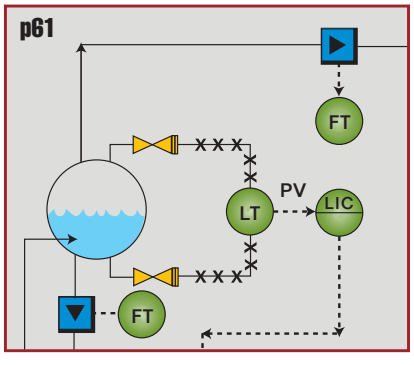

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Chemicals & Allied Products	15,926
Food & Kindred Products	15,821
Paper & Allied Products	4,087
Primary Metal Industries	5,341
Electric, Gas & Sanitary Services	2,707
System Integrators & Engineering Design Firms	7,933
Rubber and Miscellaneous Plastic Products	5,016
Stone, Clay, Glass & Concrete products	2,580
Textile Mill Products	2,088
Petroleum Refining & Related Industries	3,361
Tobacco Products	167
Total circulation	65,027

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Reasons to Buy -
It's more DURABLE

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Here are just a few of the reasons C-more is one of the most durable touch panels you can buy

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- Wide range of color gradients
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More high end features
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More durable
- 50,000 hour bulb life (all sizes)
- User replaceable bulbs on 8,10,12,15-inch
- Replaceable bezel with touch overlay
- Wide operating temperature range

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- FTP data exchange over Internet
- Built-in Ethernet PLC drivers
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6-inch STN, grayscale	$425
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** Does not include features requiring Ethernet, audio, Compact Flash (CF).

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PUTMAN MEDIA INC.
555 W. PIERCE RD., SUITE 301
ITASCA, ILLINOIS 60143
630/467-1300
FAX: 630/467-1124
WWW.CONTROLGLOBAL.COM

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EDITORIAL TEAM

EDITOR IN CHIEF: WALT BOYES
wboyes@putman.net

EXECUTIVE EDITOR: JIM MONTAGUE
jmontague@putman.net

DIGITAL MANAGING EDITOR: RICK PEDRAZA
rpedraza@putman.net

SENIOR TECHNICAL EDITOR: JOE FEELEY
jfeeley@putman.net

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The Summertime Blues

We're getting into high summer. The User Group Madness has died down, temporarily, to resurge once again in September, October, and clear into the beginning of December. I'm going to Seville, Spain, and to Bahrain in November to speak to user groups. User groups are great! Honeywell, Emerson, Siemens, Rockwell, Invensys, and all the others are trying very hard to walk that fine line between empowering their customers and letting their customers walk all over them.

While it's true that there's a fair share of "Evil Vendors," there's a whomping, big number of "Evil Customers" too.

How about customers who deliberately write vague or misleading specs, so they can get what they want through no-charge change orders, and keep costs down? How about customers who don't buy hot spares, and then blame the vendor for the damage that occurs when the device fails and their plant has to shut down? Evil enough for you? How about customers who insist that products have to be 10% cheaper every year?

There is an adversarial relationship between vendors and end-users that's very hard to combat. Yet it must be resolved if we're ever going to get best-value cooperation between vendors and end-users.

Some end users are working to make sure that standards bodies, user groups, and the like are increasingly devoted to solving end user problems, as opposed to just presenting vendors' gee whiz solutions in a glitzy and powerful forum. Some vendors are working, with no strings attached, to empower end users of their products, so those users will develop the type of powerful brand loyalty that used to make plants "a [insert name of vendor] plant."

True brand loyalty can't be bought, faked, or forced. It's a measure of how well you walk your walk, how well your actions are congruent with your messaging, coupled with a measure of how valuable what you sell is to your end users. Just dragging a group of users to a user group meeting doesn't improve your branding, vendors. But because every vendor wants a user group now, the user community is becoming quite fragmented, and it's difficult to get them to arrive at a consensus for action. This means that the vendors' "divide and conquer" strategy for user group marketing may be working.

What I'd like to see is a user group run by end users of many different kinds of products by many different vendors.

Oh, wait! There is one, or there used to be. It was called the ISA. But ISA lost its way somewhere back when all its executive board wanted was to have a great tradeshow. When the end users figured out that there were no big benefits, they voted with their feet. This was perhaps not the best thing to do. It paved the way for competing user groups, and captive customers.

Well, end users, here's another challenge. If you think going to each vendor's user group isn't a good use of your time, tell them. If you think it would be better if there was one unbiased user group, then I concur. If we agree there ought to be a master user group again, then let's create one. If you think it can be saved, let's go back to ISA, and turn it into the huge unbiased user group it was in the beginning.

One final thing. If you'd like to be a technical editor for Control, email me and we'll talk.

***Walt Boyes,* Editor in Chief**
wboyes@putman.net

Rick's Picks of the Web

What's New This Month on ControlGlobal.com

www.controlglobal.com/thismonth

by Rick Pedraza, digital managing editor

OPC: A Decade of Success by Any Standard

New to ControlGlobal.com this month is OPC Connection, a site resource for the OPC community, where specialists in the field pass on the latest in open-connectivity technology, specifications, expert advice, and thoughts on where OPC in industrial automation is headed.

In this first column, subject-matter expert Eric Murphy takes a look at how OPC has grown since the first specification was introduced 10 years ago. Murphy speculates on the future of OPC, noting that the *de facto* standard for industrial automation data transfer and device interoperability is staying true to its principles with the evolution of the OPC UA specification. Follow the latest trends and discussions about this widely accepted standard at ControlGlobal.com/OPC.

Also online this month is a link to highlights from the recently held Honeywell Users Group Symposium. Headlines from the three-day event include the unveiling of Honeywell's secure industrial wireless roadmap, as well as excerpts from presentations by Honeywell Automation and Control Solutions' Roger Fradin, who shares his company's process automation vision; ExxonMobil's Johan Nye and Michael Sarli on security risks and tomorrow's plant operators; and Genentech's Chris Schreil on his company's joint development with Honeywell of a new control system. Go to ControlGlobal.com/HUG for complete coverage of the event.

How Secure is Your Process Control System?

It's no surprise that control system security is the hot topic this month on ControlGlobal.com. During the Spring meeting of the Process Control Security Forum, (PCSF) an ad-hoc working group convened to discuss getting a security testing consortium/foundation off the blackboard and onto the street. Cheri McGuire, director of the CIP Cybersecurity Program for the U.S. Department of Homeland Security's National Cybersecurity division, addressed the audience and fundamentally indicated, without really saying so, that the government will impose standards on process control systems unless it likes what public and private partnerships like PCSF are doing. If you weren't lucky enough to attend, then check out our series of blogs about the proceedings and other industrial security content at ControlGlobal.com/SoundOff/security..

Talking MOLA on the Process Automation Radio Network

For those of you who've discovered our ever-expanding collection of audio podcasts on ControlGlobal.com (and those of you who haven't yet), the Process Automation Radio Network lets you turn on to new technology by offering audio programs you can listen to or download to a music player or PC. This month, get the story behind the story as editor-in-chief Walt Boyes revisits his old career at Texas Nuclear (TN) to talk MOLA (moisture online analyzer) with David Faulkner, product manager with Thermo Electron, the successor to the old TN. Faulkner discusses a new instrument for moisture detection and control that's the first new product to come from Thermo since its consolidation. Hear the entire interview at ControlGlobal.com/MOLA.

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630/467-1300
FAX: 630/467-1124
WWW.CONTROLGLOBAL.COM

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PUBLISHING TEAM

GROUP PUBLISHER/VP CONTENT:
KEITH LARSON
klarson@putman.net

MIDWEST AND SOUTHEAST REGIONAL SALES MANAGER:
GREG ZAMIN
gzamin@putman.net
630/551-2500, Fax: 630/551-2600

WESTERN REGIONAL SALES MANAGER:
LAURA MARTINEZ
218 Virginia, Suite 4, El Segundo, California 90245
310/607-0125, Fax: 310/607-0168
lmartinez@putman.net

NORTHEAST AND MID-ATLANTIC REGIONAL SALES MANAGER:
HARVEY WARREN
10 Marggraff Court
Oradell, New Jersey 07649
201/261-1550, Fax: 201/261-1876
hwarren@putman.net

INSIDE ACCOUNTS MANAGER:
DORIS DEUTSCH
ddeutsch@putman.net

AD TRAFFIC SUPERVISOR:
ANETTA GAUTHIER
agauthier@putman.net

SUBSCRIPTIONS/CIRCULATION:
JERRY CLARK, JACK JONES
888/644-1803

REPRINTS SALES MANAGER:
FOSTER REPRINTS
NANCY ROTHMAN
1-800-382-0808 ext. 134
4295 Ohio Street
Michigan City, IN 46360
nrothman@fostereprints.com

ADMINISTRATIVE TEAM

PRESIDENT & CEO: JOHN M. CAPPELLETTI

VICE PRESIDENT: JULIE CAPPELLETTI-LANGE

VP, BUSINESS DEVELOPMENT: MARK HARLING

VP, CIRCULATION: JERRY CLARK

SIS: No Unscrambling Here

The cover story, "Unscrambling Safety" [*Control*, May '06, p. 40], doesn't really unscramble much of anything. Rather, it continues to raise the specter (in questionable fashion) that plants really don't know what's going on internally, that they're way out of step with the new SIL/SIS standards, and that lack of compliance is a major culprit hurting operations.

I think you do a disservice to process and power industries by not fully explaining the realities behind the SIS/SIL standards. They're not just contradictory or overlapping (as is pointed out in the article), but the application of any given SIL level or other related standard is hardly uniform even within the standards organizations.

Further, just buying rated gear doesn't eliminate the need to apply the experience and expertise of staff or consultants. This is an even more important aspect of any safety program. Safe systems and safe plants can't simply be built from equipment. The National Fire Protection Association (NFPA) recognizes that virtually any system can be applied to boiler safety, so long as the design engineer is tasked with the responsibility, and has sufficient expertise, knowledge and effort. It specifically avoids prescriptions for hardware platforms and ratings.

Safety can't be bought from a box of parts, no matter the rating. Every year or so in the news, we see a petrochemical or refinery plant go up in flames, despite years of investment by the industry in TMR solutions, massive volumes of safety codes, etc. How often do we see similar episodes in the utility power world? I think this dialog would be better served with more disclosure and closer scrutiny, and by comparing and contrasting with other industry practices.

Tim McCreary, president
HF Controls Corp.

Immigrants are Essential

I read your editorial, "Immigrants Are Us" [*Control*, May '06, p. 11], and I must tell you it was right on target. If the general public realized the true value of immigration to our economy, and its positive impact on our culture of tolerance, then I think people wouldn't fall as easily for the "flavor-of-the-day politics of expediency."

The post-World War II economy in the U.S. created the greatest wealth mankind has ever seen. Despite all the criticism, we've managed well as a society. And, more importantly, quality of life for the general population has improved greatly. So, I'd say there are more reasons to view our immigrant-based society as a strength, and fewer reasons to suddenly blame it for all our misfortunes.

Knowledge and appropriate exploitation of our workforce is to today's economy what fertile land was in the days of feudalism. And we all know that many battles were fought over fertile lands—not to get rid of them, but to acquire them. Why would we not want to acquire more of what now is the most significant factor in creating wealth—a highly motivated workforce?

Specifically and closer to our industry, I hear manufacturing executives tell me that "flight of experience" is on their top 10 list of challenges. I was part of an ISA panel here in Texas last year, and one of the most discussed issues was the knowledge gap created by retiring employees. This challenge is compounded by the noticeable drop in the ratio of students enrolling in engineering schools in the U.S. compared to developing countries, specifically China and India.

So, with all that said, I guess you can tell that I might have some bias toward progressive immigration policies that can continue to help sustain our phenomenal comparative prosperity growth.

Eddie Habibi, CEO
PAS

Raising the Bar for Control System Security

Even the most casual computer user is aware of the many security vulnerabilities being found regularly in commercial operating systems and office software. If you don't patch your personal computer on an almost daily basis, these flaws will soon let it fall prey to the hackers, viruses, and worms that run rampant in the IT world.

What most of us don't realize is these vulnerabilities can be just as prevalent in hardware and software in programmable logic controllers (PLC) and distributed control systems (DCS), remote terminal units (RTU), or human machine interfaces (HMI). Just as in commercial software, human beings design and implement control and automation systems, and the mistakes they make allow unscrupulous attackers to take advantage of them, too.

Currently, there are no efforts to establish procedures for the security testing and certification of products used for control.

Most of the vulnerabilities discovered in control systems aren't widely publicized. However, in the course of research over the past five years, we've seen many major brands of controllers that are so flawed they could be easily exploited by the most unskilled attacker. For example, one PLC faulted while being scanned with a standard security tool, indicating a serious Transport Control Protocol (TCP) implementation issue. Further investigation showed this device's behaviour violated the TCP specifications so badly that it was a security hazard, not just to itself, but to every device on its control network. In another case, an installation of OPC client software forced an otherwise secure Windows HMI to become so poorly configured that hacking it took a matter of minutes.

Security: Afterthought and Ad Hoc

The problem is that today's automation systems weren't originally designed with security in mind, but were intended to perform control simply and efficiently. The thought that someone might deliberately attack a PLC or DCS system was never part of many manufacturers' design specifications or in the system development plan. To make matters worse, most control system vendors have had limited capacity to rigorously test new products for possible security flaws. As a result, the owners and operators of critical control systems have had little knowledge of their systems' robustness in the face of a cyber attack—until disaster strikes.

Many control system owner/operators, equipment vendors and government agencies are well aware of need for security and are attempting to address it through a number of operations-focused standards such as ISA SP-99 and NERC CIP-2-9. While these efforts certainly help the end-users ensure that their control systems are managed in a secure manner, currently there are no efforts under way to establish procedures for the security testing and certification of the actual products used for control. This lack of a foundation for secure products could put these other efforts in jeopardy.

If this seems a bit harsh, think of the engineering involved in building a brick wall. Certainly a good structural design is critical, but implicit in that design is the understanding that the bricks will have certain material properties and will act in a predicable manner. Build a wall out of bricks that crumble in the first rainstorm and the best design in the world won't save the wall.

In the absence of any formal product security standards, some end users have started developing their own specifications, often with little technical understanding of how to properly evaluate a product's security. Those companies that do have the knowledge for proper evaluation are expending significant resources to do so, and are creating conflicting requirements that make compliance difficult for the vendors. Even the government and university labs that are currently testing the security of control system products have been unable to show consistency in these efforts, making product comparisons extremely difficult.

Building Consensus

To sort out this current chaos, industry leaders from a number of major control system operators and manufacturers, including BP, Exxon-Mobil, Shell Oil, Honeywell, and Emerson Process Management have proposed that an organization be formed that would create a set of well-engineered specifications and processes for the testing and certification of critical control systems products. Similar in concept to the internationally accepted TÜV certification for Safety Instrumented Systems (SISs), control system

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100,000th Proline® Promass F Coriolis sensor delivered to Nature Works® LLC in Blair, NE
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Promass F Coriolis: Advantages and applications
Page 4

US manufacturing capabilities - service and support arm of a global company - near you.
Page 6

Find a local Endress+Hauser representative in your area
Page 8

Measuring solutions for your process

A Coriolis meter that can simultaneously measure and control multiple variables with no additional sensors

Endress+Hauser
People for Process Automation

Issue 6 ■ Volume 9

Our commitment to the future

Flow manufacturing facility expansion

The following are excerpts from the ground breaking speech by Hans-Peter Blaser, General Manager, Endress+Hauser Flowtec AG, Division USA:

"Ladies and gentlemen, the ground breaking for this new production facility is a very special and a very significant milestone in the history of Endress+Hauser USA. Since we have moved from Beverly, MA to Greenwood in 1974, the company has been continuously growing and expanding its services to the US customers.

For Endress+Hauser Flowtec in particular, today is a very happy day. We started production of our flowmeters 10 years ago in January 1996 and since then achieved some remarkable successes together with our colleagues from the sales organization. Successes that allow us to break ground today for a 92,000 square feet facility, an overall investment of over 17 million dollars.

I view this as a team effort of all Endress+Hauser companies on the campus, our sales organization, our IT support entity, our level & pressure entity as well as our temperature entity. Only together are we able to serve our US customers in the best possible way.

In the past days, I was asked by many different people, why is Endress+Hauser investing this kind of money in Greenwood, Indiana? My answer always was, why not? We have an excellent and central location, we have a city and community that supports us and most of all we have and we find well educated, highly motivated and dedicated employees. Only with this kind of employees are we able to be successful.

Artist's rendering of the new Flowtec manufacturing facility located in Greenwood, Indiana. Estimated completion date is April, 2007.

Why do we not move our manufacturing to countries like Mexico, China and India? We have a simple answer to that too. Our owners, the Endress family, and we as a management team believe in customer closeness, flexibility and loyalty. How can we possibly serve our loyal North American customers fast and according to their specific needs, if our plants are located in China or India? We believe in decentralizing manufacturing and to serve the different market places locally. This keeps us close to our customers and their needs; it helps us to reduce logistics and delivery times, and also helps the environment."

For more info go to: www.us.endress.com/flowtec-expansion

Hans-Peter Blaser, General Manager, Endress+Hauser Flowtec AG, Division USA (left) is joined by local dignitaries for the ground breaking of the new production plant.

Every Flowtec employee took part in the ground breaking ceremony for the new plant on April 7, 2006.

Made in the USA - for the USA

Endress+Hauser is one of the world's leading suppliers of measuring instruments and automation solutions for industrial process automation.

Committed to continual research and development efforts, Endress+Hauser invests a large portion of annual sales in the development of new products. This dedication has established us as a trendsetter and market leader with an unparalleled offering of innovation and quality instrumentation.

Around the world, Endress+Hauser has made tremendous investments in manufacturing excellence. Here in the United States we have ISO 9001-2000 certified manufacturing facilities and ISO 17025 certified calibration capability to help us deliver the highest quality measurement devices to our customers in North and South America.

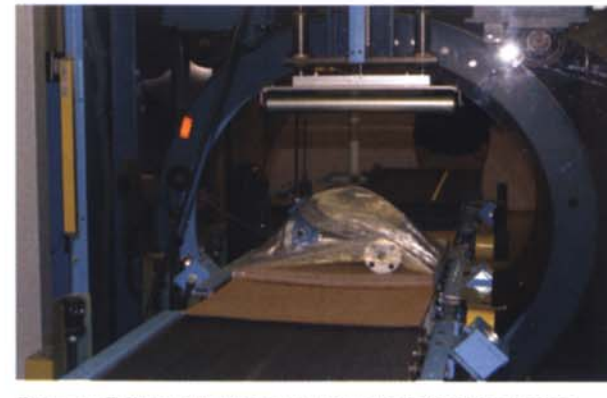
Flow calibration rig is accredited to international standards and helps us respond to the fast delivery times that are sometimes required by our customers.
www.us.endress.com/calibration

Coriolis, electromagnetic, vortex, ultrasonic - Endress+Hauser has local production to serve the needs of our local customers. Local production is critical in giving us the flexibility to assist our customers with special application requirements and with quick delivery - at times, within 24 hours.

- Electronic production, welding, logistics and calibration - all in the US production center
- Calibration is accredited according to ISO/IEC 17025, NIST traceable
- Sophisticated robotics systems in a fully automated processing center before production, the system detects the liner material, size and flange standard
- Electron beam welding - all process parameters are computer controlled and monitored - induction welding - under high vacuum for preparation of the measuring pipes
- KANBAN logistics enables reduction in stock cost and improvement in delivery times

Achieving a high level of production efficiency requires high-tech equipment and an educated, skilled workforce.

Automated welder (shown above) assures repeatability and quality of the flange welds by eliminating human error.



Promass F flowmeter being prepared for shipping. Being close to the market allows us to respond to our customer's needs with quicker delivery times.

Proline Promass F

The innovative Coriolis flowmeter - often copied, never equaled!

Field-proven in use for over 10 years
With an installed base of more than 100,000 instruments, Promass F from Endress+Hauser stands out above the rest. Many imitations have tried but haven't even come close to the long term and zero point stability offered by Promass F.

Liquid applications
- Liquids in general (example: water)
- Very low flow rates (< .01 gpm)
- Non-conductive liquids
- Viscous liquids (> 50 cP)
- Cryogenic fluids (example: liquid O_2)
- Food and beverages (example: milk, beer)

Gas/steam applications
- Gas flow in general (example: air flow)
- Low flow rates (< .7 ft^3/min) (< 20 l/min)
- Steam*

Special applications
- Slurries, suspended solids*
- Liquid/liquid mixtures (oil/water)
- Liquid/gas mixtures (water/air)*
- Corrosive liquids (acids, alkalis)
- Corrosive gas flows (example: HCl vapor)*
- Bidirectional metering (forward/reverse)
- Custody transfer

Ranges of application

1/24" to 10" (DN 1 to 250)
max. 5,076 psi (5,800 psi) (max. 350 bar (400 bar))
-60° to 660°F (-50° to 350°C)

***Conditionally suitable (depending on application, meter design/materials). Please contact Endress+Hauser for product details.**

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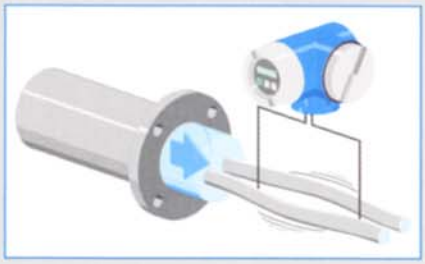


Coriolis mass flowmeters

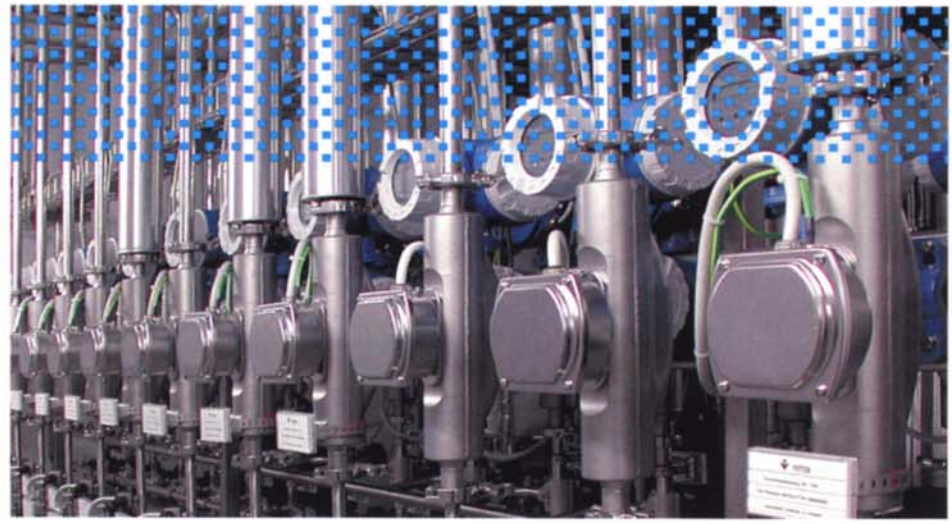
Simultaneous measurement of mass, density, temperature and viscosity

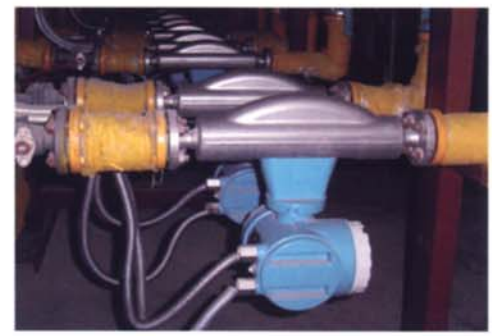
Measuring principle

If a moving mass is subjected to an oscillation perpendicular to its direction of movement, Coriolis forces occur depending on the mass flow.
A Coriolis mass flowmeter has oscillating measuring tubes to precisely achieve this effect. Coriolis forces are generated when a fluid (= mass) flows through these oscillating tubes. Sensors at the inlet and outlet ends register the resultant phase shift in the tube's oscillation geometry. The processor analyzes this information and uses it to compute the rate of mass flow.
The oscillation frequency of the measuring tubes themselves, moreover, is a direct measure of the fluids' density.
The temperature of the measuring tube is also registered for compensating thermal influences. This signal corresponds to the process temperature and is also available as an output signal.

Advantages at a glance

- Universal measuring principle for liquids and gases
- Simultaneous and direct measurement of mass flow, density, temperature and viscosity (multi-variable sensors)
- Measuring principle is independent of the physical fluid properties
- Very high measuring accuracy (typically ±0.1% o.r.)
- Not affected by flow profile
- No inlet/outlet runs necessary
- No straight run requirements
- Onboard diagnostics which allow for continuous monitoring of corrosion, erosion or build-up





The advantages of Coriolis mass flow measurement are self-evident. It comes as no surprise to find that this principle is used in a wide range of industry sectors, including pharmaceuticals, chemicals and petrochemicals, oil and gas, food, and also in custody transfer applications in general.

Virtually all fluids can be measured: cleaning agents and solvents, fuels, vegetable oils, animal fats, latex, silicon oils, alcohol, fruit solutions, toothpaste, vinegar, ketchup, mayonnaise, gases, liquefied gases, and more.

One for all: multi-variable metering
The ability to measure several process variables all at the same time opens up completely new application fields. Mass flow, density and temperature (the primary measured variables) can be used to derive other variables such as volume flow, solid contents, concentrations, and complex density functions.

Promass 83I is the world's first Coriolis mass flowmeter that can measure four primary fluid properties in-line: mass, density, temperature and viscosity!

When asked about the importance of a single vendor for flow, level, pressure, temperature, and liquid analysis, Mr. Marek responded, "It certainly adds value and makes things easier, but more important since my name is on all the selected devices, I really have to look for the best possible instrument to solve my application."

Marek adds, "There are several reasons we selected Promass for our applications. It sounds simple, but the biggest reason is that is works. We have used other Coriolis meters in the past and had mixed results. Promass is reliable, accurate, and easy to install. Since Nature Works is a young company producing a new product, downtime and waste in our processes is not an option for growing the business. We need a meter that is proven in real world environments."

"It was critical to Nature Works that Promass F Coriolis meters were fully drainable. Due to the ease with which bio-polymer will solidify, the meters and process piping are purged with nitrogen gas to ensure the system is free of product. Promass F sensors provide the most accurate measurement for mass flow and density in the process with no compromise in long term performance after repeated blow down with nitrogen. As an insurance policy, the meter displays, outputs, and alarms on the measuring system excitation current. With the use of Fieldtool™ software, we were able to graphically monitor the process and derive a set-point for the meter operation that warns us if the product is becoming too dense or on the verge of solidifying."

About the application:
Nature Works® PLA is a form of proprietary bio-polymer. Carbon dioxide and water form cornstarch that is further processed to dextrose. PLA is made from lactic acid that has been fermented from dextrose. Endress+Hauser Promass F Coriolis mass flowmeters are used in the process to measure the bio-polymer. The bio-polymer is a white, crystalline substance derived from lactic acid through a distillation process. The production of bio-polymers requires strict observance and control of process temperature. At 93° Celsius or 199.4° Fahrenheit, the viscosity of bio-polymer is about 2 Centipoise. At 92° Celsius or 197.6° Fahrenheit however, the process solidifies and must be mechanically removed from the process piping and flowmeter measuring tubes.

"The experience with Endress+Hauser has been very positive. With the assistance of Endress+Hauser and Miller Mechanical Specialties, We've been able to use the W@M Applicator® tool to size and apply flowmeters for our applications easily. Our technicians are able to keep the reports for our future use and consideration when planning for replacements or new applications. The fact that Endress+Hauser is manufacturing the flowmeters in Indiana is also a plus for our site. We can expect a quick delivery for most meters and even count on special handling when it's needed. We've had Promass meters ship the same day they were ordered."

"It certainly adds value and makes things easier, But more important since my name is on all the selected devices, I really have to look for the best possible instrument to solve my application."
- Don Marek, UNICCO Instrumentation and Electronics Technologist at Nature Works

Nature Works PLA products provide a bio-degradable, environmentally friendly approach to food packaging, beverage containers, service ware, and consumer goods such as wraps and films. For more information go to: www.natureworksllc.com

Case study

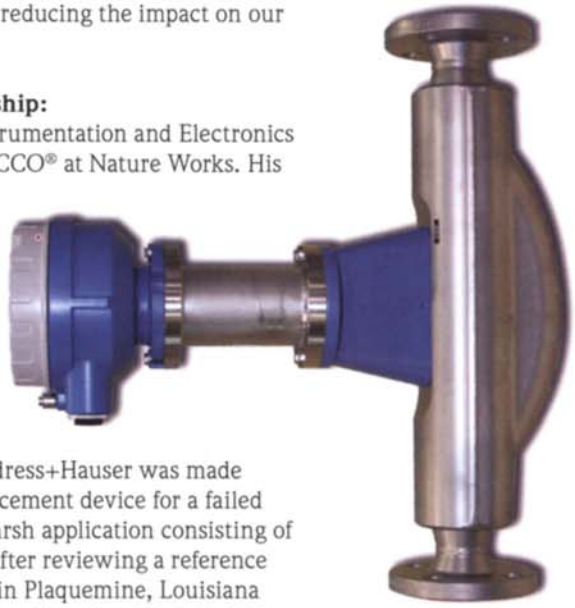
Endress+Hauser's 100,000th Proline® Promass F sensor installed at Natureworks® LLC

Nature Works® LLC, formerly Cargill Dow LLC ordered the 100,000th Promass F meter on January 19, 2006. The meter was sold by Dave Morris of Miller Mechanical Specialties, Inc. headquartered in Des Moines, Iowa. Morris provides single point support to NatureWorks' installed instrument base that includes Endress + Hauser flow, level, pressure, and temperature measurement devices.

Proline Promass currently in its second generation, is the successor of the ground breaking M-Point™ Coriolis meter. Introduced in 1986, M-Point was the first dual, straight tube sensor design for measuring mass flow based on the Coriolis principle. Promass F provided end users with the first fully drainable, dual bent tube Coriolis flowmeter design. In 2000, Frost & Sullivan, an international strategic market consulting and training firm awarded Endress+Hauser its prestigious Market Engineering Award for demonstrating great insight into customer needs and product demands in the flow sensor market.

About Nature Works® LLC:
Using the motto, "Your food comes from nature, now so does your container," Nature Works offers a family of polymers derived from annually renewable resources such as corn, with the cost and performance to compete with petroleum-based packaging materials and fibers. The company uses an innovative and unique process to create a proprietary bio-polymer marketed as Nature Works® PLA and Ingeo™ fiber brand names. Nature Works PLA products provide a bio-degradable, environmentally friendly approach to food packaging, beverage containers, service ware, and consumer goods such as wraps and films. The Ingeo™ fibers provide a high performance, alternative product to apparel, bedding, carpet, and personal care product manufacturers while reducing the impact on our environment.

About the relationship:
Don Marek is an Instrumentation and Electronics Technologist for UNICCO® at Nature Works. His first contact with Endress+Hauser was made while seeking a replacement device for a failed Coriolis meter in a harsh application consisting of Hydrochloric acid. After reviewing a reference from Dow Chemical in Plaquemine, Louisiana USA, Nature Works purchased and installed the Endress+Hauser Promass H, a single, bent tube meter design with zirconium wetted parts. This meter has been in process for 18 months validating the mass flow technology from a new vendor in a harsh and challenging application. Subsequently, Promass was evaluated to replace the existing Coriolis meters used at Nature Works. Today, Nature Works' installed base includes Promass Coriolis meters in a variety of applications with sizes ranging from 1/2" to 6".



Cover photo:

Jerry Stevens, Endress+Hauser (holding instrument on left), and Dave Morris of Miller Mechanical Specialties (behind instrument) present a "Golden Promass" award to the Natureworks Instrumentation group in recognition of being the recipient of the 100,000th Promass manufactured by Endress+Hauser.

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vendors would be able to offer products that are proven to meet a standard set of security requirements. In addition, the organization would work closely with existing standards groups, supplying them with draft documents that can be formulated as standards.

For end users, this certification process will result in significantly reduced costs and time commitment in product selection and acceptance. It will also help ensure that products are more secure out of the box. For vendors, the organization will provide a single testing framework and an industry stamp of approval, resulting a faster time to market and lower development and integration costs. Finally, for the largely volunteer-based standards organizations, this new organization would help them draft documents and supporting research, so they make informed decisions on security standards.

However, before an effective control system security testing and certification program can be created, it's essential that a members-based organization be founded to manage and fund the process. This organization needs to be global in scope, include vendors, system integrators and end-users and be open and inclusive to all, except those that clearly pose a threat to security.

Of course there are many logistical and organizational questions that need to be answered before the organization can be setup. For example, would it be most effective as a non-profit foundation or a member-controlled, for-profit company? How would membership benefits and fees be structured? What is the start-up model versus the long-term, sustainable model? How would legal issues like intellectual property rights, liability and warranty be addressed? These are not difficult questions, but they need to be carefully investigated and a plan of action needs to be developed.

To help answer these and other questions, security researchers at Wurldtech Analytics (formerly the BCIT Security Research team) are partnering with ISA to conduct a feasibility study, and develop an operations model for this security-certification organization. Companies interested in becoming charter members of the organization are being asked to supply $2,500 to help fund this portion of the study. Companies wishing to be involved should contact Joann Byres at jbyres@wurldtech.com.

Eric and Joann Byres are principals of Wurldtech Analytics in Vancouver, BC, Canada.

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Why Users Need One Twisted-Pair Bus

Is the two-wire hurdle going to hold back wide applicability of discrete devices in fieldbus? Clearing the hurdle of two-wire technology—having power and signal on the same pair, or at least the same cable—is an obstacle for many instruments. Coriolis, magmeters, pH, and similar measurements usually require "four-wire" solutions, which was true even before fieldbuses became available for process control. In most cases, we grudgingly accept four-wire solutions because we have little choice, if we want to get the benefits of, for example, a Coriolis flow measurement. But what if that flow was a "critical" measurement? What are the alternatives for that "other" pair?

Only one supplier has a two-wire package with temperature specs for our climate in Ohio. We'll be testing it this winter.

The beauty of a "loop" or segment-powered device is that you can do all your work for robustness back in the house. You can have redundant power conditioners connected to auctioneered DC supplies, which in turn ride on redundant UPS, if you have the money and the space.

But what do you do about the four-wire device? Do you run UPS all the way out to it? Do you have little satellite, dual-auctioneered DC power boxes? UPS circuits, like any AC circuit, at some point can have so much cable that the 20-amp breaker in the house panel is no longer effective.

Thankfully, fieldbus signal diagnostics are comprehensive enough that surviving a "bad" measurement from Coriolis or pH is usually manageable. Most host systems that support fieldbus also can propagate signal status—although some more innately than others—which in turn can at least ensure that no control valve goes full-open or full-shut.

The two-wire hurdle becomes more of a barrier when you seek to apply discrete quarter-turn actuators using fieldbus. To use segment power and achieve an elegant two-wire solution, current offerings have to use a "piezo" solenoid, which can turn air on and off while consuming little power. They have some disadvantages, however, and; the most glaring is that their tiny orifices and flow paths can plug easily. They also have "memory," and so they'd rather not sit around energized for many months, which is normal practice in process plants. They also don't like it when it gets cold.

Nearly all makers of fieldbus-capable, two-wire, discrete positioners have addressed this "memory" issue. We who use them can attest that the present time is much better than the early days, but we're still concerned, for example, about what can happen after two years. Has anyone had a piezo-actuated solenoid energized for two years straight? Let us know if it closes. We don't have any that have been "on" for that long, but I haven't seen any exhibit the "stuck on" behavior we saw in the early offerings.

The plugging issue also is fixable. Users can install an in-line sintered filter, or the manufacturer can make it integral.

However, we're most concerned about the *cold* issue. While not a huge worry for users in East Texas, California, or the Persian Gulf, a low-end temperature rating of 10 °F or more isn't going to cut it in many places, such as Canada, most of the Eastern U.S., and continental Europe. So far, only one supplier has a two-wire package that has reasonable temperature specs for our climate in western Ohio. We'll be testing that package this coming winter.

The alternative—using a separate DC power source for the solenoid—brings forth all the issues discussed above. In this case, we're dealing with a final element, a valve, where our automation system interacts with the physical process. So, the forgiveness afforded by thoughtful fieldbus and device diagnostics on the sensor/measurement end of the loop are too-little, too-late on the valve end. When it fails, in all likelihood the process is impacted immediately as the actuator drives to its fail position, or even worse, if it fails to go to its fail position.

If I have to run a separate pair out for power, why not just use a conventional discrete output point? In my mind, it's easier and more robust than trying to achieve the same integrity on a field-originated DC power system. Am I going to install satellite UPS systems also? We tried this on some critical analyzers (yes, analyzers) used in a SIL 2 shutdown system. That power scheme has been replaced and I would discourage others from imitating what we did.

Four-wire technologies have serious issues for critical final-element applications. Existing two-wire solutions for solenoid-actuated valves also are less than ideal. As users consider candidates for their fieldbus projects, we need to encourage our suppliers to strive for the robust two-wire technology we apply happily elsewhere.

John Rezabek is process control specialist for ISP Corp. in Lima, Ohio. E-mail him at jrezabek@ispcorp.com.

Automation Federation Joins Security Project

Joining end-user companies such as Shell and ExxonMobil and automation vendors like Emerson and Honeywell, and *Control* magazine, ISA's new "Automation Federation" will participate in the formative study being conducted by Wurldtech Analytics (www.wurldtech.com) toward forming a Control Systems Security Foundation (see "Other Voices" in this issue, p. 16). Wurldtech will evaluate the feasibility of creating specifications and processes for the security testing and certification of critical control systems products.

Industry standards like those arising from ISA-SP99 and NERC CIP-2-9 and the work of the OMAC MSMUG group have been addressing this issue from an end-user prospective, but this new initiative aims to help define methods suppliers can use to validate that their products enable necessary levels of secure operation. With this program, control system suppliers would be able to offer products that are proven to meet a standard set of minimum security requirements.

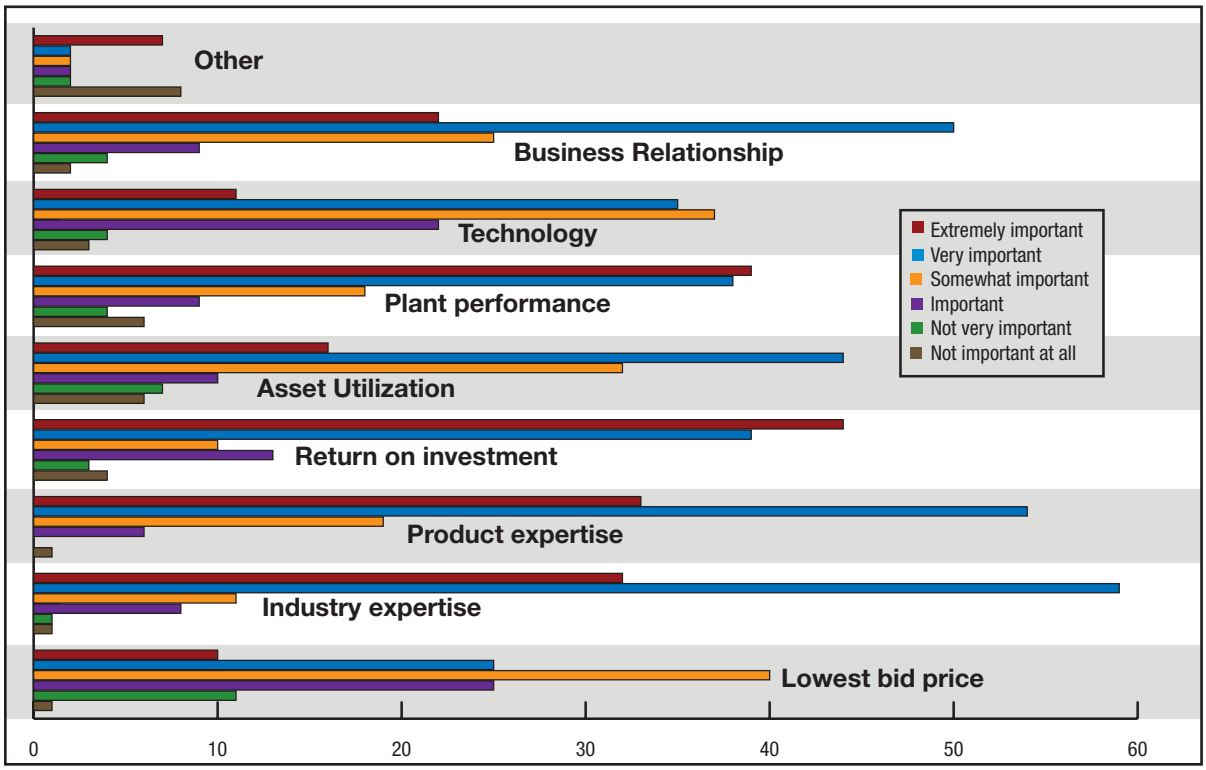
Joann Byres, Wurldtech's applied research director, says a proposel for setting up the organization will be completed by September 2006, and that it could be launched in early 2007. "Our vision is that any certification organization that arises will work very closely with existing standards groups. We'd give them draft documents that can be formulated as standards and supporting research to enable informed decisions on security standards," says Eric Byres, Wurldtech's director. "We welcome the Automation Federation's support, especially because of ISA and OMAC's work in the security standards arena, and we're looking forward to a close partnership."

Users Embrace Automation Services

A new survey by Larry O'Brien, process automation research director at ARC Advisory Group (www.arcweb.com), shows "a clear shift away from purchasing automation services from large engineering and construction firms toward purchasing services from suppliers or independent third-party system integrators. This is consistent with ARC's observation that more of the large EPC firms are downsizing their automation departments and leaving more of the responsibility for the automation phase of large projects to subcontractors such as suppliers and smaller SIs."

FIGURE 1.

ROI AND PERFORMANCE RULE



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At the same time, O'Brien reports, "Users are increasingly looking to suppliers to provide them with a central point of responsibility for executing automation projects, and suppliers must be prepared to meet this demand. More suppliers are therefore taking on the responsibilities of main automation contractor (MAC), main automation vendor (MAV), and main instrument vendor (MIV). These roles can mean different things to different suppliers, but they all connote increased responsibility for managing automation projects by users, who either did it for themselves or outsourced the task to systems integrators or engineering firms."

O'Brien also claims his respondents are "placing less emphasis on lowest bid price for services and more emphasis on preexisting relationships and expertise of the supplier or service provider." (Figure 1)

"We asked respondents to rate the likelihood of future outsourcing of service functions to automation suppliers that were previously performed inhouse," says O'Brien. "Key categories that respondents said they would increasingly outsource to suppliers include training, installation, system configuration, system and device maintenance, and engineering and design. Other areas include remote system monitoring, mechanical and electrical condition monitoring, and spare parts and replacement. Areas where end users are likely to outsource at the same level they do today include commissioning and checkout, loop tuning, and operational services. There were very few categories where user respondents said they would not outsource more or the same."

Emerson, Siemens Coordinate Offerings to End Fieldbus War

Emerson Process Management and Siemens Automation and Drives (A&D) have apparently begun declaring peace in their decade-old fieldbus war. The companies will exchange technology and engineering support to offer customers expanded global interoperability and greater functionality. First products are expected to be available mid-2007.

Their cooperation grew out of the work both companies have done with user-supplier consortiums to deliver fully open and interoperable field instrumentation and electric drives using the the newly enhanced, global standard Electronic Device Description Language (EDDL).

Siemens will support Foundation fieldbus (FF) standards, in addition to the Profibus and HART standards it supports now. This will enable Siemens to connect to FF instruments and valve controllers from Emerson, as well as devices from other companies that support FF.

Emerson will similarly support Profibus DP and Profinet standards, in addition to the Foundation fieldbus and HART standards they support today.

OPC Foundation Releases Long-Awaited Unified Architecture

OPC Foundation formally launched its long-awaited OPC Unified Architecture (UA) standard at a June 27 press conference during ARC Advisory Group's 14th annual forum in Boston, and the organization's manufacturers and other members are lining up to support it.

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uses Microsoft .NET and web services. The new standard upgrades OPC communications from Microsoft COM/DCOM, and reportedly features enhanced reliability, security and robustness, while maintaining a common data model, interoperability, and integration. The new standard reportedly provides the framework for platform neutrality, enabling real-time and historical information interoperability between systems and OPC users can integrate OPC UA into existing OPC implementations.

"OPC technology can eliminate expensive custom interfaces and drivers traditionally required for moving information around the enterprise," says Tom Burke, OPC Foundation's executive director. "It promotes interoperability, and, therefore, cuts costs, speeds development, and promotes increased operating efficiency."

Several vendors have announced support for OPC UA, and their continuing plans to integrate their technologies with it and build applications around it.

ISA Website Peeks into Standards Development

ISA's website at www.isa.org is allowing users to preview ISA standards. Overviews, tables of contents, scopes, definitions, and other informative sections of the standards are now available to visitors.

"We think it's important to make the scopes and overviews of our standards available to the public. This feature gives automation professionals worldwide a sneak peek at ISA's standards, and provides a framework for understanding the variety of standards that we offer," says Ian Verhappen, ISA's standards and practices vice president.

ISA's noteworthy standards updates include:

- The Enterprise-Control System Integration committee, ISA SP95, is conducting a final review and voting on Part 5 of its standards, focusing on business-to-manufacturing transactions.
- The Manufacturing and Control Systems Security stan-

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dards committee, ISA SP99, has completed a first round of voting on the first and second parts of its ISA-99 series of standards, and is reviewing input comments.

- ISA recently formed its new SP101 standards development committee on human-machine interface (HMI) following input from industry users asking for HMI guidelines and standards.
- ISA's SP100 committee for Wireless Systems for Automation was formed in 2005. See this issue's cover story.

ISA Announces First Marketing and Sales Summit

The Management Division of ISA's First Annual Marketing & Sales Summit will be held 13-15 September 2006 at the Omni Austin Hotel Downtown in Austin, Texas. The summit, entitled "Best Practices in Customer Acquisition and Retention," will help marketing and sales executives in the automation markets meet the strategic and tactical challenges of the 21st century. Keynotes and plenary sessions from Jane Lansing, VP Marketing, Emerson; Dan Miklovic, Managing Vice President, Gartner, Inc.; Mark Taft, Senior VP Global Systems Marketing, ABB; and Brian Chapman, Director of Corporate Marketing, Honeywell Process Solutions, will address the state of the manufacturing industry, the changing role of the senior marketing and sales executive, and the shift to service selling. Breakout sessions will provide practical information on search engine marketing, solution selling, public relations, branding, and more.

"This forum will provide a venue where we can discuss and more fully understand the realities of our business and how to better create customer-focused organizations," said Shari Worthington, president of Telesian Technology, and Event Program Chair.

Control's Walt Boyes serves as General Chair for the summit. Boyes has more than 25 years of experience in automation marketing, including Executive Committee experience in three companies, as well as experience as an editor and industry analyst. .

Canon Co-locating 22 Tradeshows, Including NMW

Canon Communications LLC, an Apprise Media company, reports it's implementing a co-location strategy for six of the eight specialty manufacturing tradeshows it acquired from Reed Elsevier in June 2006.

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"Co-location is an important element of our strategy to integrate all of the Reed Elsevier shows into Canon's existing platform of tradeshows servicing the $1 trillion advanced technology manufacturing sector," says Charles McCurdy, Canon's chairman and CEO. "Co-location has been a key driver of Canon's growth strategy for a decade, and has created the four largest, annual design and manufacturing tradeshows in the country."

Canon adds its co-location strategy provides OEM buyers with access to "mga-events" in four geographical regions—the West, Midwest, Northeast, and Southeast—and it addresses industry concerns about flagging attendance and the proliferation of increasingly specialized tradeshows. Canon shows reportedly bring as many as 2,000 exhibitors and 50,000 design and manufacturing professionals together under one roof.

"Canon's decade-long commitment to co-locate allied advanced technology manufacturing shows, geographically positioned in the center of their served markets, is virtually unique in the exhibition industry," says Kevin O'Keefe, vice president of Canon's events division. "With co-location of as many as six design and manufacturing events under one roof, exhibitors gain access to thousands of additional sales targets across multiple industries."

Already under development for three months, Canon's co-location strategy will result in:

- The Assembly, Quality, and National Manufacturing Week shows will be co-located near Chicago in late September 2007. The combined exposition will occupy all 800,000 square feet in the Donald E. Stephens Convention Center in Rosemount, Ill., making it one of the nation's largest design and manufacturing events. It's expected to attract 1,800 exhibiting companies and 50,000 attendees.
- The Powder & Bulk Solids Southeast and Advanced Manufacturing Exposition (AM Expo) will be co-located with the SouthPack, Automation Technology Expo (ATX), and Design and Manufacturing South shows in Atlanta on April 24-26, 2007, to create a 300,000-square-foot event with 800 exhibitors at the Georgia World Congress Center.
- The NEPCON East/Electro show will be co-located with the OEM New England show in the Boston Convention and Exhibition Center in Fall 2007.

For information about Canon's events, visit www.canontradeshows.com.

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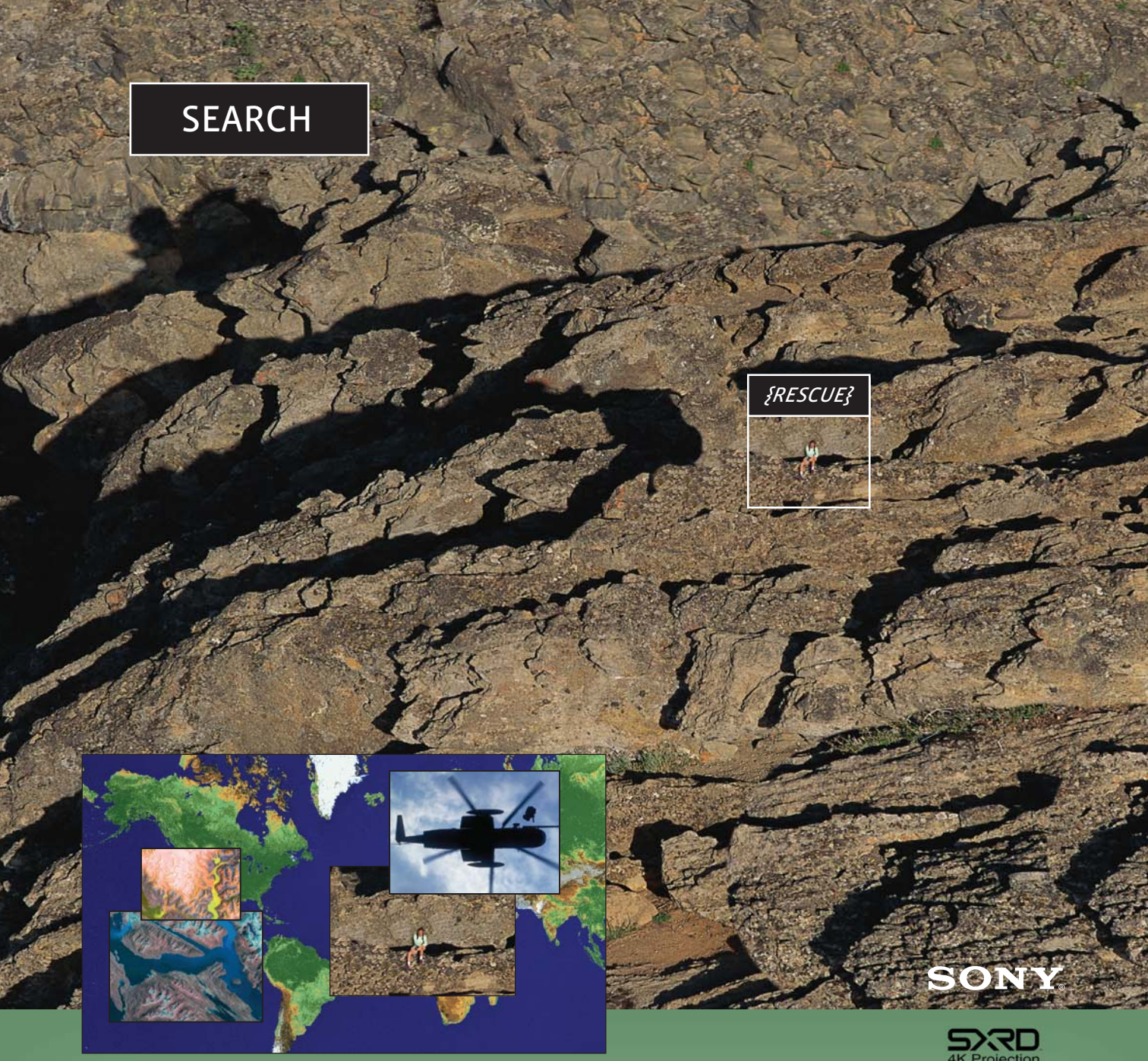

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Seeing Deeply

If I pay attention, almost every day I'll see something that I've never seen before. One winter day, for example, I saw a walled snow fort covering the top of a red minivan. Another time, while flying in a 12-seat airplane up California's coast, I noticed that my empty soda pop can was revolving in its shallow, tray-table holder *without anyone touching it*! I thought the plane was haunted until I realized its vibrations were causing the can to rebound imperceptibly within the slightly larger circle, which created torque and motion, like a slow-moving rotary engine. In one long-ago police blotter, I wrote about a routine search for some stolen tools, which led to an inventive young boy, who had boosted the sideboards from an entire outdoor ice rink, and reassembled them into a ramp for jumping his bicycle!

What's truly new may be a deeper understanding and appreciation of what we thought was so familiar.

Truth isn't just stranger than fiction, it's often incomprehensible until it occurs in reality. That's why I enjoy covering process control—more and better surprises.

Still, the drumbeat of routine can make it hard to find what's new and interesting as the weeks and years pile up. Maybe it's one of the occupational hazards of experience—mistakenly thinking you've seen everything—like the 19th Century U.S. patent official who famously said everything that could be invented had been invented. Naturally, the same fatigue can overtake control engineers, technical professionals, and journalistic hangers-on like me, especially after writing about my 2,000th terminal block or its 20th version, whose only "innovation" seems to be that its jacket now comes in sea foam green. Ho freakin' hum. Many improvements—like bottled water, energy-wasting SUVs, and water-logged hams—aren't really improvements.

So, I must confess that I wasn't expecting much when I recently traveled to Boulder, Colo., to visit Emerson Process Management and its Micro Motion division for their Perspectives 2006 event. Of course, I was wrong, as I usually am when I anticipate the future. I was wrong because Emerson and Micro Motion unveiled multivariable digital (MVD), sensor-transmitter, and self-diagnostic improvements that make their Coriolis flowmeters more accurate, stable, and widely applicable, especially in entrained-gas measurement. However, beyond these technological advances, I was wrong to anticipate for an even more important reason. I forgot that just getting out and talking to people about what they care about is what lights me up personally, and many of the folks at Micro Motion were particularly energetic sources of the specific input I find interesting and hope is useful to readers.

For example, while showing how Micro Motion's new temperature and rain chamber is used to test flowmeters, engineering technologist Dana Bates reported that she and her colleagues still effectively use an old method of sprinkling baby power inside flowmeter housings, so any clumping will indicate where there's a moisture leak.

Rock Tanner, supplier quality director, reported that Micro Motion is updating its X-ray weld-inspection equipment from film to digital, but added this conversion was held up until he could find a way to calibrate the new devices. The old X-ray used strips of metal, but the digital camera's multi-dimensional capabilities meant that Tanner had to investigate and eventually use special ball bearing as calibration guides.

Similarly, principal design engineer Mark Bell demonstrated how Micro Motion developed and fine tunes its multi-phase fluid stand, which separates and measures water and oil levels. And metallurgists Gerri Berry and David Murdock showed how Micro Motion updates its corrosion guides, and uses forensics to conduct failure analyses.

Finally, Joel Weinsten, an intern from the University of Colorado, showed demonstrated how another Micro Motion test stand uses two-phase flow to test for entrained air in fluid samples supplied from end users' own applications.

All of these professionals showed how they use technological expertise, traditional common sense, and inspired thinking to create, test, and support Micro Motion's solutions. While product launches usually get most of the corporate glory, it's the people behind them and their intangible contributions that make them shine brightest.

So, the next time you get flooded with new and improved non-improvements, get out and talk to some engineers directly. They reminded me that, even if there's nothing new under sun, what's truly new may be a deeper understanding and appreciation of what we thought was so familiar.

Jim Montague, executive editor
jmontague@putman.net

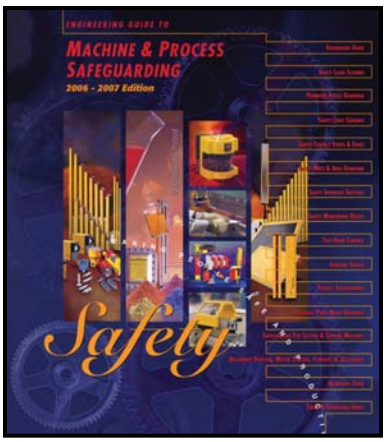
Serious Safety Sources

***Control*'s Monthly Resource Guide**

Every month, *Control*'s editors take a specific product area, collect all the latest, significant tools we can find, and present them here to make your job easier. If you know of any tools and resources we didn't include, send them to wboyes@putman.net, and we'll add them to the website.

ENGINEERING GUIDE ON PROCESS SAFEGUARDING

The 2006-2007 Engineering Guide to Machine and Process Safeguarding has 125 pages of educational resource material, including articles on safety regulations and directives, risk assessment, types of protective measures, and background on safety technologies. The more than 850-page guide also has products for safety, including safety laser scanners, perimeter access guards, safety light curtains, resource modules, safety mats and area guards, safety interlock switches, and safety monitoring relays. For a copy, send an e-mail to russ_wood@sti.com. **Scientific Technologies; 800/479-3658; www.sti.com**



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"Voltage Warning Indicators Add Safety to Lock-Out, Tag-Out Procedure" whitepaper discusses how to use voltage indicators to provide electrical maintenance workers with the ability to pre-verify isolation and the presence of electrical power in a panel or enclosure, without exposure to hazardous voltage. It also helps avoid arc flash problems. Download the paper from www.grace-eng.com/newsletters/VoltageVisionPR/voltage-indicators-10-2005.shtm. **Grace Engineered Products; 800/280-9517; www.grace-eng.com**

CIP SAFETY PROTOCOL PUBLISHED FOR ETHERNET/IP

The CIP Safety protocol for EtherNet/IP meets the requirements of IEC 61508 up to Safety Integrity Level 3 (SIL 3/Cat 4). CIP Safety on EtherNet/IP will complement CIP Safety on DeviceNet, which was released in 2005. CIP Safety provides a single protocol solution for functional safety applications, and eliminates the needs for purpose-built safety networks. ODVA published the protocol in the latest edition of the CIP Safety Specification, Edition 1.1. For a copy, send an e-mail to ameyer@odva.org. **ODVA; 734/975-8840; www.odva.org**

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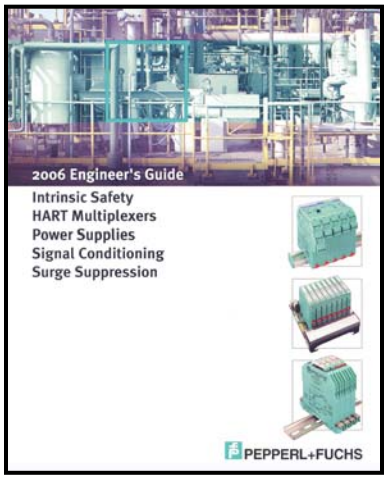

of educational resource material, including fully illustrated articles on the basics of intrinsic safety, hazardous locations, safety system application theory, and practical solutions. Products include hazardous location isolators and barriers. For a copy, send an e-mail to chris@longrenparks.com. **Pepperl+Fuchs; 330/486-0002; www.am.pepperl-fuchs.com**

FREE SIS COURSES

Free courses on safety instrumented systems at PlantWeb University's online learning center (www.PlantWebUniversity.com) provide fundamentals and practical tips for SIS planning, selection, and implementation. The courses include concise information on basic SIS concepts, design and installation, operation and maintenance, safety standard compliance, and new "smart" SIS technologies. Each course takes about 15 minutes. Content is devoid of marketing hype, and gives practical insights into applying new technologies. **Emerson Process Management; 512/832-3089; www.emersonprocess.com**

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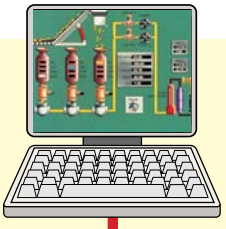

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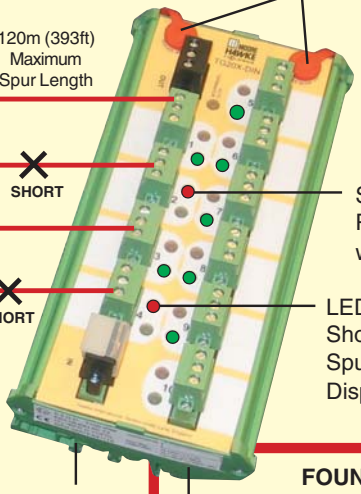

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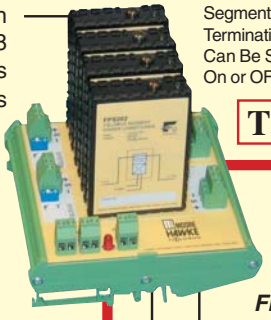

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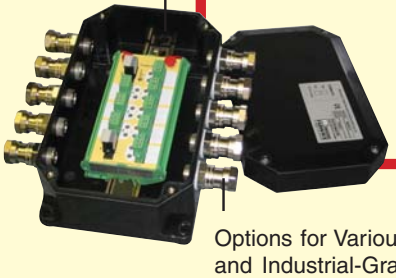

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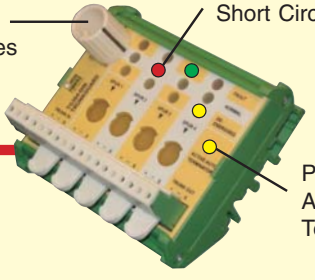

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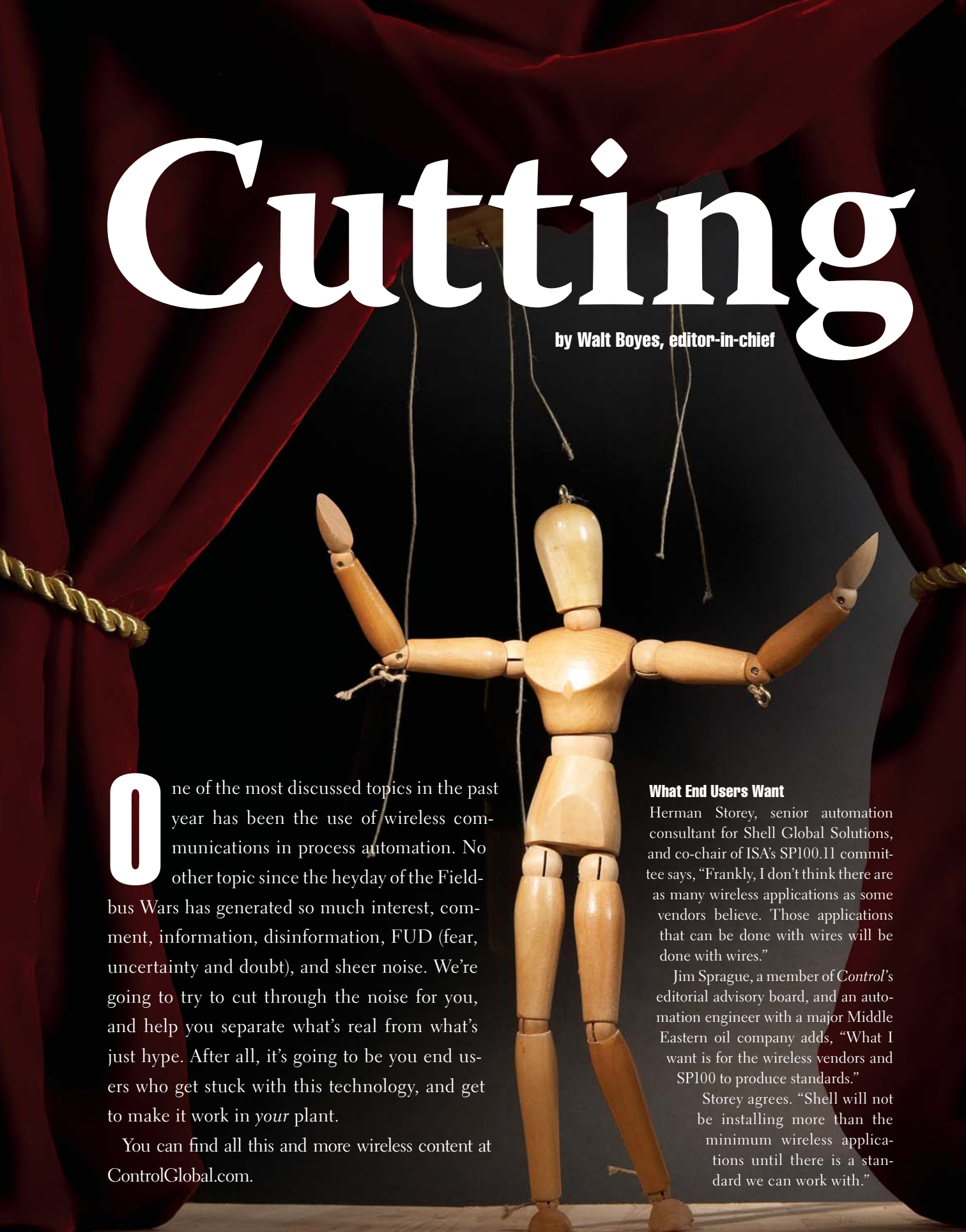
Cutting

by Walt Boyes, editor-in-chief



One of the most discussed topics in the past year has been the use of wireless communications in process automation. No other topic since the heyday of the Fieldbus Wars has generated so much interest, comment, information, disinformation, FUD (fear, uncertainty and doubt), and sheer noise. We're going to try to cut through the noise for you, and help you separate what's real from what's just hype. After all, it's going to be you end users who get stuck with this technology, and get to make it work in *your* plant.

You can find all this and more wireless content at ControlGlobal.com.

What End Users Want

Herman Storey, senior automation consultant for Shell Global Solutions, and co-chair of ISA's SP100.11 committee says, "Frankly, I don't think there are as many wireless applications as some vendors believe. Those applications that can be done with wires will be done with wires."

Jim Sprague, a member of *Control*'s editorial advisory board, and an automation engineer with a major Middle Eastern oil company adds, "What I want is for the wireless vendors and SP100 to produce standards."

Storey agrees. "Shell will not be installing more than the minimum wireless applications until there is a standard we can work with."

the Wires

Users want wireless, vendors want to sell wireless, so what's the problem?

Even so, many end users have told *Control* they believe wireless automation systems can be made reliable enough to be used even for loop control. Polls conducted by B&B Electronics (*Control*, November 2005), and by the SP100 committee itself (see "Tell SP100 What You Think" sidebar) indicate there's widespread interest in using wireless sensors even for loop control in "non-critical" applications. A few end users even believe wireless can be reliable enough for critical, closed-loop control and safety instrumented systems (SISs).

The bottom line is that end users want a standard or a set of standards that they can depend on before they'll use wireless in their plants other than in demonstrations, or where they absolutely have no other choice.

Such "no other choice" applications abound, at least apocryphally. That is, everybody has a story, but nobody will identify where the application actually was located. There's the steel plant that uses wireless temperature sensors to protect the water bath around an electric arc furnace. There's the tank farm with all wireless level sensors. There are many others, and many vendors and even some end users will talk about them, but won't name them. So, it's hard for end users to figure out what's FUD and what's not. But on July 14, the SP100 Committee issued two *Calls for Proposals* for a standard. The rubber is about to meet the road.

The Wireless Landscape

As we reported nearly a year ago ("Users Want an Industrial Wireless Standard" *Control*, Sept. '05, p. 103), wireless isn't like wired communications. "It's not enough to say, as if we were Rodney King, 'Can't we all just get along?' " says Dr. Peter Fuhr, of Apprion Corp., and chair of SP100's interoperability committee. The wireless landscape is tricky, and littered with standards and applications (Figure 1) that have little or nothing to do with process automation, with the exception of one glaring fact—they're all in use in process plants from refineries to biopharma installations.

In fact, as a senior ABB Inc. official told me recently, using wireless in discrete automation and robotics is far advanced when compared to the use of wireless in continuous process or batch process automation. "We're even delivering power to the machine wirelessly," he said.

Many plants, whether discrete, process, or hybrid are already using wireless technologies such as radio frequency identification (RFID) and WiFi (802.11a-g) both in offices and on the plant floor.

Everything is easier in the wired world, where all you must have is interoperability between devices on the same bus. So, you can have Profibus, HART, Foundation fieldbus, Modbus, and other devices in the same plant without worrying that adding a device or a network will cause the rest of the communications buses in the plant to go dead.

Once you go wireless, you're in a different world. Interoperability of devices on the same network is still an issue, but it pales in comparison to the problem of *coexistence* between wireless devices on the same or adjacent networks. Now you have to decide if adding a device or another network node will work, work part of the time, or maybe even bring down the entire network. Wireless communications are such that you may not even know the answer until you turn the new device on.

The SP100 committee has defined five different classes of wireless automation, ranging in complexity from Class 5 to Class 0.

Class 5: Monitoring without immediate operational consequences.

The ISA SP100 *Calls for Proposal* state, "This class includes items without strong timeliness requirements. Some, like sequence-of-events logs, require high reliability; others, like reports of slowly changing information of low economic value, need not be so reliable, since loss of a few consecutive samples may be unimportant."

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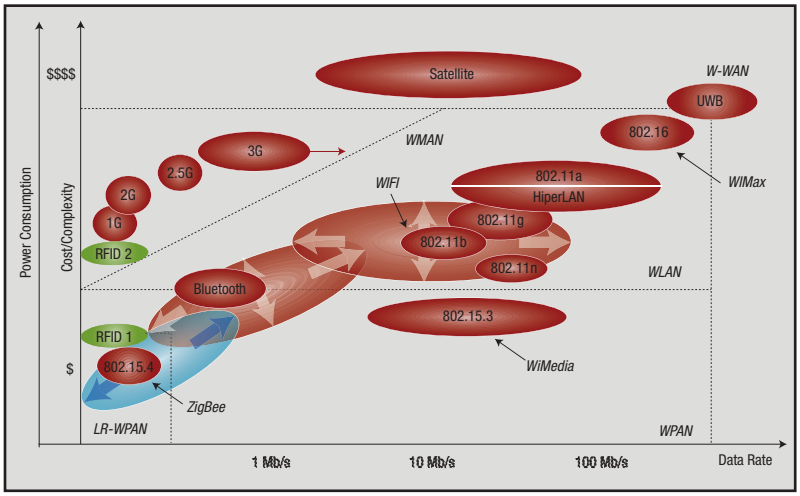
FIGURE 1.

ALL THE DIFFERENT WAYS TO WIRELESS



Many standards can mean tough decisions for process control engineers.

Hesh Kagan, Invensys executive and president of the Wireless Industrial Networking Alliance (WINA), believes this will be the largest category of wireless applications in the near future. "The biggest bang for the buck, the low hanging fruit," he says, "is probably going to be in the area of maintenance and condition monitoring, followed very quickly by an opportunity for incremental process measurements."

In the old days, the only process variable measured was the one controlling the loop: flow, temperature, level, pressure, density, pH, and conductivity.

The huge relational databases optimized for process control, such as Invensys' Wonderware historian, Honeywell's PHD, or OSIsoft's PI, make it possible to use information from process sensors not directly involved in loop control. Wireless sensors have a great deal to offer in these applications.

Class 4: Monitoring with short-term operational consequences

"This class includes high-limit and low-limit alarms and other information that might instigate further checking or dispatch of a maintenance technician," read the *Calls for Proposal*. "Timeliness for this class of information is typically low (slow), and measured in minutes or even hours."

These applications includes monitoring inventory level of tanks, as well as alarms designed only to actuate if the process goes out of whack, but that don't require an immediate response. Alarms, however, are a special case. They can be of any class. For example, a Class 0 alarm could be a toxic gas sensor which triggers automatic containment. Class 1 alarms would be those alarms that would cause immediate shutdown of the process, what the SP100 committee calls, "high-impact process conditions." Class 2 alarms would include any normal automated response to a normal change in process conditions, like a low level alarm causing switchover from one storage tank to another. A Class 3 alarm would require the intervention of a human operator, such as deciding whether to divert from Reactor 1 to Reactor 2. Class 4 alarms would describe an equipment condition with a short-term response requirement, such as to go find out why the low pressure alarm is going off for no apparent reason. A Class 5 alarm might be to order spare parts to repair an analyzer that isn't critical to the process.

Class 3: Open-loop control

This class includes actions where an operator, rather than a machine, 'closes the loop' between input and output. Such ac-

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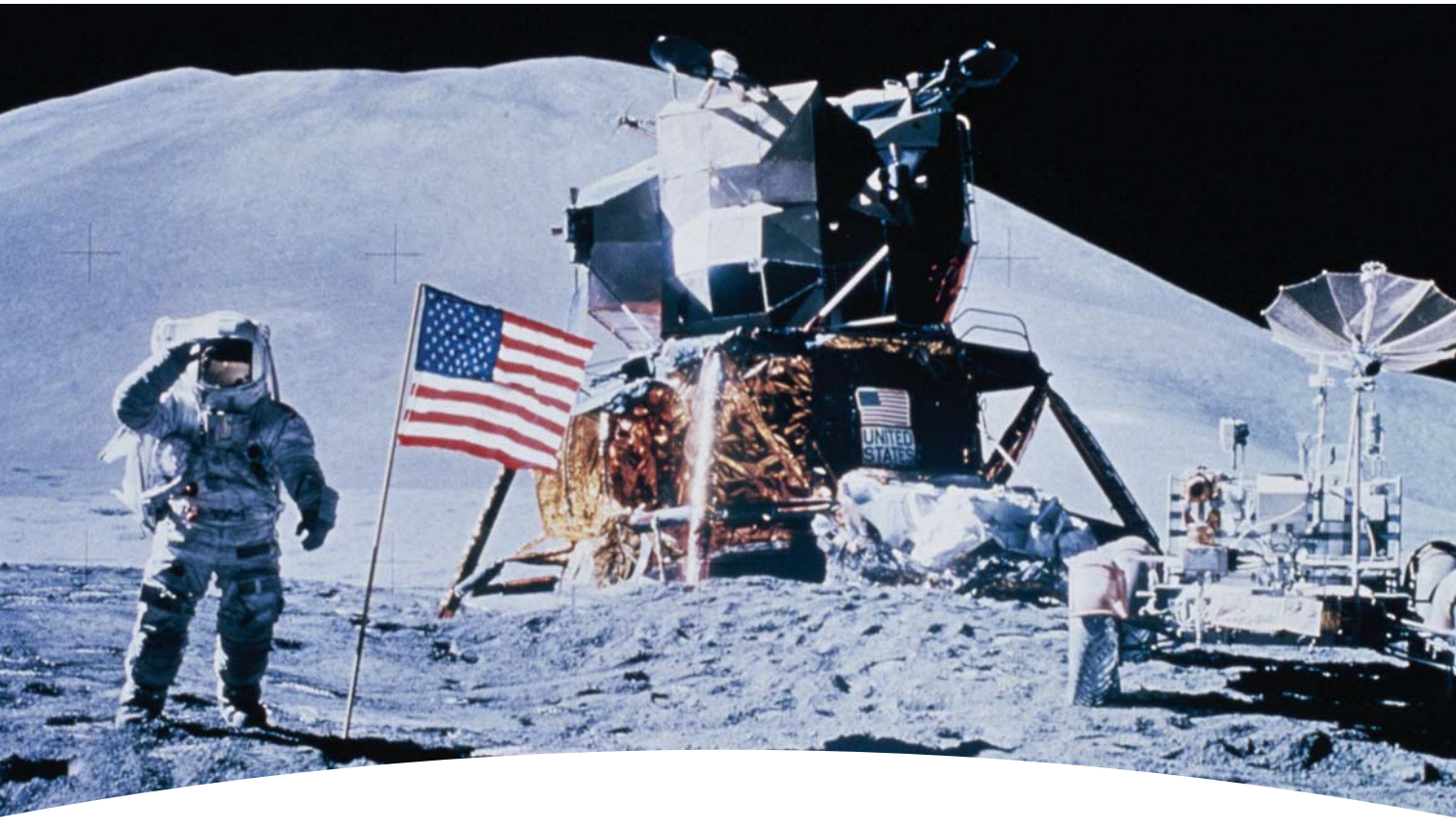

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TELL SP100 WHAT YOU THINK

What do you want to use wireless for? Where would you like to deploy wireless? What are the benefits you expect to get in using wireless?

For once, it's easy for end users to pipe up and tell the industry what they think. The SP100 committee continues to solicit end user opinions on these questions. We encourage you, as end users, to visit http://www.isa.org/MSTemplate.cfm?MicrositeID=1134&CommitteeID=6891 and take the end user poll.

tions could include taking a unit offline when conditions so indicate. Timeliness for this class of action is human scale, measured in seconds to minutes.

Class 2: Closed-loop, supervisory control

"This class of closed-loop control usually has long time constants, with timeliness of communications measured in seconds to minutes," adds the *Calls for Proposal*. "Examples are batch unit and equipment selection." The quintessential examples of closed-loop, supervisory control are oilfield pipeline and water/wastewater SCADA systems. These have been wireless for two decades, already.

Class 1: Closed-loop, regulatory control

"This class includes motor and axis control as well as primary flow and pressure control," says SP100. "It is a matter of semantics," adds Kagan.

"We get a lot of head-nodding," he continues, "when we ask end users if they want to use wireless for closed-loop control. However, when we peel the onion's layers, we find they're really talking about using wireless for incremental process measurement. They're not looking for closed-loop control. There's no user I'm aware of who's interested in wireless for closed-loop control."

Class 0: Emergency action

This class includes safety-related actions that are critical to personnel and plant. Most safety functions are performed through dedicated wired networks to limit failure modes and susceptibility to external events or attack. Examples include safety-interlock, emergency-shutdown, and fire control. However, Angela Summers, Ph.D., president of SIS-TECH consultants and chair of ISA's TR84.02 and TR84.04 committees says, "Wireless sensors have not demonstrated the reliability necessary for SIS applications in the petrochemical and refining industry."

Interoperability vs. Coexistence

It isn't enough, however, to limit your thinking to just these six classes because they don't define all the wireless applications you'll find in your plant. There are a lot of different standards designed for one or more distinct applications. For example, RFID is becoming more and more prevalent, and is used for worker-location systems, inventory management, and even batch-management applications. So, when you're writing specs for your plant's wireless systems, you need to take into account the effects of the RFID transmitters in your wireless control system, and vice versa. Perimeter security is another significant wireless application, and workers in your plant are already using walkie-talkies and cell phones, as well as wireless laptops, PDAs, and other wireless-enabled tools. So, what will happen when somebody keys a microphone right next to your wireless loop controller?

"There are lots of pieces of equipment in the 'wireless cloud' that surrounds a plant," says Dave Kaufman, Honeywell's new business development director (Figure 2). "So, you have to make sure that not only are all your pieces of wireless equipment compatible with each other, but also that you have some way of assuring that the *next* piece of wireless equipment you install won't bring down the existing networks."

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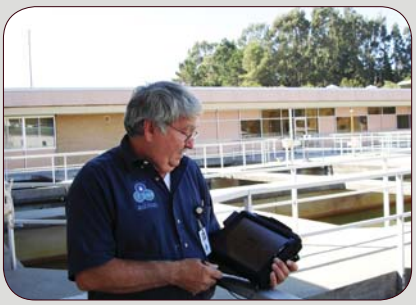
WIRELESS WHILE WE WORK

At the Harry Tracy Water Treatment Plant in San Bruno, Calif., San Francisco Water Department plant operators have cut the cord and become wireless workers, using tablet PCs, and HMI from Wonderware.

"My operators are more efficient," Fong says. "Before, if they had an alarm, they had to go to the control room to acknowledge and identify the alarm. With the tablet PCs, they can acknowledge and address the alarm wherever they are if they're in contact with the WiFi network."



Fong says that many two-man jobs are now one-man jobs. "For example," he continues, "an operator can log in that he's working on the chlorine residual analyzer, so when the operator in the control room sees the chlorine residual alarm goes off he can also see when the alarm is acknowledged. Calibration of our field instruments has become very easy with the tablet PCs."

The tablets are on the plant's WiFi LAN, so if they are in range (the Tracy plant has some physical-location issues with hills and tunnels), the operators can monitor and operate the plant from wherever they are.

Kaufman continues, "When you have wired networks, you're most interested in making sure that every device on the wire is 'interoperable' with every other device, and that there are few if any incompatibilities. This is magnified in the wireless world."

What happens when you try to install too many wireless sensors in what Emerson Process Management's director of technology, Bob Karshnia, and John Berra, Emerson's president, have called, "canyons of steel?"

According to Graham Moss, president of Elpro Technologies, "what you get is increasing latency." Latency is the time lag between the time the message is sent, and the time it's received at the address to which it was sent. Currently, as congestion or interference increases, latency also increases. So, if you want your signal to be received quickly (a critical alarm, perhaps), you may be in trouble. "This means that just installing one huge network with 10,000 sensors isn't likely to work," says Moss. He predicts "multiple networks, arranged in layers."

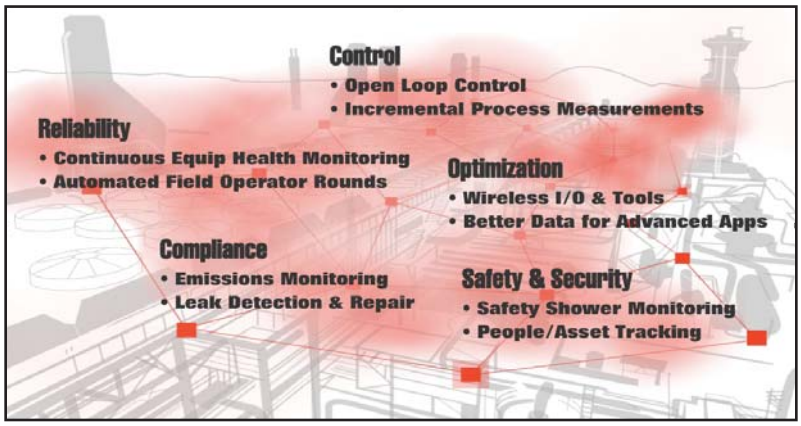
This is why the SP100 committee included coexistence and quality of service as two of its main goals. How does this system coexist with other systems operating in the same band, and how does it coexist with other systems in the facility? How does the system deal with quality of service (latency)? How does adding other services affect the latency of existing services in the network?

A Hacking We Will Go

And then there's the problem of security. Most process control systems aren't directly connected to the Internet, but a wireless network theoretically can be hacked by anybody who can pick up the signal. There's more than one reported case of wireless tampering with process systems. Any industrial wireless standard will need to have a significant security component. In addition, those wireless components will need some sort of certification that they won't die when somebody attacks them. See this issue's "Other Voices" column (p. 16).

FIGURE 2.

THE WIRELESS CLOUD



There are lots of pieces of equipment in the 'wireless cloud' that surrounds a plant.

HART Wireless Update: Draft Spec by End of 2006?

Based on work done at the recent meeting in Minneapolis, most observers believe, along with WINA's Kagan, that "HART Wireless is once again on track, and may well meet your (*Control* editor-in-chief Walt Boyes') challenge to have a draft specification by the end of 2006."

There might even be an interoperability demonstration of HART Wireless at ISA Expo this coming October.

With approximately 20 million devices already installed, this is good news for people who want to start using their HART-enabled devices for more than just smart calibration.

SP100 Standard—Where Do We Stand?

ISA SP100 has split into several working groups. The most important of these groups are the SP100.11 and SP100.14 committees. There's some overlap, but fundamentally SP100.11 is working with process measurement and control networks, while SP100.14 is working with long-term monitoring and asset management applications.

Throughout this summer, SP100 is working on the *Calls for Proposal*. A proposer conference is planned for early September, and final proposals for the standard are due in early December. Selection of the final proposal is scheduled by the end of January 2007, and draft specifications are expected to be written by August 2007, with a standard recommended out of the committee by the end of June 2008.

Assuming an end to the infighting that has dogged ISA standards committees, ISA should approve an S100 standard by October of 2008. Tom Phinney, of Honeywell, who is the convenor of the equivalent IEC standard committee, estimates a similar time scale for IEC approval. Recently, the signs have been good that the infighting will end. Emerson and Honeywell have publicly pledged to turn over any patents and other proprietary intellectual property to the HART Communications Foundation and SP100 committee. "We believe in standards," say Emerson's Berra and Honeywell Process Systems' CEO, Jack Bolick, in nearly the same words. Peace is being made.

Where's All This Going?

What does all this mean for you? You'll be able to confidently purchase wireless networking gear for your HART-enabled field devices by early 2007, and know that they're being built to a standard. By sometime in 2007, some vendors may be producing "draft S100 standard" compatible products, and by 2008, there should be a single standard for industrial wireless in process automation.

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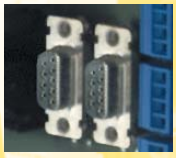

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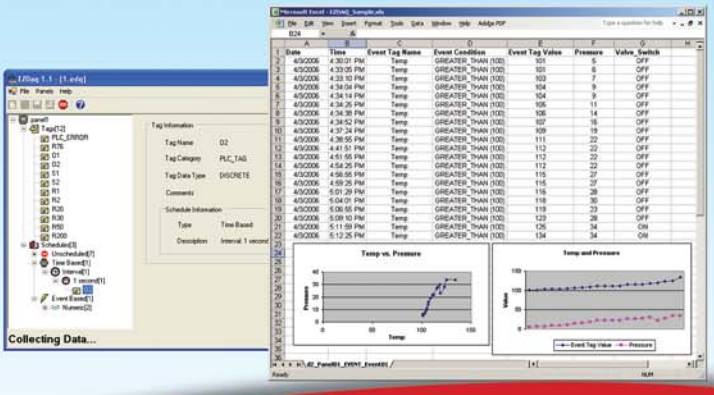

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HUMAN MACHINE INTERFACES EVOLVE IN PROCESS CONTROL

When we moved to digital controls, we forgot the operators.

by Ian Nimmo, president, User Centered Design Services LLC

Our understanding of human machine interfaces (HMI) has evolved. In the early days of automation, the pneumatic panel became the first process industry HMI. These panels (Figure 1) were well planned and broken down into task-specific sections, optimizing the panel operator's performance. Without incorporating task analysis and forethought into the optimal instrument layout, the panel operator would have had to run backward and forward and up and down the panel.

Early panels allowed operators to manage specific equipment from one position with clear understanding and separation of controls. For example, a fuel gas system and boiler feed water were grouped into a logical concentration. The ignition system also was in a logical location with appropriate alarms and interlock mechanisms within easy reach. These panels often had an overview mimic of the process, which was very comprehensive, but often out of date.

In the 1970s, these panels were replaced with an early DCS which didn't have an overview feature, and only allowed the operator access to a group of eight controllers (Figure 2). This initial DCS had many pages of these groups, and operators managed to memorize which controller was on which page. Less familiar operators relied on special keyboards to navigate to relevant groups. Plus, alarms within a group caused the keyboard button to annunciate the alarm.

FIGURE 1.

TRADITIONAL PANEL

Early control panels only indicated equipment status (lights at left), annunciated alarms (two-deck component at right), displayed process variables (windows at center), and relied on human perception (operator in foreground).

Each controller was a software representation of the pneumatic or electronic equivalent controller located on the panel. The group was configured based on layouts on the panel, and no new task analysis was done. Sometimes due to the limit of only eight controllers per page, a necessary grouping was lost, and operators had to navigate to other pages to resolve abnormal conditions. Though operators lost the trend that was with the controller, they could navigate to a detail display to retrieve this information. This was only done during the diagnostic process, so normal operation lost a valuable insight to changing variables and direction of change. The big picture provided by the mimic panel soon became outdated, leaving operators with only a keyhole view to the process. The only overview the system provided was an Area and Unit overview display, which provided a summary of the group display.

Without incorporating task analysis and forethought into the optimal instrument layout, the panel operator would have had to run backward and forward and up and down the panel.

The discipline of task analysis was also lost in this transition, and operators became adept at finding information, but often suffered from tunnel vision. Hence, the need for additional alarms was conceived. Though the system had some minimal trending provision, it was difficult to use and not always user friendly, so operators became reluctant to use it. Some managers tried to reuse some independent trending equipment, but because there was no service on the trending equipment, they too became superfluous.

The next DCSs allowed any single point the ability to configure multiple alarms per controller. Unfortunately, without enough discipline or management of change, plant alarms went from 75 hardwired alarms to 24,000 alarms with some duplication. A single event could initiate several alarms such as level, plus rate of change, and sometimes

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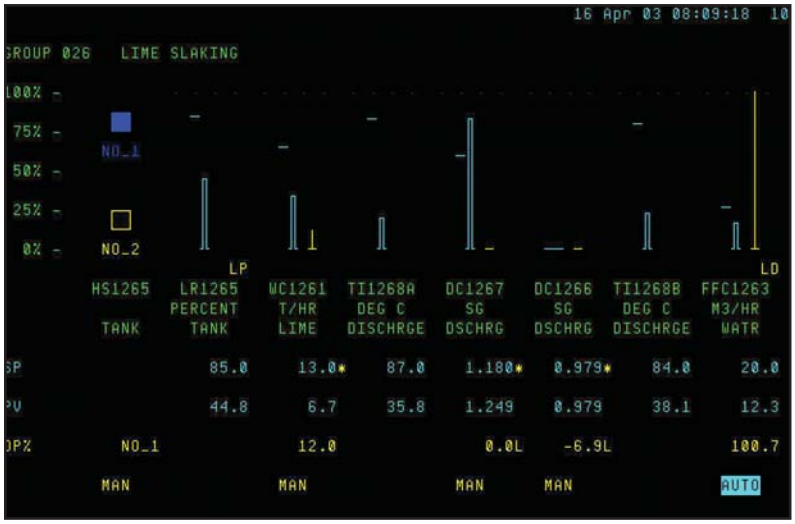
FIGURE 2.

DCS: THE EARLY YEARS



Originally, DCSs didn't have an overview feature, and only gave operators access to a group of eight controllers.

bad process variable as the transmitter went out of range.

This generation of DCSs (Figure 3) also provided simple graphical representations of the process. These early graphics were crude and fairly low resolution, and their design was based on simple segregation of the plant P&ID. This was the control engineers' perception of the process, and how they related to it. Before long, operators complained about the graphic or schematic designs because they didn't reinforce the operator's mental model of the plant, and this loss of task organization required the operator to redesign the graphics.

The redesigned graphics improved, but were still problematic because they went from a logical interpretation of the P&ID to an ergonomist's worst nightmare. They had major human-factor issues, too many colors, and multiple uses of color. Rather than reserving colors like red and yellow purely for alarms, all color in these graphics was highly intense. The objects had unnecessary detail; object sizes and position on layout weren't consistent with relative importance; direction of flow was inconsistent for feeds and products; symbols and lines were poorly implemented; text and numbers were small; and fonts were difficult to read. Numeric format also was difficult to read, some variables also had inappropriate levels of detail, and engineering units were inconsistently applied, especially as a space-saving exercise to cram more information on the screens.

Operating Display Problems

Graphical shortcomings have contributed to human errors and incidents, some with severe consequences. For example, regulatory inspectors identified several HMI-related issues as contributng to the July 1994 explosion and fire at the Texaco Pembroke refinery in Wales, U.K. Under the fluid catalytic cracking unit (FCCU) control system that existed on the day of the incident, any imbalance in liquid flow through the FCCU could lead to liquid accumulation. Therefore, it was important that any imbalance in liquid flow be detected, so the mass flow of the unit could be returned to a balanced position. The plant was well equipped with alarms, which showed where liquid was accumulating, but it was more difficult to assess the relative flows through the vessels and the overall mass balance of

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the unit. The process of fractional distillation requires that one raw material be divided into many fractions. While it was easy to assess the unit feed rate, the various outputs of the process were spread over five product streams. This caused a practical problem because the accumulated outputs of the system may be spread across several different control display units, and the overall output of the unit wouldn't be readily apparent unless the control system were configured to meet this need.

Unfortunately, this need reportedly wasn't met at Texaco Pembroke. There were no displays providing an overview with an appropriate time scale on the FCCU. Therefore, it was difficult to obtain a complete picture of the whole or large sections of the process. In a mostly display-screen-based operating system, the provision of good overview displays is especially important because the operator doesn't have a continuously available set of panel indicators.

During the incident, no one from the operations department had a complete picture of the FCCU. The actual FCCU graphics on the operating displays weren't best designed or configured to help operators control the process. The operating graphics on the FCCU contained limited amounts of process data per graphic, and didn't use color and intensity to highlight process data. Some graphics contained details of the internal structure of plant items. However, displaying the structure of plant items is only useful if measurements or derived information, such as pressure, temperature and flow, also are displayed to give the operator information relevant to plant status. At times, the text was unnecessary in the FCCU graphics. Text takes up large amounts of space on a graphic, and there were instances where the same information could have been better indicated by color change, according to U.K. Health and Safety Executive (HSE) investigators, reporting on the disaster in "Investigation into the Explosion and Fires on the Pembroke Cracking Co.'s Plant at the Texaco Refinery."

The HSE investigators' recommendations on human factors are of immediate interest in designing HMIs, especially the report's Recommendation #3: "Display systems should be configured to provide an overview of the condition of the process including, where appropriate, mass and volumetric balance summaries."

Other studies into these types of human errors have identified that 30% of errors are attributed to the operator not becoming aware of an abnormal situation due to information overload, vague or misleading information, inappropriate levels of detail and lowered operator vigilance. Further research identified an additional 20% of errors

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FIGURE 3.

STICK FIGURES

Initially, DCS-based process representation graphics were crude and fairly low resolution, and their design was based on simple segregation of the plant P&ID.

attributed to the operator's inability to identify the root causes in a timely manner due to insufficient knowledge, lack of operator experience, conflicting priorities, inaccurate labeling or information presentation, and excessive mental tasks. Many of these issues can be resolved by a well designed HMI.

Many of these traditional problems were initially introduced by poor guidelines from DCS manufacturers and even worse project implementation practices by vendors. Fortunately, vendors have continued to evolve the DCS, and now the graphics are high resolution, and are becoming web based. Unfortunately, the vendors again haven't invested in human factor education (Figure 4), and now promote 3-D objects that take over 60% of the screen. This means that 60% of the screen's fixed data is unfortunately in the foreground, while real data useful for detecting and diagnosing abnormal situations is shoved to the remaining 40% of space in the graphic's background.

Ergonomics Can Resolve Operator Errors

In addition, the process industry hasn't responded to the human-error problem by following guidance that's now available for alarm management. EEMUA has also produced a similar guideline for graphics called "Process Plant Control Desks Utilizing Human-Computer Interfaces: a Guide to Design, Operational and Human Interface Issues, Publication 201." This document hasn't had the same publicity or exposure, and needs further detail, but it's a great start to improving the standards and current HMI or HCI practices.

Other industries, such as the nuclear field, have good guidelines for HMI developers, and many of these are directly relevant to the process industry. Some are available in the U.S. Nuclear Regulatory Commission's "Guildeines for Control Room Design Reviews," published in 1981.

Likewise, ISA produced recommended practices for fossil fuels on HCI and task analysis, which appeared in the organization's RP 77.60.05-2001 report, "Fossil Fuel Power Plant HMI Task Analysis."

Display systems should be configured to provide an overview of the condition of the process including, where appropriate, mass and volumetric balance summaries.

In fact, the latest recommended practices for developing graphics includes starting with, not just a task analysis as was done 50 years ago for the control panel layout, but more in-depth analysis using a variety of techniques. These formats include:

- Hierarchical task analysis
- Tabular task analysis
- Timeline analysis
- Link analysis diagram
- Tabular human error analysis

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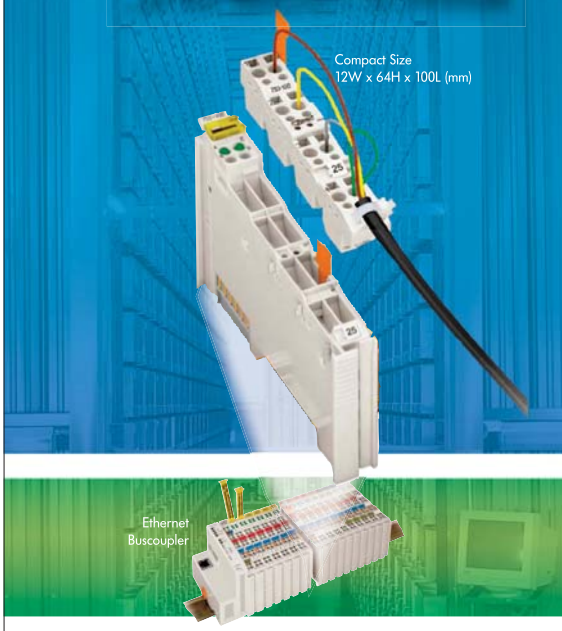

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Better Graphics

Some of the latest practices associated with HCI development directly address these human factor issues by first providing a good graphic style guide. The background of HMI graphics also has changed from the traditional black to a grey background to mitigate glare. Most lighting schemes will produce glare and contrast issues when first delivered through glass VDUs. This problem was dramatically reduced as technology changes allowed LCD screens to be implemented. Black backgrounds facilitated operators turning the control room lights out, but use of multiple bright colors on these black backgrounds added to their complexity.

Graphics also have become easier to read as developers moved to brighter grey backgrounds, and reduced use of added colors by using high and low-intensity *hues*. Glare and reflection were reduced, but potential problems can still arise. Moving to brighter background color can introduce new problems if a control room's lights are still turned out. Ideally, the control room should be designed to facilitate operator vigilance, especially with 12-hour shifts, by illuminating the control room between 500 and 800 LUX.

As mentioned above, the practice of using grey on grey for lines and vessels has allowed developers to put the fixed information into the background of the graphic, and place the variable information in the foreground.

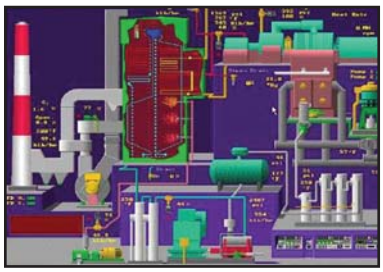
Reserving colors and limiting color adoption in the graphic enables easier identification abnormal operation using grey on grey as normal and color as an attention getter. When a high-priority alarm annunciates on these displays, it becomes very visible on the graphic. And when the alarm is acknowledged, the color intensity is reduced to half, making the alarm still visible, but not the highest attention. This allows priority to become part of the graphic design.

Some investigations have been done about using graphical analog instruments, rather than just using digital numbers. Level and flow gauges are easier to read and make estimates of than relying on purely digital information. These new graphic systems use objects to represent mathematical solutions, such as mass energy balance and polar star techniques to represent change, and how it impacts other physical variables. This provides a better understanding of how to correct problems.

The technology now allows integration of information outside the DCS, such as Microsoft Word and Excel documents to be integrated into the graphic. By using pulldown menus, one of these IT based documents can

FIGURE 4.

HMI DESIGN MISTAKES REDUX



HMI makers mistakenly fixed 60% of a screen's data in the foreground, while data for diagnosing abnormal situations shoved into the remaining 40%.

be launched to provide calculations, lookup tables, procedures, and diagnostic information.

As the HMI or HCI world continues to change, we have the power, but we need the discipline and knowledge of how to use this technology. We need to recognize that the problem we're solving is called "situation awareness," and alarms, trends and graphics are an integral part of the awareness solution.

This solution requires a true overview display that allows operators to monitor the important equipment within their scope of control, and provides clear indication of the highest level of alarms. The display needs to provide production information that will help the operator monitor the most important key performance indications (KPIs). Some solutions may include a graphical barchart showing normal, the ± off-normal indication, and relevant alarm data.

The graphic system of the future will have trending embedded in the displays; operators again will use trends to predict off-normal; and they'll be less dependent on alarms. The new system will exploit new navigation techniques, such as hierarchical navigation, which moves from overview to unit view to detail view, and finally down to diagnostic views and controller change zones. This form of navigation allows use of yoking techniques. This will allow automatic screen setup based on selecting a point, and populating the other displays with the relevant information associated with that point from overview to unit to detail to trend and alarm data.

Industry needs to free up and encourage engineers to address this topic. Research should be done looking back at how operators and supervisors used the panel, how operators adapted to group displays and the new transition to cool graphics that provide good situation awareness. As an industry, we also need understand the research and knowledge that the nuclear, fossil fuels and aircraft industry have identified as best practices in HMI development.

The final solution will not be driven by operators or control engineers, but by a management team that understands the issues with the existing culture, is brave enough to follow the leaders in industry, and invest in good task management and up-front project loading. Investment is needed in graphical style guidelines suitable for all DCS vendors systems and project-specific requirement specifications. This specification is a how-to guide to implementing the style guide on a specific system and/or project. Investment in graphical object libraries is also required to provide more than traditional valve, pump, and compressor objects, so the new systems will have diagnostic modules that help the operator understand more advanced information, such as material mass and volumetric balance.

Driving human error out of systems and improving safe work practices are management responsibilities, but management still lacks knowledge and experience to prevent those errors. They have a poor understanding of human factors, and are therfore complacent. The control engineer can do a service to his company by raising awareness of this topic at the highest levels in an organization, and pursuing best practices on the control system upgrade projects.

Ian Nimmo, PE, IEng, MIET (CEI), is president of User Centered Design Services LLC, and can be reached at inimmo@MyControlRoom.com and at www.MyControlRoom.com.

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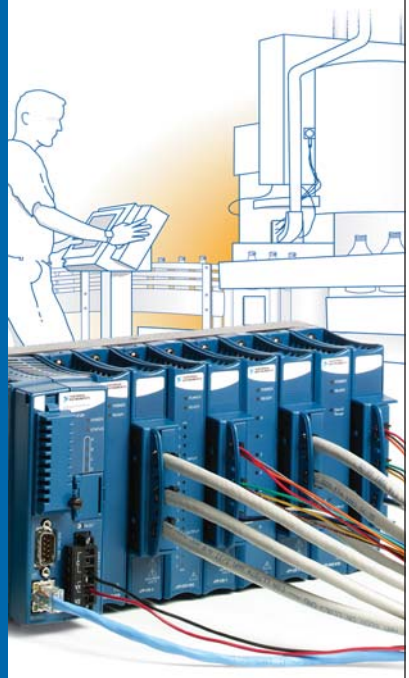

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BOILING WATER IS EASY AS ONE, TWO, THREE, FOUR

How many elements does good boiler level control need?

by David W. Spitzer

An analysis of boiler level controls shows the importance of the interrelationships between the process, measurement instruments, and control strategy used to achieve the control objective. In short, the boiler burns fuel to convert water into steam, and makeup water is added to replace the steam (water) that leaves it.

Operating at a high water level in the boiler can carryover water into the steam header. Operating the water level too low can damage the boiler. Neither of these conditions is desirable, so the boiler should be operated so its water level is such that neither the high nor low-level switches shut it down. The overall control objective is to replace the water that leaves the boiler in such a way that its water level remains constant.

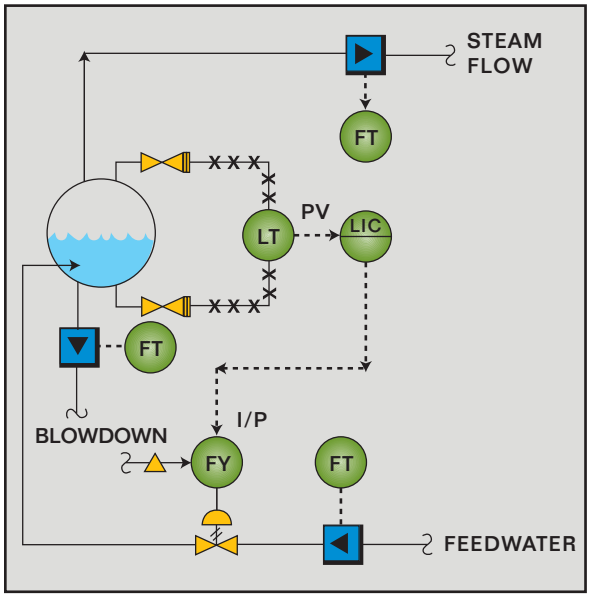
A single-element boiler control strategy (Figure 1) achieves this objective by measuring boiler level, and using a controller to manipulate the makeup water feed valve. Consequently, increasing steam demand causes more steam to leave the boiler, which lowers its water level. The controller senses the low level (feedback), and reacts by increasing the makeup water flow to increase the level to its setpoint.

The nature of this control strategy is such that changes in the makeup water flow are initiated when the boiler level isn't at its setpoint. Because makeup water flow will vary with demand and firing rate, the boiler level will deviate from its setpoint during normal operation. This strategy may be adequate for many boilers, especially in applications where steam demand is constant or varies slowly.

Complicating the issue is the transient inverse response of the water level in the boiler. When more fuel is burned, the water's temperature increases, and it expands, thereby raising its level. On the other hand, more water is boiled when more fuel is burned, thereby lowering the level. In actual operation, the boiler level initially increases in response to expansion followed by a drop in level due to the loss of water. Therefore, the level controller can react incorrectly to a fuel increase by initially reducing makeup water flow in response to the increasing level measurement. More so-

FIGURE 1.

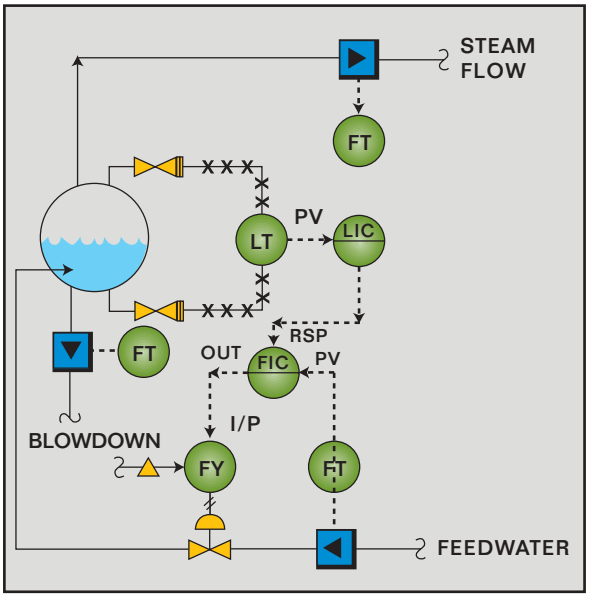
SINGLE-ELEMENT BOILER LEVEL CONTROL



A single-element strategy keeps a boiler's water level constant by measuring level, and using a controller to manipulate the makeup water feed valve.

FIGURE 2.

TWO-ELEMENT BOILER LEVEL CONTROL



Two elements add a cascade-makeup flow loop to reduce level variations, and makeup water fluctuation. This minimizes flow loop disturbances, and better tunes the level loop.

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phisticated controls may be necessary to compensate for this effect.

For instance, implementing a two-element boiler control strategy by adding a cascade-makeup, water-flow control loop to the boiler level control (Figure 2) can improve control by reducing level variations from setpoint and makeup water flow fluctuation. This strategy cuts the effect of disturbances in the flow loop, and allows the level loop to be tuned for better performance.

To reduce the effect of changing steam demand on boiler level, a three-element boiler control strategy can be used (Figure 3). In this strategy, steam flow from the boiler is measured, and is used to adjust the output of the level controller, so that a change in steam flow results in a like change in makeup water flow. This control strategy operates as feed-forward because it manipulates the makeup flow before the boiler level has time to change.

Finally, four-element boiler level control (Figure 4) can be implemented by taking boiler blowdown into account.

Fossil-fuel boiler designs are more similar, so shrink-swell varies with the density difference between the water and steam. In higher pressure boilers, the density difference is less than in lower pressure boilers. Therefore, boilers operating at higher pressures generally exhibit less shrink-swell than lower pressure ones.

Alex Klemptner, lead instrumentation and control engineer at the DTE Energy Fermi 2 Nuclear Power Plant in Monroe, Mich., says, "The distributed control system that controls the reactor vessel's water level uses single-element control for the plant startup before automatically switching to three-element control at about 20% power. The system also has a special design feature to compensate for a level swell following reactor scram (shutdown). The control system temporarily reduces the water level setpoint to supply more feedwater to the vessel. This is done to ensure that the water level stays above the emergency cooling initiation setpoint."

"The boiler burns fuel to convert water into steam, and makeup water is added to replace the steam (water) that leaves it...The overall control objective is to replace the water that leaves the boiler, so that its level remains constant."

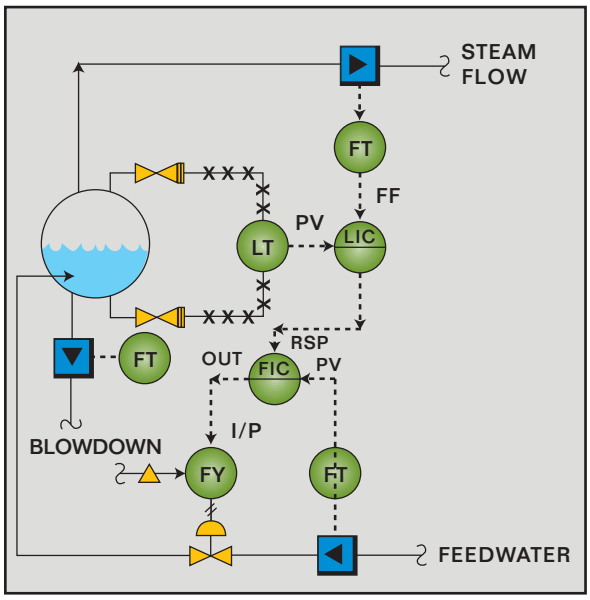
Greg Shinskey, a process control consultant from North Sandwich, N.H., says, "The magnitude of shrink-swell (inverse response) varies with the type of steam generator. In nuclear power plants, pressurized water reactors exhibit much more shrink-swell compared to boiling water reactors where large volumes of water surround the nuclear fuel." He suggests using a non-linear filter in the controller to let the feed-forward do most of the work in pressurized water reactors (see "Taming the Shrink-Swell Dragon", *Control*, March '04, p. 35).

Klemptner adds, "Reactor water level is the most important input into the automatic control system, so high instrument accuracy is needed. Nuclear generating stations would like to take advantage of the latest level measurement technology by using smart transmitters. However, smart transmitters aren't qualified for nuclear applications where stringent temperature, humidity, and radiation requirements must be met."

John O'Malley, control system engineering at Consolidated Edison

FIGURE 3.

THREE-ELEMENT BOILER LEVEL CONTROL



In three-element boiler control, steam flow from the boiler is measured, and used to adjust the output of the level controller, so a change in steam flow results in a like change in makeup water flow.

(New York, N.Y.) advises, "There are additional issues that should be considered when designing drum-level controls for heat-recovery steam generators (HRSG)."

O'Malley continues, "The HRSGs we installed use exhaust heat from a gas turbine and supplemental gas-fired duct burners to provide heat to multiple economizer, evaporator, and superheater sections. The large steam drum, rapid startup, and wide load range created difficulties in maintaining drum level, so that four-element boiler water level control with metered blowdown was required.

"The differential pressure level transmitters are pressure-compensated to take the density of water and steam in the drum into account. In addition, the level setpoint is varied based on pressure during startup."

To ensure accuracy, O'Malley uses "properly sized, insulated, heat-traced, stainless-steel impulse lines for freeze protection that are properly installed as close as practical to the taps to reduce cost and potential leakage. In addition, the large steam drum was shown to have distinct internal waves due to rapid heat input during startup and ramping, so level transmitters are located with independent taps near the middle of the drum and at both ends. The distributed control system bases its control actions on the median of the three level transmitters or the average of two level transmitters if one transmitter is not operational."

Sarah Parker, level products applications manager at Emerson Process Management in Chanhassen, Minn., says, "Differential pressure and displacer level transmitters are

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reliable, and have been widely used to measure boiler level. The use of capillary tubing with diaphragm seals, as opposed to using a wet-leg differential pressure transmitter design, is primarily a matter of user preference."

Parker cautions that level measurement systems are susceptible to accumulation of solids, and should be blown down occasionally. "Susceptibility depends, in part, on its connections. For example, 0.5-in. tubing would be more likely to plug than a 3-in. diaphragm seal," she says. "The accuracy of these devices is affected by the water and steam density changes caused by the operating pressure and temperature. In particular, water density decreases and steam density increases as steam pressure and temperature increase. The water density alone can change by as much as 10-20% where higher pressures create greater density changes. This change impacts the level measurement, and can be significant where users want to control within 1-2 in. with a 30-in. range. Transmitters are typically calibrated to compensate for the water and steam density at the nominal operating pressure, and can compensate for pressure changes in their control systems."

Johnathan Rowe, pressure instrument marketing manager at Invensys Foxboro in Foxboro, Mass., adds, "Multivariable transmitters can use the steam pressure measurement to calculate and compensate for water and steam density changes. Also, error can be introduced by ambient temperature variations of the impulse lines or capillary tubing. In typical applications, transmitter calibration is calculated assuming that the impulse lines and capillary tubing are at a fixed nominal average ambient temperature. Multivariable transmitters can be used to measure and compensate for the ambient temperature of the impulse lines or capillary tubing."

Meanwhile, Parker reports, "There's an increase in the application of guided-wave radar level transmitters for this application. They aren't affected by density changes, but they are affected by the increase in dielectric constant of steam as its operating pressure increases. However, guided-wave radar level transmitters can be calibrated to compensate for nominal operating conditions, and compensation for actual operating conditions can be performed in the user's control system."

Implementations beyond single-element control require the measurement of feedwater, steam, and/or blowdown flow rates. "Both differential pressure, such as orifice plates, and vortex shedding flowmeters have been used in this application," says Steve Pagano, senior product manager

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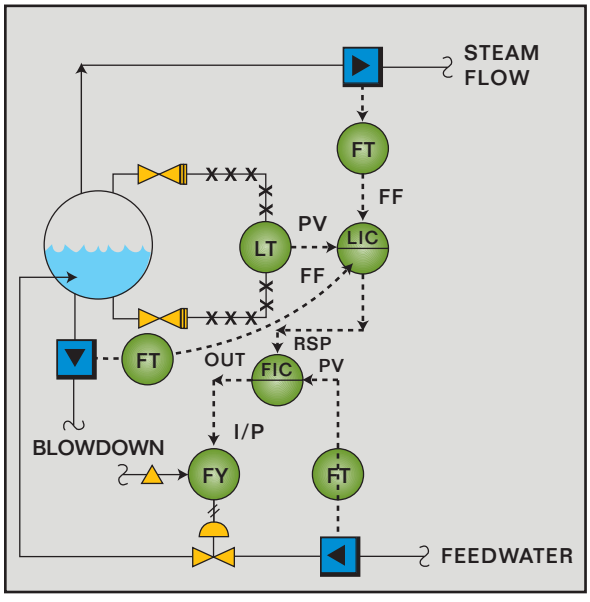
FIGURE 4.

FOUR-ELEMENT BOILER LEVEL CONTROL



Four-element boiler level control is implemented by taking boiler blowdown into account.

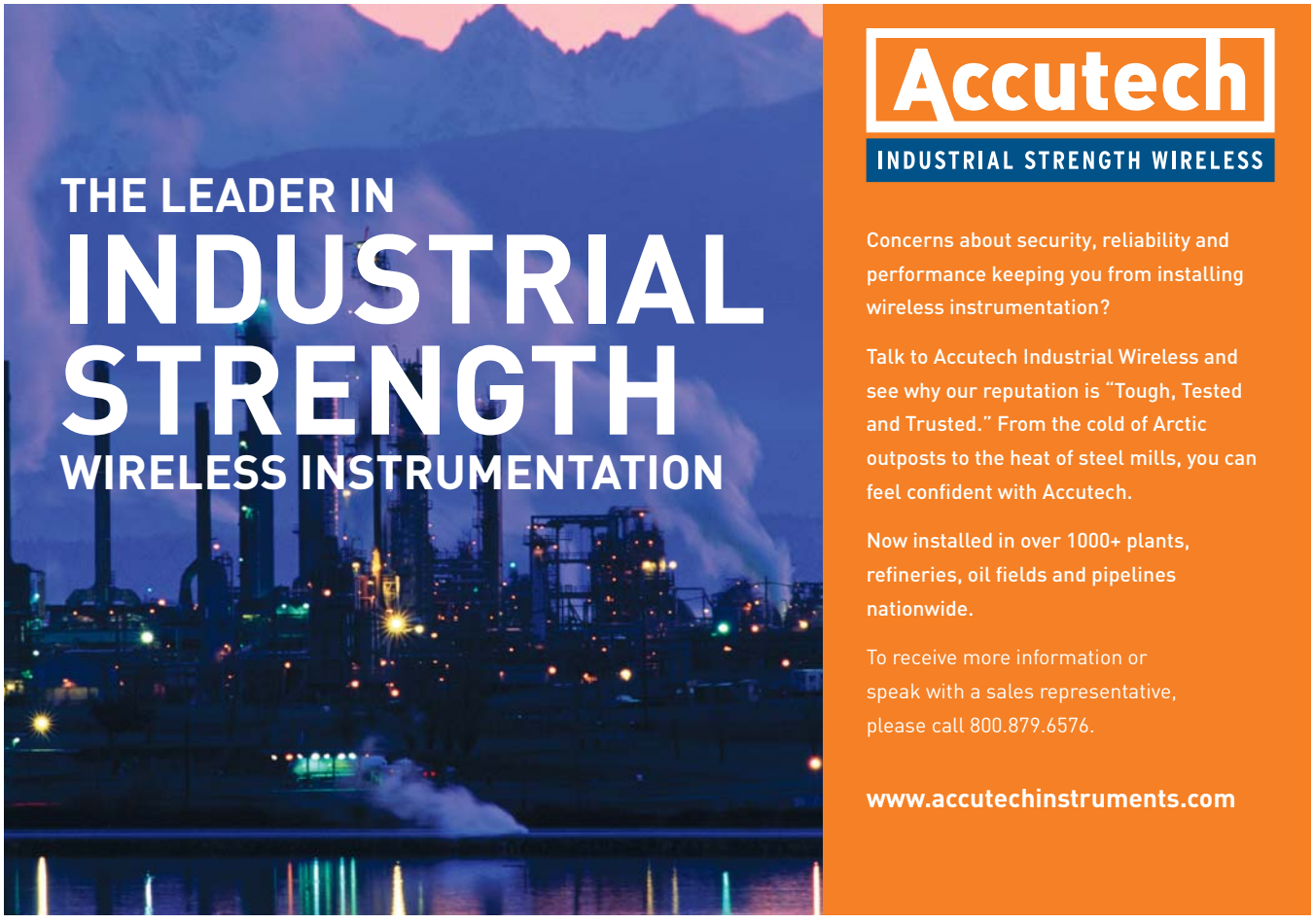
at ABB in Warminster, Pa. "However, the distinct trend has been toward increased application of vortex shedding flowmeters that offer high accuracy with good flow turndown within relatively short straight run requirements.

"There is a similar trend towards vortex shedding flowmeters in steam flowmeter applications where mass flow is typically measured. Though both technologies offer pressure and/or temperature compensation in their respective multivariable transmitters, or in a separate flow computer, the trend is toward multivariable vortex shedding flowmeters that have integral pressure and/or temperature measurements.

"In general, it's felt that vortex flowmeters are more forgiving than differential pressure flowmeters, when measuring blowdown flow that can be subject to two-phase flow conditions."

So, there's much more to boiler level control than simply measuring level and adjusting a feedwater valve. Improved measurements and inverse response are just a few of the influences on the effectiveness of the control loop and the ability to operate boilers in a reliable manner.

David W. Spitzer is a principal in Spitzer and Boyes LLC, and can be reached at 845/623-1830 or at spitzer@spitzerandboyes.com.



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NESSI LOCKS ON TAMING COMMUNICATIONS MONSTER

Despite recent fieldbus-related growing pains, the new sampling/sensor initiative (NeSSI) is moving steadily to fulfill its initially promised efficiencies and savings, and eventually deliver micro-analytic capabilities.

by Jim Montague, executive editor

Maintaining and replacing the tubes, pipes, and brackets used in traditional process analytical sampling systems can take a lot of time. Just describing it can be a mouthful.

Technicians have to check drawings, disassemble the system, remove worn or faulty parts, bend new tubes, rethread pipes, check fitting unions, coordinate I/O components, remount brackets, and complete other application-specific tasks. These jobs typically take two to four hours or more, and even then breaking and reinstalling seals can damage their surfaces, and cause chronic leakage problems.

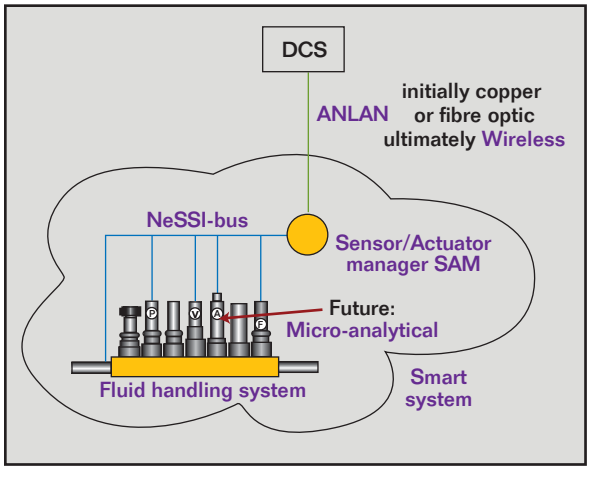
So, it's no wonder that process analytics users long sought a way to make sampling systems easier to repair and reconfigure. However, even though some found and developed a modular solution that allows them to switch out a regulator in 60-90 seconds and doesn't leak later, making it a mainstream reality has produced some unforeseen delays and technical snags on the way to the desired happy ending.

Sampling History

Analyzers were first added to industrial process systems in the 1940s, and they've progressively evolved from bathtub-sized enclosures into smaller, less intrusive, intrinsically safe packages. However, it wasn't until the late 1990s that some process analytics users, reportedly from U.S. refineries on the West Coast, observed uniform, surface-mount components being used in semiconductor processes, and reasoned that their sampling systems could use a similarly standardized strategy. These semiconductor components include 1 1/8-in. gas sticks and metal o-rings used that help measure gases used in semiconductor manufacturing.

FIGURE 1.

SAM ON THE BUS



The New Sampling/Sensor Initiative (NeSSI) uses a software-based sensor/actuator manager (SAM) to help it handle network communications and complete fluid handling tasks.

These "metal Lego blocks" allow users to switch out components from different manufacturers without having to switch out a whole sampling system.

These users formed the New Sampling/Sensor Initiative (NeSSI) with help from the Center for Process Analytical Chemistry (www.cpac.washington.edu/NeSSI/NeSSI.htm) at the University of Washington in Seattle. The group's initial, surface-mount strategy was adapted for oil, gas, and chemical processing applications by the ISA's SP76 committee, and became the basis for the international ANSI/ISA-76.00.02 standard for sample flow control, conditioning, and analysis, which was released in 2002.

The modular standard requires compliant sampling systems to be built onto a standard substrate from closely-joined, 1.5-in-square manifolds with 0.110 sample-transfer ports, which are attached with four Allen wrench screws. These "metal Lego blocks" allow users to switch out components from different manufacturers without having to switch out a whole sampling system.

Another of NeSSI's physical advantages is that it can perform all same functions of a 30x60-in. legacy sampling system in a 16x20-in or 12x16-in space. This can add up to a lot of savings in refinery that may have dozens or hundreds of sampling systems, and where a square foot might be worth $1,000, depending on its location. In addition, a 1-sq-ft legacy valve system can be replaced by a 1.5x3x3-in NeSSI valve.

"NeSSI is cheaper to build and cheaper to own. Its reliability is greater, and it delivers more operating and safety credits than traditional analyzer systems," says Jeff Gunnell, lead specialist for process analytics at ExxonMobil in the U.K.

"I've been working with sampling systems since 1969, and NeSSI is the biggest advance in process analytics in the past 50 years," says Bob Sherman, an ISA fellow and industry specialist for process analytical systems at Circor Tech, which manufactures modular, NeSSI-compliant devices. "Whenever people move from legacy sampling systems to modular substrate systems, they find they can optimize filters much more easily, for example, and achieve new, more efficient capabilities."

In fact, Sherman adds that several hundred NeSSI-compliant components have been installed, tested, and serving in sampling systems in some West Coast refineries for up to four years, and the maintenance they require is 30-70% less than the maintenance required by legacy systems. Gunnell adds this is why NeSSI's organizers can still stand behind their initial estimates that NeSSI can save 30% on building costs and save 40% on ownership costs over a legacy-based solution during the 15-year lifecycle of a complete sampling system.

The Communications Race

Since it was first conceived and started development, NeSSI has been organized into three successive generations. Generation 1 covers the mechanical components, which were first available in 2001-02. These include flowmeter switches, regulator valves, filters, and other devices. Dave Veltkamp, senior research scientist at CPAC, says NeSSI's emergence has helped reinvigorate fittings companies, such as Swagelok, Parker Hannifin and Circor, which modified traditional valve systems to serve in surface-mount settings, and also began making the substrate onto which the surface-mount components fit.

Generation 2 covers and is seeking to define how electrical signals and power are delivered to NeSSI devices. These tasks include how NeSSI should build multidrop networks and how to communicate temperature and pressure signals. However, the biggest stumbling block has been how NeSSI should define its communication network. Several candidates have been evaluated, and each is developing the necessary modifications to be used in NeSSI sampling systems.

"When we released Generation 2's specification in 2002, we thought CPAC's NeSSI steering committee could simply state what communications network we wanted to use, and then all NeSSI developers would have to use it," says Veltkamp. "However, once we looked at what was available,

we found that nothing met all of NeSSI's requirements. So, we're being more pragmatic now, and we're letting the users and vendors decide what bus they want to use. The component vendors have agreed to support whatever bus the users request when working with NeSSI or when they order a system.

"We got a bit ahead of ourselves by saying that NeSSI systems had to be intrinsically safe, but we didn't look hard enough at what we had to do make this a reality. So, NeSSI has languished some because of this lack of an available, intrinsically safe network for NeSSI to specify."

So far, Veltkamp reports that CANbus does much of what NeSSI's end users want to do, but end users want an intrinsically safe network, and CANbus isn't intrinsically safe yet. So, CAN in Automation (CiA) is working with NIST on developing an intrinsically safe bus for NeSSI. The two groups are testing a draft specification, but it will be six months to a year before they can release an intrinsically safe bus for NeSSI.

Similarly, Foundation fieldbus (FF) is intrinsically safe, but Veltkamp adds its chipset was too big for NeSSI's existing, physical footprint. Profibus' hardware also was too big because it uses the same chipset as FF. Emerson Process Management recently offered to release its chipset for NeSSI to vendors that want to use FF for their NeSSI systems.

Meanwhile, Siemens reportedly is planning to use i^2C in its NeSSI devices. This is an open, short-haul, chip-to-chip communication protocol that's usually used to communicate within individual devices. However, i^2C may actually need less renovation than other communications buses to work with NeSSI devices, according to Veltkamp.

Though it might seem like the large process control vendors are seeking to establish proprietary holds on NeSSI's communications methods in the same way they perpetuated proprietary fieldbus standards in recent years, Veltkamp says its unlikely that NeSSi will succumb to these forces because its end users have already decided what functions they want, and it's up to the vendors to meet those demands. For example, NeSSI's mechanical requirements already are settled in Generation 1, and vendors will likely have to comply with the open standards that users will want to see included in Generation 2.

However, another potential snag in Generation 2 is what to do with data signals once they're on the communications bus. Veltkamp says Generation 2 specifies using a software-based sensor/actuator manager (SAM), which comes in two versions, one for PCs and another for hockey puck-sized embedded computers (Figure 1). This is expected to help gas chromatograph (GC) vendors, for example, as they decide how they want to implement NeSSI-based

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data acquisition and control. In fact, less than six months ago, three major GC vendors, ABB, Emerson, and Siemens, began cooperating on designing automated NeSSI systems for GCs, and plan to have devices available in about one year, according to Veltkamp.

Besides including electronic device description language (EDDL) communications, including FF and Profibus, in their specs, Veldkamp adds that NeSSI's developers also plan to interact with OPC and wireless technologies.

When everyone finally settles on implementing Generation 2's communications, NeSSI's developers plan to work on Generation 3, which is a "wish list" for building a platform for getting micro-analytical devices on NeSSI's substrate. Though no specification has been developed for Generation 3, it eventually will allow lab-on-a-chip capabilities, bolt-on NeSSI spectrometers, and other advanced devices and capabilities.

Gaining Momentum

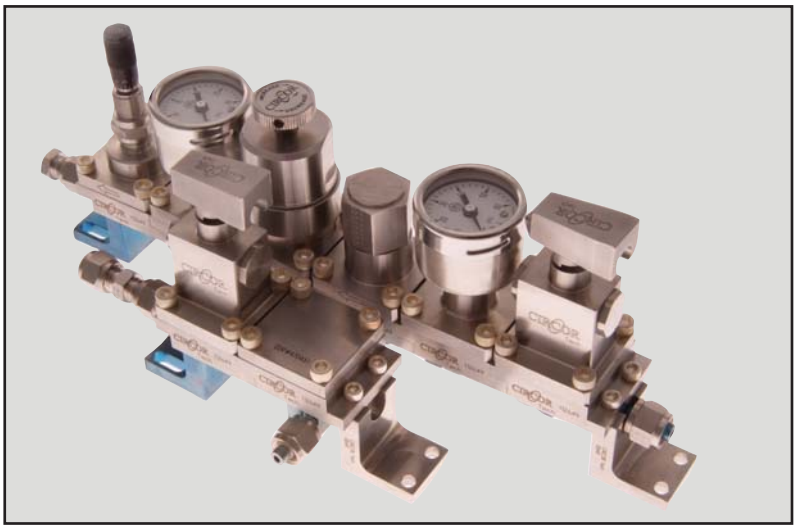
NeSSI's supporters say the good news is that sales of its components are more than doubling every year, but the bad news is that they expected much larger, faster sales increases given NeSSI's huge benefits and potential savings—even though NeSSI's components initially cost about 20% more than traditional tubing and brackets. Despite all of NeSSI's demonstrated advances and benefits, however, many potential end users reportedly remain reluctant to try it. This might be the result of users' traditional "if it ain't broke, don't fix it" attitude, and that fact that many of them (and most of the rest of us) are still like the two little boys in the legendary Life cereal commercial—"I ain't gonna try it. You try it."

End users' hesitance on NeSSI also may come from the feeling that its benefits are just too good to be true. Efficiency gains of a few percent or even a fraction of a percent can produce big savings in process applications, and so users can probably be forgiven for viewing comparatively huge, double-digit improvements as somewhat unreal.

"I owned and operated a system integration company several years ago, and the first modular system Circor Tech put in was for cleaning up samples for Reid vapor pressure analysis on a gasoline blender. The refiner's technician said his application was just going to break

FIGURE 2.

STUDENTS' MODULAR SAMPLING SYSTEMS



Student teams had 45 minutes to use a box of substrate components, including Circor Tech's μMS³ modular substrate basic manifold system, to assemble this and other functional modular sample conditioning systems during ISA Expo 2005's International Student Games in Chicago.

our new modular components, but eight weeks later he was asking 'How many more can we get?' These components became a *de facto* standard for them," explains Sherman. "The question that potential users have is does NeSSI have staying power, or is it just another flash in the pan? Over the last six months, I think we're beginning to reach critical mass."

In fact, NeSSI may be on the verge of making exactly the big leaps in adoption its fans desire.

Sherman adds that one major refiner previously required its technicians to go out and read rotometers and pressure gauges each shift, and they couldn't use legacy flow controllers and pressure transmitters because these components were too large to fit into the sample conditioning enclosures. This refiner now uses mass flow controllers and NeSSI-based pressure transmitters that report directly to its control room, which means technicians only need to go out if an exception report is generated because the system isn't operating within range.

In fact, NeSSI may be on the verge of making exactly the big leaps in adoption its fans desire because there are reportedly several requests for proposals (RFPs) for several hundred to several thousand NeSSI systems in projects likely to be built soon. For example, ExxonMobil reportedly plans to add 150 GCs and 300 continuous gas analyzers equiped with NeSSI and i^2C to a nine-plant chemical complex that it's building in Singapore. Likewise, a major, unidentified, Gulf Coast-based hydrocarbon producer reportedly also will soon issue an RFP for 3,500 to 5,000 modular sampling systems.

"In the first quarter of 2006 (1Q06), ExxonMobil specified that NeSSI is the preferred way to do new construction in plants with sampling systems, which take uniform sample from a process stream, present them to an analyzer, and return them to the process," add Veltkamp.

Still, even though hundreds of NeSSI devices have been operating successfully in the field for several years, many users won't give up their doubts. "Even when users are putting in a NeSSI system, they still say, 'We know it's going to fail, so we'll wait and see what the failure mode is,' " adds Sherman. "And, two years later when the NeSSI system hasn't failed, they say, 'We're still waiting to how it fails.' "

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THE ABCS OF XML, PART 2

More of what you need to know to survive in the world of connected data.

by John T. Sever, Cascade Controls

The World Wide Web Consortium's (www.W3C.org) XML Recommendation opens with a list of 10 design goals. The first goal states: "XML shall be straightforwardly usable over the Internet." Straightforward or not, eXtensible markup language (XML) *is* used extensively over the Internet, and has become a *de facto* standard for data interchange.

Because of its association with Internet applications, however, automation engineers have sidestepped XML, assuming it has little applicability to their daily work. This is a mistake. Though this technology has been used extensively for Internet-based applications, XML is an extremely simple and flexible data format with untold uses waiting to be uncovered and implemented in industrial controls and automation.

Alone, XML data is simply raw text that has little to offer automation engineers. But XML isn't alone. Developers everywhere have jumped aboard the XML bandwagon to create a seemingly bottomless reservoir of tools, applications, services, and standards all designed to create, consume, translate, store, and present XML data. This infrastructure of supporting applications is what makes XML such a compelling choice for application data. This article will introduce XML's fundamental concepts for those who have so far managed to avoid this important technology. Parts 3 and 4 of this four-part article will address XML supporting technologies. ["The ABCs of XML, Part 1" ran in *Control*, June '06, p. 55.]

Not a Typical Language

XML isn't a language in the sense that there are defined keywords, functions, or statements. XML is often compared to hypertext markup language (HTML) because it works well with HTML applications, has similar markup, and has been joined with HTML to create the XHTML specification. However, the HTML specification defines list element tags like <body>, <h1>, , and with defined behavior for HTML browsers.

XML lacks a defined set of tags and allows anyone to create their own set of tags and attributes to suit their own application needs. Instead, the XML specification defines a set of markup rules that must be followed for the marked up text to be interpreted as XML data.

10 Well-Formed Rules

XML is organized in a logical or physical structure called a "document." An XML document may be a file on disk, it may be streamed from a server, or it may be hard-coded text inside an HMI VBA application. Though the data may have many different sources, the document metaphor still applies as long as it's "well formed." To be considered well formed, an XML document must adhere to the constraints defined in the W3C's XML specification. These constraints can be distilled into 10 easy rules.

1) **XML is just plain old text.** XML is designed to be human-readable as text. This means any text editor can be used with an XML document. A simple text editor will treat an XML document just as it would an INI file, a CSV file, or any text file.
2) **XML is data.** XML is designed as a flexible self-describing data structure. By itself, XML can't do anything, nor does it define how data should be processed or handled. By contrast, HTML includes both data and a description of how it should be displayed in a browser.
3) **XML documents must have one root element.** There can be only one top-level root element in an XML document, and all other elements must be between the root element start and end tags.
4) **XML white space data is preserved.** HTML reduces consecutive white space characters to a single space character. With XML, white space is interpreted as data—just like any other character.
5) **XML naming rules.** Element names can't include white space, must start with a letter, and can't include characters that are used for markup such as <, >, ;, &, among others. It's generally a good idea to limit element and attribute names to letters, numbers, and underscore.
6) **XML elements must be closed.** An element can be closed with an end tag, or optionally with the shorthand notation for empty elements. HTML doesn't require that elements be closed and most browsers will attempt to render any HTML element whether or not it's closed properly. XML is not so forgiving. An empty element is one with no value and no child elements, though it may have attributes. An empty element may be closed with the shorthand notation /> at the end of the start tag. For example <Value /> is equivalent to <Value></Value>.
7) **XML elements must be properly nested.** In HTML, elements were allowed to overlap like this: bold and italic text italic only. This type of element

crossing is not allowed in XML. An element that starts inside a parent element must end inside the same parent before the parent element is closed.

8) **XML is case sensitive.** An XML tag <recipe> isn't the same as <Recipe>. However, HTML isn't case sensitive so <h1> is identical to <H1>.

9) **XML attribute names must be unique within an element.** An element may have any number of attributes but each attribute name must be unique. The following example is incorrect:

<People Person="John Doe" Person="John Smith" />

This example could be structured properly as follows:

<People>
<Person>John Doe</Person>
<Person>John Smith</Person>
</People>

Or as:

<People>
<Person Name="John Doe"/>
<Person Name="John Smith"/>
</People>

10) **XML attribute values must be quoted.** XML attribute values must be enclosed either in single quotes or double quotes. If an attribute value contains a double quote character, then enclose the value in single quotes. Likewise, if the attribute value contains a single quote, enclose the value in double quotes.

Comments

In addition, the markup for a comment in XML is identical to HTML. The comment opens with <!-- and closes with --> and may span multiple lines.

XML Declaration

An XML document may begin with an optional XML declaration. The XML declaration must precede all other content, and isn't considered part of the XML document. It's used to provide information to XML processors about the document's content. Because the declaration isn't an element, it must not have a closing tag. The declaration looks like this:

<?xml version="1.0" encoding="utf-8" standalone="yes"?>

If the declaration is included, version is the only required attribute, and must have a value of either 1.0 or 1.1. Version 1.1 supports special Unicode character handling functionality that's rarely needed, and, therefore, version 1.0 is used almost exclusively.

The encoding attribute defines the character encoding used by the document, so that an XML processor may properly parse the document. The default encoding used by XML processors is UTF-8.

A SAMPLE OF XML

The sample fragment of XML software code (below) was lifted from a recipe exported from Rockwell Automation's RSBatch product. Rockwell defined the element and attribute names for describing an RSBatch recipe. Other system vendors may define a separate set of elements and attributes to describe a batch recipe. If you're a system integrator that works with batch recipe software from different system vendors, you may choose to define your own system-agnostic batch recipe XML data structure (or schema) for internal development purposes, which can easily be converted to/from a vendor specific structure.

```
<!-- This is an XML comment -->
<Step XPos="600" YPos="600" AcquireUnit="true">
 <Name>FEED:1</Name>
 <StepRecipeID>FEED</StepRecipeID>
 <UnitAlias>UNIT500</UnitAlias>
 <FormulaValue>
  <Name>AGIT_SPEED_SP</Name>
  <Display>true</Display>
  <Value />
  <Real>75</Real>
  <EngineeringUnits>RPM</EngineeringUnits>
 </FormulaValue>
</Step>
```

The first thing to notice about the code fragment is the angle brackets (< and >) which mark an XML "tag." An XML "element" includes a start tag like <Step>, an end tag like </Step>, and everything between the two. Notice that the Step element contains four child elements: Name, StepRecipeID, UnitAlias, and FormulaValue. The FormulaValue element contains five child elements. This parent/child relationship between elements shows that XML can support hierarchical data structures.

XML is designed to be self-describing. A document's data is stored in element values and attribute values, so that element and attribute names describe the data they hold much like data is described in a relational database by table names and field or column names.

Data stored as an element value is the text between start and end tags. In this sample, the value for the element EngineeringUnits is RPM. Data stored as an attribute value is found on the quoted right side of a Name="Value" pair. In our example, the Step element contains three attributes named XPos, YPos, and AcquireUnit, which have attribute values of 600, 600, and true, respectively.

Processing Instructions

Any number of processing instructions may appear below the XML declaration and before the root element. It must be enclosed in <? and ?> like the XML declaration, and provides application-specific handling information. A Microsoft Word 2003 XML document may include the following processing instruction, which tells the Windows operating system to identify the XML document as an MS Word file. When double-clicked, an XML file with this processing instruction will open in MS Word, as shown: <?mso-application progid="Word.Document"?>

XML Validation

An XML document has a structure of element names, attribute names, and hierarchical parent-child relationships. As long as a document is "well-formed," it can have any structure and contain any data. This flexibility is what makes XML extensible.

However, applications that interpret XML documents have expectations that the XML will adhere to a particular structure. Validation is the process of checking an XML document for conformance to a schema. A schema can be defined in the XML document, or a reference can point to an external schema document. There are multiple standards for defining a schema including Document Type Definition (DTD), and others. An XML document that that adheres to a defined schema definition is judged to be "valid." A control system vendor that supports import of XML data into the control system will likely validate a document before the import process to prevent loading data that may lead to a control system fault.

Namespaces

XML namespaces solve a problem that can occur when an element name may have different meanings within one document. For example, the element name template is an XSLT keyword, and its meaning is different than the template element used in an MS Word XML document. All elements and attributes in an XML document are included in a namespace, even if a namespace isn't explicitly declared. When no namespace is defined in a document, content is included in the default null namespace. A namespace may be defined as an attribute of the start tag of an element with the following format:

xmlns:prefix="namespaceURI"

Where a namespace is declared for an element, all child elements with the same prefix are included in the same namespace. The element where the namespace is declared may also be included in the namespace if the same prefix is used in the element name.

```
<cc:Recipe xmlns:cc="http://www.cascon.com/Recipe">
  <cc:Step cc:XPos="600" cc:YPos="600" AcquireUnit="yes">
  <cc:Name>Feed:1</cc:Name>
  <UnitAlias>Unit500</UnitAlias>
  </cc:Step>
</cc:Recipe>
```

In the sample above, the prefix *cc* refers to the namespace http://www.cascon.com/Recipe. Elements included are *recipe*, *step*, and *name*. The element UnitAlias and the attribute AcquireUnit are included in the default null namespace.

The namespace prefix *cc* serves as a shorthand for the full namespace, which may be any string value but is meant to be globally unique. XML parsers don't enforce uniqueness, nor do they expect any particular notation such as a web universal resource identifier (URI). A web style URI is frequently used because a real web URI like www.cascon.com is guaranteed to be globally unique and minimizes the chance of colliding namespaces.

To simplify this example, the namespace can be declared as the default namespace with no prefix:

```
<Recipe xmlns="http://www.cascon.com/Recipe">
  <Step XPos="600" YPos="600" AcquireUnit="yes">
  <Name>Feed:1</Name>
  <UnitAlias>Unit500</UnitAlias>
  </Step>
</Recipe>
```

The namespace attribute *xmlns* no longer includes the prefix definition *cc*. Now, it becomes the *default namespace* for the element where it's declared. This makes the Recipe element and all its child elements members of the namespace http://www.cascon.com/Recipe. Default namespaces apply to elements only, not to attributes. Therefore, the attributes of the Step element are included in a null namespace (equivalent to xmlns=""), and not the default namespace. This special behavior for attributes can be quite confusing. You'll likely not need to bother with namespaces in documents created for internal purposes. However, you'll need to understand namespaces when working with vendor-generated XML files. You will see the importance of namespaces in Part 3, which will cover XSLT.

John T. Sever is president of Cascade Controls. Reach him at johnsever@cascon.com

RESOURCES ON XML

There are many reference books and online resources for learning more about XML. Recommended online resources include:

- www.w3c.org/XML
- www.w3schools.com/xml
- www.xml.com
- www.xml.org
- www.topxml.com

Stop Wasting Money on Backup Power

Many of you have spent and will continue to spend significant sums on backup power systems, power filtering devices, and other power-protection components. However, most of these expenditures aren't necessary because short-duration voltage sags are the real source of most power problems in North American industrial plants. Fortunately, equipment for shielding your plant from these voltage sags is inexpensive, easy to install, and extremely reliable, especially when compared to the elaborate, expensive, and hard-to-maintain systems required for protection against power outages.

It's more cost-effective to protect against frequent, short-term voltage sags, than trying to handle much rarer power outages.

Power outages and airplane crashes make the headlines, which gives the impression that they're common, when, in fact, they're actually very rare.

Hard data reveals some interesting, contradictory facts. In 2005, SoftSwitching Technologies (www.softswitch.com) conducted a study of 30 major automotive plants located in 10 states and one Canadian province. They installed a power quality monitor for 12 months at each site, and captured 472 power quality events.

Of those 472 events, 461 events (98%) were less than 2 seconds in duration. The other 11 events were all over 15 minutes, and were classified as power outages. Seven of these 11 events occurred within one week of July 4, and may have been related to planned shutdowns.

There was an average of 16 events per site per year, with an average of less than one event per year longer than 15 minutes per site (an outage). These data clearly show that most power-quality events are short-term interruptions, not relatively longer term outages. The geographic distribution of the 30 automotive plants covered the U.S. and Canada, and there's no reason to believe that a more widespread study would generate different conclusions.

What exactly are these short-term disturbances? In almost all cases, they're voltage sags. A detailed study of one U.S. automotive plant by SoftSwitching showed that 94% of voltage sags were less than 2 seconds. Based on these and other data collected over the last few years, SoftSwitching reports that 98% of power quality events in the U.S. and Canada are, in fact, these short-term voltage sags of 2 seconds or less.

What causes voltages sags? In technical terms, a short-circuit fault on the three-phase utility power grid. When a relatively conductive material comes into contact with one of the individual three-phase lines and ground, or with two of the three-phase lines, then a short-circuit condition occurs.

When this happens, an instantaneous current demand is placed on the power source, which is the supplying power plant. Because the power plant can't deliver current to the fault quickly enough, the voltage on the lines starts to collapse. The effect is voltage sag on the lines that can be seen for up to 400 miles.

Typical causes of voltage sags include lightning strikes or failed utility equipment. Any disturbance that connects a phase to ground or a phase to phase also can cause voltage sag. The Electrical Power Research Institute (www.epri.com) reports that a typical industrial manufacturing site experiences an average of 12–14 voltage sags each year.

So, what do short-term voltage sag do to your plant's operating equipment? SoftSwitching's study of 11 automotive plants showed that about eight of the average 15 short-term voltage sags per site per year would have had no effect on typical production processes. The other seven short-term voltage sags per site per year "usually cause typical automation equipment such as relays, solenoids, contactors, PLCs, and HMIs to error out or trip off during these types of events," says Bill Brumsickle, SoftSwitching's engineering director.

The conclusion is clear. The most cost-effective strategy for dealing with power quality events is to protect against frequent, short-term voltage sags as opposed to much rarer power outages.

Protecting against outages by installing and maintaining on-site backup power generation is extremely costly, especially since most outages are longer than 15 minutes, which is a long time to provide backup power.

Luckily, it's a lot less expensive and easier to protect against more common short-term voltage sags with equipment from SoftSwitching and others. For example, SoftSwitching sells a 480 VAC, three-phase, 100 amp unit that will protect against voltage sags down to zero volts for up to 2 seconds.

Dan Hebert, PE, senior technical editor
dhebert@putman.net

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Converting Pneumatics; Summing D/P Signals

"Ask the Experts" at ControlGlobal.com is moderated by Béla Lipták (liptakbelaieh4@aol.com), editor of the *Instrument Engineer's Handbook* (http://a9.com/bela%20liptak?c=1&src=amz). He is former chief instrument engineer at C&R (later John Brown), and received ISA's 2005 Life Achievement and *Control*'s 2001 Hall of Fame awards. In this column, he and his co-authors welcome the questions of their colleagues working in the fields of process control and automation.

Question:

I'm working at the Esfahan refinery, and we're upgrading our aged pneumatic control system to a hybrid type (DCS and fieldbus). Which type of control loops are recommended to be converted into fieldbus type subject to safety and reliable operation of the control loop?

Name withheld

Answer:

Golden Advanced Control Opportunity

I hope that others will offer recommendations about whether to implement the counterpart to existing pneumatic control loops in fieldbus or the DCS—that is, where to locate the PIDs. My response is along different lines.

In the existing pneumatic control system, there is probably very little use made of advanced regulatory control, other than a few cascade loops and perhaps a few ratio loops. With DCS and fieldbus, there's a chance to do much more at the advanced, regulatory control level because there are many more cascade, ratio, feedforward, override, and decoupling loops. Hence, it's a mistake to think only in terms of "converting the existing loops," when there is such a golden opportunity to rethink the entire control strategy.

You may also have, or intend to have, an advanced control package (model-predictive control). Then it becomes a question of how much of the advanced control strategy should be implemented in DCS or fieldbus, or left to the advanced process control (APC) package? There are differences of opinion here. Some take a "push up" philosophy, and have as much of the advanced control done in the APC package as possible. Others take a "push down" package, and implement as much of the advanced control in function blocks in the DCS (or fieldbus) as possible. I take a somewhat middle ground, and say that even after the APC is implemented, one can look for additional opportunities for advanced regulatory control for DCS or fieldbus implementation.

Harold Wade

Fieldbus Just as Safe as Pneumatic

There are several examples of customers moving from pneumatic to fieldbus. For example, a platform from Petrobras has been operating for several years now (http://ethernet.industrial-networking.com/ieb/articledisplay.asp?id=83).

Just about any control loop can be moved from pneumatic to fieldbus. Fieldbus may not be fast enough for surge control, but neither was pneumatic. Fieldbus is just as safe and reliable as pneumatic. Prolonged communication loss can trigger a fail-safe state, much like a loss of supply air does for pneumatic. Fieldbus power is more reliable than compressors, filters, and dryers associated with pneumatic, so spurious trips are likely to reduce. Fieldbus can also be intrinsically safe.

SMAR even makes a bus-to-pneumatic converter, which may be used with some valves and dampers that remain pneumatic (https://www.smar.com/products/fp302.asp).

SMAR and other suppliers have pneumatic valve positioners based on fieldbus. Any low-cost pressure transmitter can be used as pneumatic-to-bus converter.

For fieldbus migration, read Chapter 5 of *Fieldbuses for Process Control: Engineering, Operation, and Maintenance*. It's available online (http://www.isa.org/fieldbuses).

Jonas Berge

Execute in Devices or Processors

Thanks you for sharing your comments and questions! Maybe five or 10 years ago, you could consider implementing a hybrid control system. Today, I think the ultimate control systems for this application would be Delta V from Emerson Process Management, or Plantscape from Honeywell, or Freelance from ABB.

The single loops with one PID or cascade loops are candidates for implementation in the Foundation fieldbus devices. I think it would be better to have more complicated control strategies execute in the control processors.

Perhaps, if you have some plants that present great risk to personal or equipment, you should make an SIL analysis to have safer and more reliable operations.

David Reyes Peña
ICA Fluor, Sistemas de Control

FF Fits Pneumatic to Fieldbus Conversion

Because Foundation fieldbus supports "control in the field" and because doing so requires input and output devices to be on the same segment, it's a natural fit for this type of conversion. The design can be based on directly replacing the tubing for the air supply with fieldbus cables, and then combining multiples of the resulting "tubing run replacements" into fieldbus networks.

Depending on the timing of the conversion, it will be possible by the end of 2007 to migrate everything, including the safety loops, to fieldbus technology.

Ian Verhappen

Question:

Summing D/P Sensor Signals and Converting to Mass Flow

I've been given the task of trying to put together a package for measuring flow rates on a plant stabilizer. I have two 3-in. runs and one 2-in. run already in place with orifice plates installed. These rates were measured at one time with Barton meters. Flow rates will be done on Y-grade product. These runs are pretty close to each other. Is there some product that I could use to tie the data from all three runs into one unit? What information do I need from the product itself? Pressure on the lines is 360 psig. Would I need to install a densitometer? I think this project started out as something small, but, as I do more research, it doesn't seem that way. Your input would be appreciated.

Alex

Alex_cstnd@yahoo.com

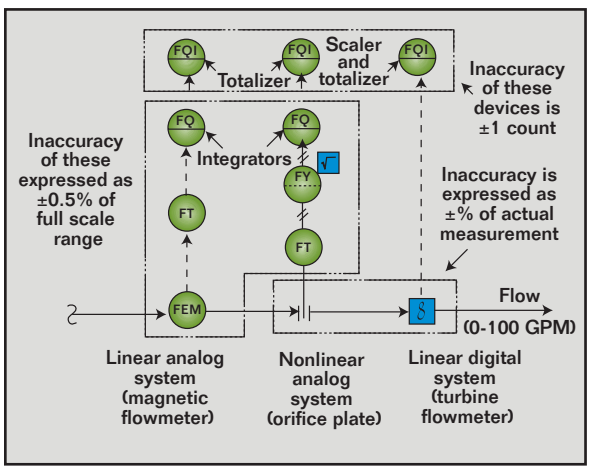
Answer:

Use D/P Transmitter Cells, Densitometer

Knowing only as much about your installation as you wrote, I'd be inclined to install three intelligent, close-coupled d/p transmitter cells. And, if the properties of the three flowing streams are identical, I'd install a single densitometer to be used to calculate the total mass flow.

FIGURE 1.

FLOW LOOP COMPONENT ACCURACY



Using intelligent d/p cells provides higher rangeability and better accuracy.

I'd also convert the volumetric flow signals to mass flow by multiplying them with density and sum the three flows digitally. This can be done by a digital summer/multiplier, which can be mounted locally or remotely.

I'd further suggest using intelligent d/p cells because they can automatically switch their spans between a high (10 to 100%) and a low (1 to 10%) d/p span. This gives you higher rangeability and better accuracy. Conventional d/p cells will have an error of about 2% full scale (0.5% of rate) over a 3:1 range, while intelligent ones can operate at an error of 0.5% over a range of 10:1. For the accuracy of the flow loop components, see Figure 1.

Naturally, this performance assumes that the plates aren't worn, are clean, and without any buildup. The tap holes should enter at right angles, and should be slightly beveled. If there's the slightest likelihood of lead line plugging, they also should be sealed or purged.

As to density correction, because density affects the d/p measurement in a square root manner, head type metering actually has an advantage in applications where density varies. This is particularly important in gas flow measurement, where considerable density variations are likely, and where the densitometer accuracies are low (1% or less). This square root relationship tends to reduce the density error to about 0.5% of the detected variation.

Béla Lipták

I/O and Terminal Blocks Appear Healthy

Product development, R&D and competitiveness are all back.

This month's roundup on I/O and terminal blocks demonstrates that the industry seems to be recuperating, product development money is flowing again, companies are getting competitive, and no one technology has a lock on the I/O and terminal block market.

We presently see a wide range of I/O and interfaces for everything from Ethernet and wireless to Profibus, ArcNet, and Modbus. On the terminal block and connector side, innovations and enhancements prevail.

Other than a new emphasis on wireless and Ethernet connections, we don't see any hard trends in I/O, connectors or terminal blocks. Instead, now that companies are recovering from their economic doldrums, we're seeing a healthy mix of products of all types coming down the new product chute.

—Control Staff

POE SPLITTER

PSD14 Series 14-W, DC-DC, power-over-Ethernet splitter provides 12.5 W from standard IEEE 802.3af power sourcing equipment (PSE). The splitter is a power supply that can be mounted

next to an Ethernet device, and is compliant with IEEE 802.3af standards. Typical applications include IP telephones, wireless network access points, Bluetooth access points, security cameras, and IP print servers. More info at controlglobal.com. **Phihong; 510/445-0100; www.phihong.com**

WIRELESS WAN ROUTER

WAN IA wireless WAN cellular router provides secure cellular GSM or CDMA connections for reliable primary and backup network connectivity to industrial automation devices such as SCADA devices, PLCs, and RTUs. It has industrial hardware/software components including a terminal block for power, DIN rail mounting, Class 1, Division 2 certification and Modbus protocol support. More info at controlglobal.com. **Digi International; 877/912-3444; www.digi.com**

SERIAL-TO-ETHERNET DEVICE SERVERS

Models ET-DS-1 and ET-SDS-1 industrial, serial-to-Ethernet device servers offer real-time secure performance including an auto-detecting 10/100 RJ45 Ethernet port and a universal EIA-RS232/422/485 serial port for serial data

tunneling across an IP network. Both have back to back (paired mode) operation, port buffering, and IPv6 support. More info at controlglobal.com. **Sixnet; 518/877-5173; www.sixnetio.com**

ETHERNET-COMPATIBLE M12 CONNECTORS

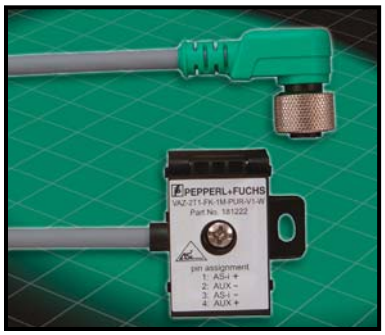
Unnamed M12 Ethernet plug connectors are available as overmolded cordsets, field-wireable, and panel-feed-through connectors. This Ethernet connection system is IP67 rated for harsh environments and meets CAT5 requirements of 100 Mbit/s. **Phoenix Contact; 800/322-3225; www.phoenixcon.com**

WIRELESS I/O

NETio I/O and wireless communications products carries data from serial and IP devices along with analog and discrete signals. It supports direct protocol addressability of connected I/O from a library of industrial protocols, including Modbus Rtu, Modbus TCP, DNP3 or EtherNet/IP. Wireless expansion modules can be as far as 1,000 feet away. **Microwave Data Systems; 585/241-5539; www.microwavedata.com**

AS-INTERFACE CONVERTER

VAZ-2T1-FK AS-interface, flat-to-round-cable converter can be used in narrow and low-profile cable channels and other space-critical locations. In most applications, installation takes less than a minute. The converter is suitable for use where AS-interface and 24 VDC external power need to be on one M12



connector, while drives are with motor starters. **Pepperl+Fuchs; 330/486-0001; www.am.pepperl-fuchs.com**

PC/104 MODULE HAS EIGHT SERIAL PORTS

Emerald-MM-8P PC/104 serial port module has eight multi-protocol serial ports on one PC/104 module with software configurability. The board works in PC/104-expandable embedded computer systems running Linux, Windows, QnX, VxWorks, and DOS. Each of the serial ports can be individually selected for RS-232, RS-422, or RS-485 (both local echo and no echo) under software control. More info at controlglobal.com. **Diamond Systems; 650/810-2508; www.diamondsystems.com**

ARCNET INTERFACE

USB22 series of network interface modules provide connectivity between desktop or laptop computers and peripherals. It includes a high-speed microcontroller that transfers data between Arcnet and USB. It has up to 128 Kbytes of Arcnet receiver buffering to handle reception of broadcast messages or monitoring networks with high traffic levels. More info at controlglobal.com. **Contemporary Controls; 630/963-7070, x124; www.ccontrols.com**

PROFIBUS INTERFACE GOES THROUGH DOOR

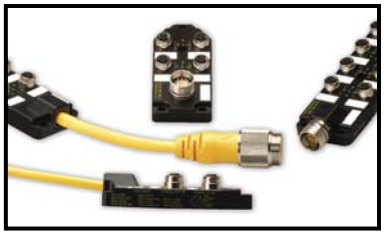
GracePort D3-Series interface gives users a convenient way to temporarily connect to a Profibus network without opening the control panel door. It supports 12-MB data transfer speed, and includes integral end-of-network termination resistors and switch. This lets users install a Profibus network tap anywhere along the Profibus cable run. More info at controlglobal.com. **Grace Engineered Products; 800/280-9517; www.grace-eng.com**

NETWORK CONCENTRATOR GOES MODBUS

Enhancements to the Net Concentrator System process control and distributed I/O network include Modbus RTU (RS-485) communications. With Modbus, the concentrator is able to use twisted-pair wiring, Ethernet, wireless, fiber-optic, and telephone/modem data links to send monitoring and control signals over rugged, long-distance, normally impassable, and hazardous environments. More info at controlglobal.com. **Moore Industries-Int'l.; 818/894-7111; www.miinet.com**

LOW-PROFILE JUNCTION BOXES

Unnamed, low-profile, M12 eurofast junction boxes provide quick-disconnect wiring. Available in 4 or 8-port with single or dual output, they offer multiple home-run connector and integral cable options. Home-run connectors can be mounted on the top or side of the housing. Integral cables with a 12-pin, 1 1/8-16 UN minifast connectors also are available for single-output junction boxes. **Turck; 800/544-7769; www.turck.com**



DIGITAL I/O MODULE

Adam-4150 digital I/O module has seven input channels and eight output channels with over-current/short-circuit protection. It has 1 kV surge protection on power inputs, 3 kV EFT, 8 kV ESD protection, power input range of approximately 10 to 48 VDC, an LED display to monitor the status of the address settings, a digital filter function, digital output support pulse function, and Modbus RTU protocol. **Advantech; 513/742-8895, x333; www.advantech.com**

Stay in Control of Diagnostics and Tuning

Intune Version 5 is a non-intrusive plant management and process control predictive maintenance tool that combines process control monitoring with diagnostic and tuning features to help companies 1) increase plant productivity, 2) reduce energy consumption, 3) improve consistency of product quality with less waste, 4) increase plant engineering and maintenance productivity, and 5) support continuous improvements, such as with Six Sigma.

Intune Version 5 still retains its core structure of tuning tools, including its IMC auto tuning, heuristic-based adaptive tuning, offline data tuning, and offline model-based tuning. Also included is a PID comparison tool, along with a variety of process and PID simulation features.

Ali Haji-Valizadeh, ControlSoft's technology director, says, "Intune v5 helps companies use a variety of process analysis and problem correction tools, including comprehensive statistical analysis, customized scripting for plant-specific indices, automatic, adaptive, model-based, and offline tuning, and web-based reporting and event notification."

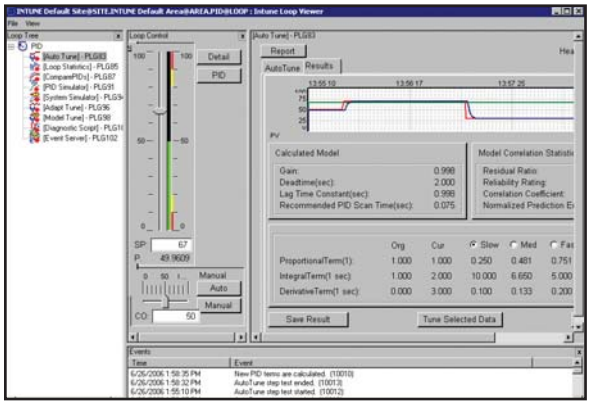
Statistics Standard

New to Intune v5 is its inclusion of more than 50 standard statistics with real-time, moving-window analyses and condition-based assessment. Haji-Valizadeh adds, "Much more than just some number cruncher that pulls data from a historical file, Intune v5 statistics feature allows you to start and stop calculations based on shifts or batches, reset error calculations based on operator mode changes, or trigger entire sets of calculations based on alarm states—all in real time."

The conditioning triggers can be based on any Intune or I/O tag addresses that can be read via Intune's OPC client. This real-time analysis uses moving windows customized to user-specified time periods, ensuring that calculations use the most up-to-date information. Since Intune v5 also is its own OPC server, it connects other OPC clients to the software's assorted assessment indices.

FIGURE 1.

BIG PICTURE



50 standard statistics let users start/stop calculations, reset error calculations, or trigger entire sets of calculations.

Diagnostic Scripting

Intune v5 also provides a diagnostic scripting tool for creating user-defined metrics and key performance indicators (KPIs). This flexible scripting tool allows users to turn raw data into useful metrics for optimizing process control.

With the scripting feature, users can differentiate between hardware and tuning problems, or automatically identify poorly performing loops. They can then calculate their economic impact in real dollar values, and track raw material waste, excess fuel and energy costs, or benchmark various measures of product quality and consistency based on operating conditions.

Event Server

To further aid in process control predictive maintenance, Version 5 gives users customizable Event Server capabilities. These can notify operators, engineers, maintenance personnel, shift managers, and/or plant managers via e-mail the instant an alarm is reached or when another user-defined event occurs. They also automatically record events in a user system's event log, while notifying appropriate employees. Each event is user-specified, and the initiating address can be read from any Intune or I/O tag.

Haji-Valizadeh adds, "The Event Server realizes its greatest potential when used as part of a predictive maintenance scheme that includes notifying appropriate personnel before alarm levels are reached, when loop quality becomes less than acceptable, or when the costs of energy or raw material waste exceed certain pre-defined limits."

Reporting

Tying all the performance improvement functionality together is Intune v5 reporting, which features customizable, web-based reports. The reports also can be customized to visually indicate tuning quality or plant performance, saving busy managers and operators from the hassle of decoding obscure numbers and sorting through statistics. Intune v5 also provides a web report server, which allows managers and authorized personnel access to all the reports generated for a plant or corporation, with the ability to sort reports by date and time, category, or keyword. **ControlSoft Inc., www.controlsoftinc.com, 440-443-3900**

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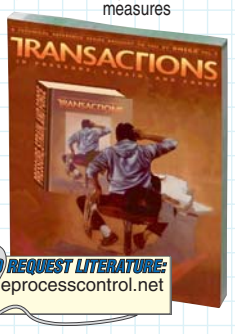

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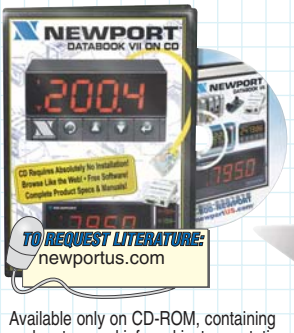

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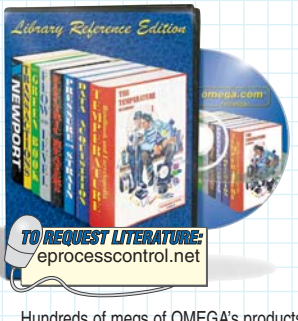

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HOT, PRESSED LEVEL TRANSMITTER

Enhancements to the 3300 Series guided-wave radar level and interface transmitter include probes to prevent leakage and enable reliable perform ance when exposed to high temperatures up to 750° F and pressures up to 5,000 psig. Applications include refinery distillation columns, liquefied gases and freons processing, power plant utility heater vessels, gas-oil production separators, oil-water separators, and compressor tanks. More info at controlglobal.com. **Rosemount; 952/949-5165; www.EmersonProcess.com**

DCS DOES DEVICE DIAGNOSTICS, DUDE

In DeltaV, Version 8.4, system software, standard HART device status information now can be used to generate PlantWeb alerts for any HART device, permitting predictive diagnostics. In addition, conditional alarming has been extended to Foundation fieldbus devices to make intelligent field devices more discriminating to reduce nuisance alarms. More info at controlglobal.com. **Emerson Process Management; 512/418-7536; www.EmersonProcess.com**

SOFTWARE DIAGNOSES LOOPS

Enhancements to PlantTriage performance supervision software include a simple text message diagnosis for every loop in the plant. The diagnosis is immediately available via a web browser. Each diagnosis is an actionable item, providing resolution of problems. The message diagnostic tells whether it's a sensor, controller, or valve-related issue. The diagnostic identifies specific control loop problems, or indicates that control loop performance is OK. More info at controlglobal.com. **ExperTune; 262/369-7711; www.expertune.com**

HANDHELD COMPUTERS REACH OUT

The JETT Wave Radio module allows the company's handheld computers to communicate wirelessly with equipment and other handheld computers at ranges up to several miles in line-of-sight applications, and up to

3,000 feet in typical indoor/urban environments. Incorporating a 900-MHz, spread-spectrum radio, this module supports various communication configurations, including point-to-point, point-to-multipoint, and multi-drop networking topologies. More info at controlglobal.com. **Two Technologies; 215/441.5305; www.2T.com**

DEVICENET TOOL CUTS PROGRAMMING

RSLogix DeviceNet tag generator automates configuration and setup steps typically associated with adding DeviceNet products to a Logix-based control system. This free configuration tool automatically defines data tags, which helps reduce the amount of programming time required. It also automatically provides status and diagnostic information for DeviceNet products. More info at controlglobal.com. **Rockwell Automation; 800/223-5354, x1916; www.rockwellautomation.com**

BLUETOOTH WIRELESS RADIOS

PSI-WL series Bluetooth radios move RS-232, RS-422 and RS-485 serial data up to 300 feet. The radios are Class I, Div 2 approved, and connect to any Bluetooth-enabled device such as

PDAs, PCs, or other devices. Modbus and additional industrial protocols also are supported. The radios can be used as a simple wire-and-cable replacement in point-to-point applications, or in point-to-multipoint configurations. One master will support up to seven slave devices. **Phoenix Contact; 800/322-3225; www.phoenixcon.com**

POWER QUALITY ANALYZERS

Models 434 and 435 three-phase, power-quality analyzers pinpoint power problems fast and in detail with on-screen display of trends and captured events. Both incorporate new IEC standards for flicker, harmonics,

power quality and troubleshooting of both three-phase and single-phase power distribution systems. Both have a safety rating of 600 V CAT IV and 1000 V CAT III. More info at controlglobal.com. **Fluke; 888/308-5277; www.fluke.com**

VIBRATING LIQUID LEVEL SWITCH

RS80 Resonator vibrating liquid level switch is unaffected by coating or build-up on the probe, allowing the

level switch to make measurements without concern for specific gravity, dielectric constant, or mounting position of the sensor. A built-in, self-test option automatically checks for fault conditions, such as excessive product build up on the sensor, and warns of potential problems. More info at controlglobal.com. **K-Tek; 225/673-6100; www.ktekcorp.com**

COMPACT CORIOLIS FLOWMETER

At approximately half the size of previous devices, Sitrans FC300 DN4 flowmeter measures 5.3x8.1x2.3 in., weighs 7.7 lbs, and has a maximum capacity of 772 lbs/hr with accuracy of better than 0.1% of rate. Sensors are available in W1.4435 (AISI 316 L) stainless-steel with a maximum operating pressure of 1,900 psi, and a Hastelloy W2.4602 (C22) model with maximum operating pressure of 5,950 psi. More info at controlglobal.com. **Siemens Energy & Automation; 215/646-7400 x2592; www.sea.siemens.com**

SOFTWARE LINKS PLANT FLOOR, ERP

Proficy Enterprise Connector 1.5 provides bidirectional enterprise integration between Proficy and ERP systems, such as SAP in real-time. It uses the ISA-95 standard to define the interface between an enterprise's business and manufacturing systems through the XML-based Business to Manufacturing Markup Language (B2MML) schema. More info at controlglobal.com. **GE Fanuc Automation; 800/433-2682; www.gefanuc.com**

PXI EXPRESS CHASSIS, CONTROLLERS

PXIe-1062Q eight-slot chassis offers up to 1 Gbps per slot of dedicated bandwidth and a mix of PXI and PXI Express slots. The chassis includes a PXI Express system slot, a PXI Express slot with system timing capabilities and four PXI peripheral slots, and two PXI Express hybrid slots that accept both

PXI and PXI Express modules. More info at controlglobal.com. **National Instruments; 800/258-7022; www.ni.com**

CONTROLS WORK WITH ARCHESTRA

Matrix v.8.0 process control system is integrated with Invensys Wonderware ArchestrA technology to manage and coordinate the development, modification, and deployment of control application components. Each of these components or templates contain all the information necessary for alarming, trending, historical documentation, and security for PID loops, motor control, I/O interfacing, and diagnostics. More info at controlglobal.com. **MTL Open System Technologies; 800/835-7075; www.mtlmost.com**

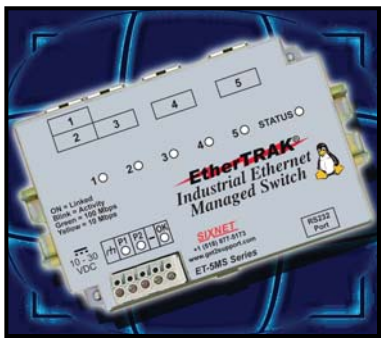
LARGE CORIOLIS FLOWMETER

Model CMU Coriolis mass flowmeter combines large line size service and high flow rate to deliver more than twice the flow rate previously available.

Suitable for line sizes from 0.25 to 12 in., it measures flow rates from 2.2 to 80,800 lb/hr. It provides true mass flow measurement, while at the same time measuring fluid temperature, density and volumetric flow. More info at controlglobal.com. **Fluid Components Int'l.; 800/854-1993; www.fluidcomponents.com**

FIVE-PORT MANAGED SWITCH

ET-5MS five-port managed Ethernet switch has industrial ratings, including a wide operating temperature range, Zone 2 hazardous locations, and rapid-spanning tree protocol (RSTP) for fast redundant rings. The switch can



detect and report network problems before they seriously affect network operations. The unit also has a self-test/alarm output signal plus authentication and encryption. More info at controlglobal.com. **Sixnet; 518/877-5173; www.sixnetswitch.com**

WEB-BASED PERFORMANCE MANAGEMENT SOFTWARE

SuiteVoyager 2.6 web-based performance management software gives customers access to real-time production and performance management information. It's integrated with ArchestrA technology, and takes advantage of Microsoft SharePoint Services technology. Plant information can be included in existing SharePoint websites, facilitating access to both business and plant information in one integrated view. More info at controlglobal.com. **Wonderware; 949/459-9696, x244; www.wonderware.com**

ANTI-FOULING LEVEL TRANSDUCERS

Series 705 non-fouling, level transducer has a wide Teflon-coated sensing area and a flush elastomeric diaphragm with a 0.90 in. surface area to prevent clogging. The transducer is available in custom level ranges from 6 ft. H_2O to 115 ft. H_2O with analog outputs of 4-20 mA or 0-5 VDC. It's

compensated over temperatures of 23° F to 122° F. More info at controlglobal.com. **Pressure Systems; 800/678-7226; www.pressuresystems.com**

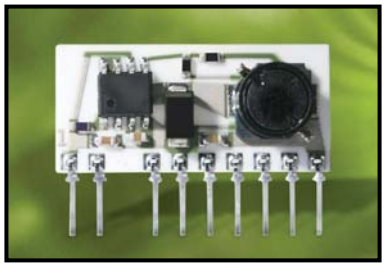
ETHERNET/IP CONNECTORS

Unnamed line of four-wire, D-coded, Ethernet connection products includes cables, switches, and conduit adapters, all compliant with EtherNet/IP standard. Cordsets are available in a variety of configurations, including pre-molded, M12, eurofast connectors, receptacles and RJ45 plugs. PVC cable

jackets meet TIA/EIA568-B.2 Category 5e performance requirements for 10 and 100 Base-T Ethernet. More info at controlglobal.com. **Turck; 800/544-7769; www.turck.com**

DC/DC CONVERTER

831 Series DC/DC converter uses a ceramic base and hybrid technology to eliminate any external heat sink re-



quirements, which reduces the overall installed cost of the device. All the device requires is two external capacitors. It's packaged in a small, space-saving, hybrid SIP configuration (1x0.5x0.225 in.) that minimizes PC board footprint. More info at controlglobal.com. **BI Technologies; 714/447-2518; www.bi-technologies.com**

ACCELEROMETER INPUT MODULE

SCM5B48 isolated and field-configurable accelerometer input module pro-

vides excitation to piezoelectric sensors with built-in microelectronic amplifiers. The module provides a constant current excitation to the sensor, then isolates, filters, and amplifies the sensor output, yielding a high-level analog voltage output. The excitation current, signal gain, and filter high-pass and low-pass cutoff frequencies are field-configurable through a set of slide switches. More info at controlglobal.com. **Dataforth; 800/444-7644; www.dataforth.com**

TERMINAL BLOCK EQUIVALENTS

Unnamed terminal blocks are identical equivalents to the Allen-Bradley 1492J series, Phoenix Contact UKN series, and Weidmuller WDU series. The color of the housings is the same, so there are no visual differences. The metal clamp-

ing bodies are made of a nickel-plated copper alloy, screws are plated to prevent corrosion, and all materials are ROHS compliant. More info at controlglobal.com. **Automation Systems Interconnect; 877/650-5160; www.asi-ez.com**

T-CONNECTOR FOR FIELDBUS

Series 870 is a 7/8-in., threaded, T-connector for fieldbus applications. It's sealed to IP67, and is available with five-poles per port. It also features two female ports and one male port. Series 870 is intended for use in ring connections in fieldbus applications. It can connect single modules without interrupting the bus or power supply. The connector's housing is plastic (TPU) with nickel-plated brass locking rings. **Binder-USA; 805/437.9925; www.binder-usa.com**

Intoxicating Answers

Greg McMillan and Stan Weiner, PE bring their wits and more than 66 years of process control experience to bear on your questions, comments, and problems. Write to them at controltalk@putman.net.

Stan Have you ever been asked a question about process control that leaves you at a loss for words? Here's your chance to amaze friends and relatives (or at least baffle them) by learning some control concepts and analogies. If you understand the ideas and their significance conveyed in the answers to the following devastating questions from students in Greg's class during his brief but illustrious career as a professor, you might just become famous by Friday. These answers never made it into a university text for obvious reasons.

Student: What is loop dead time?

Greg: Dead time is that period of time from the start of a disturbance until the controller makes a correction that arrives at the same point in the loop that the disturbance entered. The controller needs to see the upset, react to it, and send the correction to the right place. To appreciate dead time, consider when you go to a party and start drinking. The period of time between when you first take a drink and when you first recognize the effect and bypass the next round for coffee is dead time. As the dead time approaches zero, the portion of the open-loop error that appears as the closed-loop error approaches zero, and nonlinearities and high process gain (high proof and small body) become unimportant.

Student: What is the open-loop error?

Greg: Well, if you continued drinking, your state of intoxication would exponentially increase, and reach a steady state when you passed out on the floor. The open-loop error is the alcohol concentration at this point. It is the result of having the controller (your mind) in manual, or disconnecting it from the loop (your body). A smart party animal knows when to say "when," according to a famous seer.

Student: What is the closed-loop error?

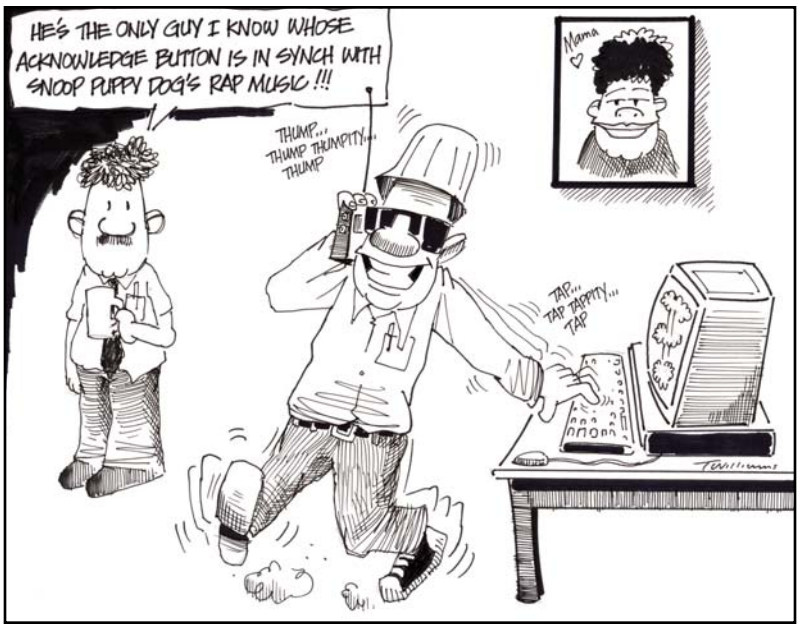
Greg: The peak excursion of the alcohol concentration until the coffee first enters the bloodstream is the closed-loop error. For well-tuned PI and PID controllers, the best you can do is limit the closed-loop error to the maximum deviation from set point that occurs at 150% and 110%, respectively, of the loop dead time. Often, the party mind doesn't behave like a well-tuned controller, and the closed-loop error reaches an unacceptable level for driving, especially if there is a non-self-regulating process.

Student: What is a non-self-regulating process?

Greg: If your liver breaks down and removes alcohol at a fixed rate, the alcohol concentration in your bloodstream can ramp, which is characteristic of an integrating process. If an increase in intoxication causes a decrease in your restraint, you might start drinking faster. This is positive feedback, and the acceleration of the concentration corresponds to a runaway process. People with such non-self-regulating processes are easy to spot at parties; they're the ones with lampshades on their heads.

Student: How can you reduce loop dead time?

Greg: When I first left home, my father



said to me, "Be as honest as the day is long, don't talk when you should listen, and don't be fooled into thinking that dead time compensators can eliminate dead time from the loop." I thought this was extremely strange in that I only planned to walk around the block.

Some professors must not have gotten the same words of wisdom because they're convinced they can mathematically cancel out the dead time term in a loop with an advanced control algorithm. Dead time cannot be eliminated without violating the basic principles of classical physics. While the theory of relativity concludes that you could contract distances and dilate time, and hence shorten transport delays as you approach the speed of light, unless you have Scotty and warp drive on your loop, you're stuck with dead time caused by the vessels, piping, and instruments in your loop.

If the loop dead time is well known and considerably larger than the largest time constant in the loop, a dead time compensator, such as the Smith Predictor, can make the loop perform better if you tune the controller with a smaller reset time. In general, your best bet is to look for the biggest sources of dead time, and work on a change in the equipment, piping, or instrument design to reduce their contribution to the total loop dead time. If the disturbance is much slower than the loop dead time, the closed loop error is small. Have you ever had guests stay too long? Just start showing a video of you at work, and simulate dead time by freeze frames. Some guests have hurt themselves in the rush for the door.

Student: What if you can't change the plant design?

Greg: Then your best bet is to use feedforward control for disturbances that can be measured when the dead time in the disturbance path is smaller than the dead time in the correction path to a common point in the process. Use cascade control for disturbances that can be isolated by an inner loop when the inner loop dead time is several times smaller than the outer loop dead time.

Do Two Top 10s Equal a Top 20?

The first top 10 list is not all that funny, but it is instructive, especially as you're trying to figure out this month's Puzzler. With the help of Mark Sowell, we conclude with a top 10 list that doesn't have much to do with dead time, but is still timely on the road to fame and fortune. Well, maybe just fame. The second top 10 list is wise beyond its years. Pay attention, grasshopper.

Top 10 Ways to Get the Least from Process Control

(10) Eliminate the process control and instrument engineering groups
(9) Ignore process technology when identifying control opportunities
(8) Rely on process technology when identifying control opportunities
(7) Don't simulate
(6) Emphasize data quantity rather than quality
(5) Look at numbers instead of trends
(4) Use so much math you can't see the obvious
(3) Confuse configuration with process control improvement
(2) Teach theory instead of practice
(1) Ignore what the Control Hall of Fame Inductees learned

Top 10 Signs You Have Too Many Alarms

(10) The console is more exciting than a pinball machine.
(9) The only time there are no alarms is when the console is unplugged
(8) Operators are deaf to the high pitch tones
(7) The operator can play along with his favorite rap song on the acknowledge button without missing a beat.
(6) A woodpecker is being trained to keep pecking the acknowledge button
(5) The alarm password is going for $50 on the plant's black market.
(4) Visiting the control room for a doughnut is not worth the noise
(3) The alarm management consultant wants to showcase your alarm log in his training class
(2) Hearing protection is required when a total download activates all your suppressed alarms
(1) "Because they're free" is the basis for your alarm design strategy

Puzzler: How do can you have a career of 30+ years at the same company in the 21st century?

This Month's Puzzler: The case of "Why Would You Want To?"

What do you think you have to do to have a career of 30+ years at the same company? Is this a good thing? What's your ideal career?

Send an e-mail with your answer to the Puzzler, CONTROL questions, or comments to controltalk@putman.net.

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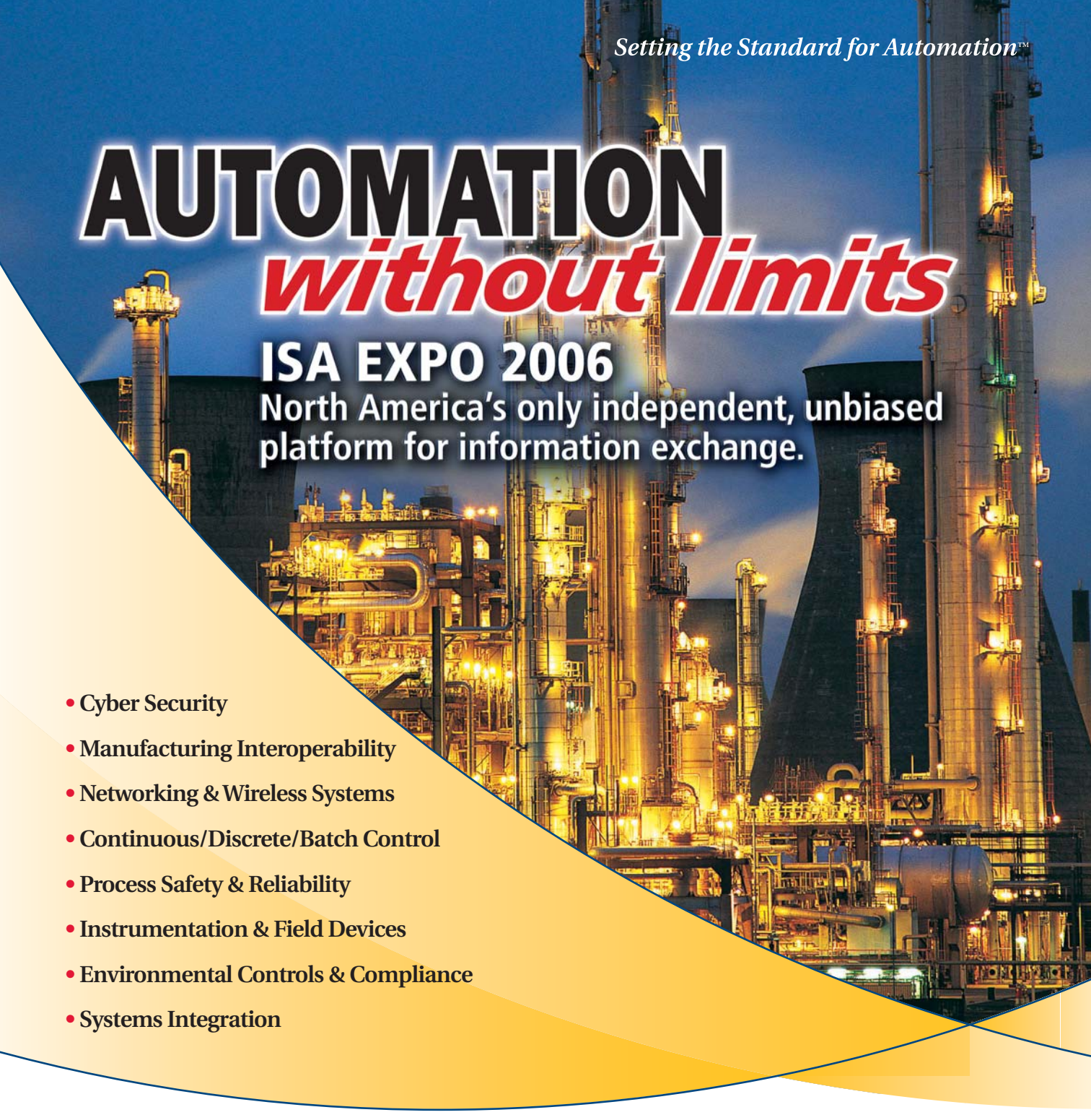

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Who Will Sign Your Next Paycheck?

The demographic pressures on the process automation profession are well-documented: an aging, retiring workforce and a diminished pipeline of new recruits. But along with the coming manpower squeeze will be a necessary shift in industry structure. Technical and organizational factors are reshaping how the work of process automation gets done—perhaps even switching who will sign your next paycheck.

Like many of you, I stumbled into process automation pretty much by accident. Sure, I took the obligatory ChE 385 lecture and lab during my undergrad years in chemical engineering, but at the time I was more interested in industrial processes themselves, the transformation of one substance into another, on an often massive scale. Controls and instrumentation were an afterthought, simply a means to a different end.

This particular brain drain is bifurcated, with process automation expertise shifting to the supplier community and to systems integration houses.

And so it is today for many young engineers—and for the end-user companies for which they would work. Among the already diminished number of engineering graduates in the U.S., ever fewer are likely to stumble onto controls and instrumentation through a career with an operating company. Why? Because operating companies are, first and foremost, in the business of making stuff, and like never before, expertise in controls and instrumentation is viewed as a non-core competency.

Who can blame young engineers if they don't choose to develop expertise in an unappreciated technical specialty? The same argument applies to seasoned automation professionals, an increasing number of whom will find their skills best valued elsewhere.

And just where will that be, you ask? Clearly, it is at those companies for which control and instrumentation *are* core competencies. Indeed, this particular brain drain is bifurcated, with automation expertise ultimately shifting to both the supplier community and to systems integration houses.

Increasingly, process control professionals with technical chops will work for these companies, and those responsible for automation at operating companies will focus primarily on managing relationships with these external vendor-partners—and far less time doing process automation work themselves.

(On a side note, even magazine publishers aren't immune to this dynamic—We've lost both an editor and a columnist to automation suppliers within the past month. Rich Merritt, long-time senior technical editor, is off to Moore Industries International, and columnist and fieldbus specialist Ian Verhappen signed on with MTL. We wish them both well!)

Even the large engineering firms are more often turning to one of the global automation suppliers as main automation contractor for the controls and instrumentation component of major projects. Options in this rarified space are now down to seven: ABB, Emerson, Honeywell, Invensys, Siemens, Yokogawa, and, if you include the hybrid/batch application space, Rockwell. And Honeywell's recent acquisition of UOP demonstrates that these companies don't just have their eyes on automation expertise, but are building process expertise as well.

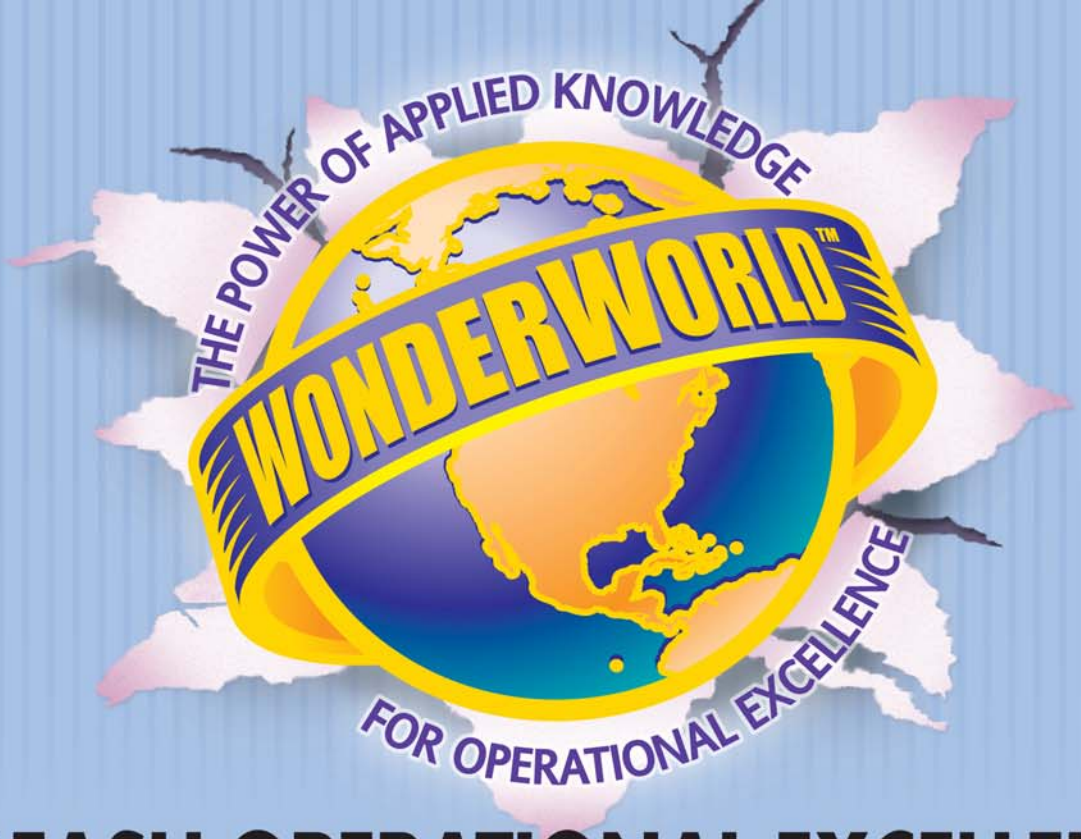
Global process companies will increasingly rely on these partner-vendors for turnkey management of their process automation needs, from system design through ongoing maintenance and support. Complementing the automation majors' global capabilities, the demand for localized or industry-focused system integration houses will continue to grow, as process companies find themselves unable to afford—or even find—the full-time expertise required when need arises.

These inevitable and ongoing shifts are both a *result of* and *response to* the overall demographic pressures on our profession—and it's all about doing more with less.

The system integrator solution is well suited to spreading automation expertise over multiple plants in one locale or multiple plants in a particular industry, and it is keeping the second tier automation vendors alive.

And on the supplier side, how better to leverage the intellectual capabilities of fewer process automation specialists than by packaging their expertise in application software and instrument firmware, and by extending their reach around the world and across companies and industries via remote access technologies?

Keith Larson,
VP Content, Putman Media



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LISTEN.
THINK.
SOLVE.

ValueStream

VOL. 4 NO. 3

FLSmidth

Remote Monitoring Cuts Phone Support Time, Ups Profits

Many manufacturers and original equipment manufacturers (OEMs) like you are providing better customer service, increasing profits and beating the competition by bundling services with their products. FLSmidth, based in Bethlehem, Pa., reduced phone technical support time and increased satisfaction for its cement industry customers by using remote monitoring for its cement unloading systems. This bundled service fills a critical need for many of its customers who don't have sufficient technical staff. The service also provides improved technical support and saves customers money.

Many other companies are taking advantage of bundled services. For example, General Electric makes a relatively small per-unit profit on jet engines, generators or

Continued on page 2

The operator interface shows data collected from all equipment in one location, so even a small staff can oversee the details of the entire pneumatic transfer.

Be Innovative To Cut Your Costs

Most of us have incorporated continuous manufacturing improvements to improve efficiency and reduce costs. We often implement these continuous improvements by making incremental changes to existing systems. This can produce significant leaps in performance.

Continuous improvement often involves incorporating new automation technology. Innovative, next-generation technology eventually replaces the technology we're currently using, not only replacing the older equipment at a lower cost, but adding significant enhancements that the older technology couldn't perform. This new technology sometimes comes from vendor upgrades, and sometimes from new inventions.

For example, systems based on our Logix control architecture can solve your control problems across your manufacturing

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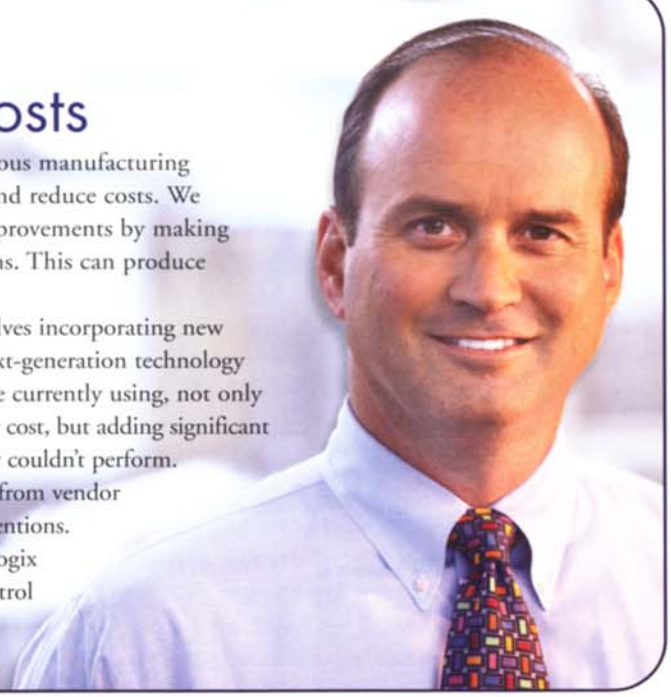

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Rockwell Automation

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floor, and even provide process and distributed control system (DCS) capabilities that traditional PLCs never could perform. These systems can provide real-time manufacturing data for a real-time look at the cost of operating each of your plants. You can use the information to dynamically shift production to the plant with the lowest current operating cost or excess capacity.

Another example of an innovative solution is to add data exchange between the plant floor and your enterprise resource planning (ERP) system. These manufacturing execution systems (MESs) can give you a real-time look at the cost of operating each of your plants so you can make informed decisions about how to improve performance.

Taking advantage of new technologies can eliminate existing problems and deliver unexpected benefits to improve your competitive position. These technologies can be applied in many areas of your manufacturing systems, from automation to electrical systems to information sharing. You benefit with lower total operations costs and higher profitability.

When looking for new and innovative solutions, a third-party point of view can provide a different perspective. Rockwell Automation is uniquely positioned to provide this perspective because we've gained experience at our own manufacturing operations. We've also gained a wealth of experience by operating across many industries and geographies. Many consultants can propose solutions, but few can draw from lessons learned in their own experience.

The articles in this issue of *ValueStream* show how we've delivered a range of solutions to help you lower the total cost of ownership in your manufacturing operations. Provide me with your feedback to let us know how we can continue to help you reduce your costs.

John McDermott
Senior Vice President
Rockwell Automation
Global Sales and Solutions
ragmsnewsletter@ra.rockwell.com

FLSmidth, *continued from page 1*

REMOTE MONITORING MAKES SERVICE PROFITABLE BY:

- Cutting technical staff travel costs.
- Reducing time required to solve problems.
- Allowing operators to anticipate and prevent problems.
- Reducing downtime.

locomotives. However, they earn much higher margins on service contracts typically sold with each unit.

REMOTE MONITORING IMPROVES SERVICE

Remote monitoring lets your experts remotely locate, analyze and anticipate problems with equipment at your customers' facilities. This allows you to make money while satisfying your customers' demands for after-sale equipment support.

FLSmidth implemented remote monitoring because the company's technicians were spending two to four hours per support call to help customers identify and solve problems. To reduce support time and increase customer satisfaction, the company applied networked and automated motor and system controls to its cement unloading systems.

The company now uses remote monitoring information to perform maintenance and troubleshooting. This service is critical because many of their customers don't have sufficient technical staff. FLSmidth engineers can remotely access the system's network, saving FLSmidth and its customers hours of phone troubleshooting and downtime. This translates to increased profits.

The heart of the Rockwell Automation remote monitoring system provided to FLSmidth is DeviceNet, a digital network that easily enables the use of remote monitoring. DeviceNet connects the Allen-Bradley CENTERLINE Motor Control Centers (MCCs) and other machinery components, such as valves, to the control system.

"The system has worked so well that we have had minimal customer support calls," says Larry Supon, FLSmidth electrical engineering and design group leader. "But, when we do get a call, we can log on remotely and immediately help our customer identify why the system faulted. We now deliver improved technical support and save our customers thousands of dollars."

"The [Rockwell Automation] system has worked so well that we have had minimal customer support calls .... We now deliver improved technical support and save our customers thousands of dollars." — *Larry Supon, FLSmidth*

Learn more about Allen-Bradley MCCs by visiting www.ab.com/mcc.

Read the full article about FLSmidth at www.ab.com/go/networkmcc.

Digital Control Network Slashes Downtime 95 Percent

Compared to traditional hard wiring, digital control networks can cut your total cost of ownership by reducing the number of wires that must be run from field devices to the central controller. For example, installing skid-mounted control systems for the printing industry can take weeks, but Eascan Industrial Controls & Automation, Inc., Winnipeg, Manitoba, cut installation of a collating machine from two weeks to three days for a North American printer by minimizing on-site wiring using DeviceNet and Allen-Bradley quick-connect devices. The modular design also helped reduce downtime by 95 percent while printing equipment was moved during the off-season.

Eascan designs and builds control systems for new and upgraded automated manufacturing processes. Though applicable to many industries, Eascan's modular plug-and-play systems are ideal for the time-sensitive printing industry.

Total cost-of-ownership savings are dramatic when printing equipment must be moved and reconfigured. "It's not uncommon for printing companies to modify and move machines every few years," says Jake Bergsma, president of Eascan. "This industry-wide requirement reinforced our need to design a modular system that would minimize downtime when modifying or relocating machines."

Premier School Agendas reduced its space requirements by 50 percent and cut labor costs by 30 percent by using a DeviceNet network to consolidate all I/O communications on a single cable.

MODULAR DESIGN CUTS DOWNTIME

Eascan designed and built a modular control system specifically for collators. At the heart of the new system is an Allen-Bradley PC-based controller from Rockwell Automation. All of the I/O, devices and drives on the collator are linked to the controller via DeviceNet.

This new Eascan control system was first installed for Premier School Agendas, North America's leading provider of full-color student planners. Because the collator was completely assembled and tested before commissioning, installation time was cut from two weeks to three days. Minimal on-site wiring was needed because of DeviceNet and quick-connects.

The new collator control system also provides additional flexibility when moving the collator. It takes technicians just three hours to disassemble, reassemble and test the control system, compared to up to four days with the old system.

"We try to coordinate machine moves during the off-season," reports Kyle Treichel, supervisor at Premier School Agendas. "However, if a machine must be moved during production, the new modular design can save us up to $10,000 in unscheduled downtime."

Learn about Allen-Bradley EtherNet/IP from Rockwell Automation at www.ab.com/networks/ethernet.html.

Read the full article about Eascan at www.ab.com/go/quickconnect.

DIGITAL NETWORKS CUT TOTAL COST OF OWNERSHIP BY:

- Reducing time required to reconfigure equipment.
- Providing extensive diagnostics.
- Reducing amount of wiring that must be installed and maintained.
- Reducing space requirements using a DeviceNet cable to consolidate I/O communications.
- Providing improved control system design and interface so operators can more rapidly input and change machine instructions.

New Plants Can Dramatically Decrease Operating Costs

Demand for wine from Australia's Yalumba Winery grew beyond its processing capacity, so management decided to build an additional wine processing plant that uses a sophisticated automation and control system. Supported by the Rockwell Automation Integrated Architecture, the automation system helped create a new winery that is environmentally friendly, incorporates the latest in winemaking technology and produces a consistent product using a unique process stream method.

The tradition of Yalumba's premium wine production and customer satisfaction continues with the help of a new Rockwell Automation control system.

BUILDING ON SUCCESS

Yalumba is Australia's oldest family-owned winery and one of the country's largest wine exporters. Its heritage-listed Angaston winery was founded in 1849 in South Australia's famous Barossa Valley.

The new wine processing plant, called the Moppa plant, took only two years to build. It commenced crushing in January 2005. An important goal for the new plant, which can process 30,000 tons of grapes per year, was to achieve a continuous production flow via a sophisticated automation and control system.

Moppa has attained this goal. The first white vintage is on the shelves in London. Yalumba Services Manager Peter Graue is responsible for maintaining the automated processing systems of both the Angaston and Moppa wineries. The secret, says Graue, is the automated process streams that ensure the grapes are fermented under optimum conditions at the required high throughput rate. It's also important to control the fermentation rate and minimize oxidation because both are highly dependent on temperature.

AUTOMATION IMPROVES PERFORMANCE

In keeping with the automation technology used successfully at the Angaston winery for more than a decade, Yalumba selected the Rockwell Automation Integrated Architecture as the foundation of Moppa's integrated control system. From a programming point of view, Integrated Architecture provides a common development environment for all applications using Rockwell Software FactoryTalk data-sharing software. The control and automation turnkey was designed by Camms Pty Ltd., a systems integrator referred by Rockwell Automation. The system performs sophisticated control of the numerous process streams while simultaneously allowing the winemakers to apply their experience to achieve the desired result.

The Rockwell Automation system helped create a new winery that is environmentally friendly and produces a consistent product.

Visit www.rockwellautomation.com/completeliar.html to learn more about Rockwell Automation Integrated Architecture. Learn more about Rockwell Software FactoryTalk at www.rockwellautomation.com/go/FactoryTalkStrategy.

HOW CAN INTEGRATED ARCHITECTURE HELP YOU?

Unlike traditional architectures, the Rockwell Automation Integrated Architecture reduces total cost of ownership by using a single control infrastructure for the entire range of factory automation applications, large or small.

This enables you to reuse engineering designs and practices to reduce development time and cost, quickly respond to customer or market demands, reduce maintenance costs and downtime, and easily access plant and production data from business systems for better management decision-making.

Reduce Electric Usage for Substantial Savings

Intelligent energy management systems and selective electrical system upgrades lower your electricity bills and provide you with quick paybacks. Rockwell Automation is uniquely positioned to offer the products and services required to reduce your electrical consumption, increase your uptime and extend your equipment life. For example, a brewing manufacturer took advantage of Rockwell Automation Power and Energy Management Services (PEMS) and achieved an annual cost savings of $360,000 with a 12-month return on investment (ROI).

An automotive manufacturer used Rockwell Automation services to track utility consumption and verify utility billing. After an initial capital investment of $166,000, the manufacturer achieved a 56 percent savings each year and a 22-month payback period.

Today's electric service market is no longer a monolithic power company with one fixed rate schedule. Knowledge is power when dealing with the power company. Your plant managers must react or risk obsolescence.

TEN REASONS TO OPTIMIZE ELECTRICAL CONSUMPTION

1. Reduce electricity usage and costs.
2. Identify best areas for equipment upgrades to cut energy use.
3. Identify rebate opportunities from electricity providers.
4. Identify inefficiencies in operations.
5. Reduce downtime by operating equipment with greater efficiency.
6. Improve power quality.
7. Reveal defects in equipment operation.
8. Provide critical design data for plant expansions and upgrades.
9. Reduce peak demand and resulting utility company penalties.
10. Determine if power should be purchased from or sold to the grid.

EXPERTISE AT YOUR FINGERTIPS

Your power company is actively trying to force you to reduce energy use and smooth demand by administering severe penalties and attractive rebates.

Effective dealings with your electricity provider require an intelligent energy management system. Such a system provides information and knowledge — information about what's happening and the knowledge to do something about it.

Rockwell Automation can provide an intelligent energy management system because they can supply the full range of products and services required.

An intelligent energy management system is the best solution for optimizing plant energy usage. However, you can realize dramatic savings by calling in Rockwell Automation experts to analyze main areas of energy usage and recommend best solutions.

Rockwell Automation software can evaluate where, when and how electrical energy is being consumed, giving your managers the ability to act on this knowledge.

SAVING MILLIONS

Andy Gagnon, Allen-Bradley motor product manager, illustrates how pinpointing problems can save millions. "One of our customers is assessed a fee of $1,500 each time they start a compressor motor," he says.

"Upgrading to variable-frequency drives will save them the entire fee amount for every compressor motor start. That, coupled with the ability to run their compressors at 10 percent lower speed with the same product output, will net them energy savings of $6.9 million over a three-year period," adds Gagnon.

Visit www.rockwellautomation.com/solutions/pems to learn how the power and energy management consulting services available from Rockwell Automation can help you reduce your electric energy costs.

Water Utility Power Management Cuts Costs

Water utilities need more efficient technologies to run their large motors and pumps. Once the water pumping system's electrical consumption exceeds a pre-set limit, energy companies must charge peak prices for the full month's billing period, even if the peak demand period only totaled half an hour during that month. This cost can contribute significantly to the total operating cost of the facility. The Monroe County Water Authority, Rochester, N.Y., achieved annual energy savings of more than $23,000, plus a $17,500 government rebate, when it used a variable-frequency drive (VFD) to automatically control the speed of the pump motor to run at full horsepower only when necessary and at lower speeds during nonpeak times.

The Allen-Bradley PowerFlex 7000 750-hp medium voltage variable-frequency drive allows easy access for maintenance.

VARIABLE-FREQUENCY DRIVES CUT TOTAL COST OF OWNERSHIP BY:

- Reducing peak electrical consumption.
- Extending life of the controlled motor and related equipment.
- Garnering utility company rebates.

The MCWA produces an average of 62 million gallons of drinking water every day. The Lee Road Pump Station pumps an average of 19 million gallons per day.

The original system at the Lee Road Pump Station had five medium voltage motors, ranging from 400 hp to 700 hp. The water authority would frequently have to run the 700-hp motor at peak times of the day. This caused an accumulation of peak demand charges over several billing cycles, as well as equipment stress.

The need to reduce energy costs and a New York state rebate program motivated the water authority to upgrade its 700-hp centrifugal pump and motor with a VFD.

VFD CUTS COSTS

The water authority now runs the 700-hp motor more consistently and at a lower speed. The new drive smoothly ramps up the motor to prevent motor wear and allows the motor and pump to run only when necessary, saving thousands of dollars a year in energy costs.

The new drive smoothly ramps up the motor to prevent wear and allows the motor and pump to run only when necessary, saving thousands of dollars a year in energy costs.

Prior to the VFD installation, the system used 590 kW per month, with electrical costs of $278,000 annually. After installation, the system used only 360 kW monthly, at a cost of $255,000 annually. The State of New York also rebated $17,500 to the water authority.

Learn about the power and energy management consulting services available from Rockwell Automation at www.rockwellautomation.com/solutions/pems.

Read the full article about the MCWA at www.ab.com/go/peakdemand.

Intelligent Control Saves Waste Company $100,000 per Day

Medical waste company Stericycle, headquarted in Lake Forest, Ill., saves $100,000 per day in downtime costs and cut $50,000 in annual travel expenses for on-site troubleshooting visits by installing Allen-Bradley IntelliCENTER Motor Control Centers (MCCs) from Rockwell Automation. The MCCs are networked for ease of wiring, which reduces installation time and maximizes uptime. They also perform remote diagnostics, reducing the number of international troubleshooting trips required by maintenance staff.

Using intelligent control systems to review and preview plant operations can help keep your plants one step ahead of the competition.

Stericycle was able to reduce the number of international on-site troubleshooting visits and save at least $50,000 in fares and other costs associated with such travel.

KEEP YOUR MOTORS RUNNING

Stericycle uses the intelligent MCCs to access operating information from its global facilities. The company's facilities move up to 100 tons per day of medical waste, and timely motor status data are the key to uptime. A typical plant includes motor-driven lines to process, sterilize and recycle medical waste.

BENEFITS OF NETWORKED INTELLIGENT CONTROL

- Decreased control unit installation time from four weeks to one week.
- Reduced troubleshooting travel from two trips per month to four per year, saving $50,000.
- Increased uptime and slashed downtime costs $100,000 per day.

Proper disposal of medical waste relies on performance-assured motor control. Motor failure means costly downtime and tedious cleanup in a biohazardous environment. Motor status information helps minimize failures and associated downtime costs. "In our industry, there's no such thing as too much information," says Rick Albright, chief electrical engineer with Stericycle.

Stericycle saves $5,000 to $7,000 per hour in downtime costs.

Intelligent MCCs and accompanying software allow operators to program predictive failure alerts. If a motor trips, users can analyze the last five minutes of operation and prevent future occurrences.

Another significant benefit gained from intelligent MCCs is remote diagnostics. Stericycle is able to view and monitor real-time system MCC status information worldwide.

"We replaced existing MCC buckets with three brand-new, different ones," says Albright. "I was able to dial in remotely and configure the new bucket nodes in 90 minutes."

Stericycle saves $5,000 to $7,000 per hour in downtime costs. Most medical waste facilities run a minimum of 10 hours and up to 18 hours per day.

Remote MCC monitoring and troubleshooting also provide a more cost-efficient solution than conducting on-site visits. "I was making two troubleshooting trips a month and, thanks to the intelligent MCCs, I'm making three or four trips a year," reports Albright. "That saves us $50,000 a year in plane fares alone, as well as reducing other logistical costs associated with international travel."

Read the full article about Stericycle at www.ab.com/go/intellicenter.

Learn more about Allen-Bradley MCCs by visiting www.ab.com/mcc.

HVAC Automation Cuts Energy Use by 15 Percent

New York's Lewis County General Hospital lowered its total cost of building automation and reduced its energy use 15 percent by integrating Allen-Bradley PowerFlex 400 AC Drives from Rockwell Automation with

The energy savings allowed the hospital to recoup its investment in six months.

building automation, energy management and HVAC systems to optimize fan and pump performance throughout its facilities. The energy savings and a government electric utility rebate allowed the hospital to recoup the drive investment in less than six months.

The total cost to purchase and install the drives on four air handlers in the HVAC system was $32,000, with New York State Energy Research and Development Authority (NYSERDA) paying for half of the total cost based on the projected energy savings.

As part of the solution, the hospital installed a computer-based energy management system to track temperatures and energy use throughout the facility. The collected data helped assess where improvements could be made.

Visit www.ab.com/drives/powerflex to learn about the range of Allen-Bradley PowerFlex drives from Rockwell Automation.

The HVAC automation system met the hospital's goal of maintaining patient care and comfort at the highest level. Additionally, the improvements balanced equal payback with energy savings.

Read the full article about Lewis County Hospital and Residential Health Care Facility at www.ab.com/go/powerflex400.

To have a Rockwell Automation representative contact you, please call (952) 946-7025 ext. 1815.
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Power, Control and Information Solutions Headquarters

Americas: Rockwell Automation, 1201 South Second Street, Milwaukee, WI 53204-2496 USA,Tel: (1) 414.382.2000, Fax: (1) 414.382.4444
Europe/Middle East/Africa: Rockwell Automation SA/NV, Vorstlaan/Boulevard du Souverain 36, 1170 Brussels, Belgium,Tel: (32) 2 663 0600, Fax: (32) 2 663 0640
Asia Pacific: Rockwell Automation, Level 14, Core F, Cyberport 3, 100 Cyberport Road, Hong Kong, Tel: (852) 2887 4788, Fax: (852) 2508 1846

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